

BUILDING INFORMATION MODELING [BIM] FOR G+ FIRST FLOOR RESIDENTIAL BUILDING

A PROJECT REPORT

*Submitted in partial fulfilment of the requirement for the
award of the degree of*

BACHELOR OF TECHNOLOGY IN CIVIL ENGINEERING

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CERTIFICATE

Certified that this is the bonafide report of the project work entitled

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IN

CIVIL ENGINEERING

During the year 2023-2024

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ABSTRACT

This abstract presents a summary of our comprehensive report on BIM, exploring its applications, benefits, challenges, and implications for the construction industry.

The report begins with an overview of BIM, delineating its fundamental concepts, components, and functionalities. We delve into the significance of BIM as a collaborative platform that facilitates the creation and management of digital representations of building projects.

Throughout the report, we analyze the multifaceted benefits of BIM implementation, including enhanced project coordination, improved decision-making, increased cost efficiency, and heightened sustainability. We also address the challenges associated with BIM adoption, such as initial investment costs, interoperability issues, and the need for upskilling within the workforce.

Furthermore, our research investigates the impact of BIM on project delivery, highlighting its role in streamlining workflows, reducing conflicts, and improving project outcomes. Case studies and best practices are presented to illustrate successful BIM implementations across diverse project types and scales.

CHAPTER 1

INTRODUCTION

Building Information Modeling (BIM) is an intelligent model-based process that provides insight for creating and managing building projects faster, more economically, and with less environmental impact.

It is a process of creating and managing 3D building data during its development. BIM is a complex multiphase process that gathers input from team members to model the components and tools that will be used during the construction process to create a unique perspective of the building process.

The 3D process is aimed at achieving savings through collaboration and visualization of building components into an early design process that will dictate changes and modifications to the actual construction process.

Over the year the industry has commercialized BIM towards architectural related professionals, however, the real purpose and benefits of BIM relate to all construction industry professionals. The 3D representation of the building and now used in roads and utilities too and is geared towards all construction professionals, and all of you are responsible for understanding the process and participate in providing input to the software.

BIM makes a reliable digital representation of the building available for design decision making, high-quality construction document production, construction planning, performance predictions, and cost estimates. Not only, that BIM can also be used by the property owners, once the construction process has ended, to carefully monitor how the building is performing and to complete repairs efficiently. The building information modeling process covers geometry, space, light, geographic information, quantities, and properties of building components. BIM can be used to demonstrate the entire building life cycle, including the processes of construction and facility operation

HISTORY OF BIM

In 1957, Dr. Patrick J. Hanratty had developed the first commercial CAM (Computer Aided Machining) program. The first CAD software with a graphical interface was Sketchpad, developed in 1963 by Ivan Sutherland.

- * During the 1960s, Hanratty himself developed DAC, a CAD system, while working for General Motors Research.
- * During the 1970s the transformation from 2D to 3D began.
- * During the 1980s the AutoCAD, CATIA, Pro/Engineer, Unigraphics and I-DEA became the leading CAD software packages.

During the 1990s the transition from 2D to 4D CAD began. That is the transition zone to the BIM.

NEED OF BIM

BIM provides a way to work concurrently on most aspects of building life cycle process.

Provides a way to change traditional architectural phases and data sharing.

- * The modeling process integrates actual construction pieces and parts BIM can be used as a tool to estimate and complete construction cost forecasting used to monitor actual building performance data
- * The software can be used to collect data on warranty, aging, defects, and installation time
- * Determine whether a temporary construction set up is needed
- * Can be used to sequencing planning or determining how the phases of a project should be scheduled.
- * Detect or avoid construction and design issues early in the game preventing change orders .

BIM APPLICATIONS

The BIM application process can be used during design and architecture process creating a clear picture used for better and more integrated designs.

The software will be used to foresee problems and coordination between different contractors and as a way to generate construction documents and process that will later be implemented during the physical process. It is ideal when there are many trades executing at the same moment or when schedules are compressed. There are multiple applications for BIM so it can be used by the following groups:

- Architecture
- Sustainability
- Structures
 - Detailed design
 - Design Analysis
- MEP
- Construction Management
 - Time Management (PERT & CPM)
- Utilities
- Road Construction
- Scheduling
 - Cost Analysis & Quantity schedule
- Property Management
- Documentation

Industry groups are trying to develop one standardized BIM model that can be used to integrate all different types of modeling systems. By doing this, they will facilitate the coordination and communication in the design-construction-operation team under one single platform. The purpose of this movement is to create a single data center, with multiple CAD and specs depending on the discipline that you are working for.

All data will then come together allowing it to be used for take-offs, analysis, coordination and important project milestones. This effort will help standardize the process and will

establish a base that can be used during the bidding process so everyone can be judged using some standard guidelines.

The Building Smart Alliance, a council of the National Institute of Building Sciences, in Washington, D.C., is leading these efforts towards a National BIM Standard.

BIM OVER CAD

One of the main advantages to introducing a design method that will resonate with everyone from project managers to contractors and developers is cost savings. With BIM being mandated in Singapore for several years, Red stack has gained significant experience implementing BIM and is seeing BIM deliver significant cost savings for design and construction projects.

In the days of 2D drawing, it was impossible to fully visualize what a project would actually look like until it was built. By then, minor issues that could be easily spotted and amended with BIM solutions have become expensive headaches instead. Not only do they cost money to fix, they take time as well. If there's anything worse than a project that's over budget, it's one that's also late.

In our experience, contractors choose BIM solutions because they can achieve a cost saving of between 10 and 12 per cent over the course of a contract.

BIM is not a software product, but rather a methodology that seeks to link all parts of the design and construction process, ensuring that any problems can be reworked before contractors break ground.

It does this through a series of steps to ensure there is consistency throughout the design process. These are:

Visualization - see how the drawings look and ensure they are viable.

Coordination - work out how the design will manifest once construction starts.

Collaboration get architects, project managers and contractors working with consistent

information.

Aside from missing out on the collaboration advantages and cost savings BIM delivered, BIM is being mandated on projects around the world and those that are not BIM ready are missing out on winning contracts. BIM has been mandated in Singapore for several years now and we have seen the early adopters of BIM thrive while those reluctant to change have struggled to remain competitive. We are seeing the same trend emerge around the world. Those who are implementing BIM now are more likely to survive and thrive in the future.

BIM MATURITY LEVELS

The concept of "BIM Levels" has become the 'accepted definition of what criteria are required to be deemed BIM-compliant, by seeing the adoption process as the next steps in a journey that has taken the industry from the drawing board to the computer and, ultimately, into the digital age.

Level 0 BIM

This level is defined as unmanaged CAD. This is likely to be 2D, with information being shared by traditional paper drawings or in some instances, digitally via PDF, essentially separate sources of information covering basic asset information. The majority of the industry is already well ahead of this now.

Level 1 BIM

This is the level at which many companies are currently operating. This typically comprises a mixture of 3D CAD for concept work, and 2D for drafting of statutory approval documentation and Production Information. CAD standards are managed to BS 1192:2007, and electronic sharing of data is carried out from a common data environment (CDE), often managed by the contractor. Models are not shared between project team members.

Level 2 BIM

This is distinguished by collaborative working-all parties use their own 3D CAD models, but

not necessarily working on a single, shared model. The collaboration comes in the form of how the information is exchanged between different parties and is the crucial aspect of this level. Design information is shared through a common file format, which enables any organisation to be able to combine that data with their own in order to make a federated BIM model, and to carry out interrogative checks on it. Hence any CAD software that each party used must be capable of exporting to one of the common file formats such as IFC (Industry Foundation Class) or COBie (Construction Operations Building Information Exchange). This is the method of working that has been set as a minimum target by the UK government for all work on public-sector work, by 2016.

Level 3 BIM

Currently seen as the holy grail, this represents full collaboration between all disciplines by means of using a single, shared project model which is held in a centralized repository. All parties can access and modify that same model, and the benefit is that it removes the final layer of risk for conflicting information. This is known as 'Open BIM'. Current nervousness in the industry around issues such as copyright and liability are intended to be resolved-the former by means of robust appointment documents and software originator/read/write permissions, and the latter by shared-risk procurement routes such as partnering. The CIC BIM Protocol makes provision for these.

TYPES OF BIM

BIM 3D-PARAMETRIC DATA IN COLLABORATIVE MODEL

BIM revolves around an integrated data model from which various stakeholders such as Architects, Civil Engineers, Structural Engineers, MEP System Engineers, Builders, Manufacturers and Project Owners can extract and generate views and information according to their needs. 3D BIM's visualizations capabilities enables participants to not only see the building in three dimensions before ground is ever broken, but also to automatically update these views along the project life cycle, from earliest conception to demolition. BIM 3D helps participants to manage their multidisciplinary collaboration more effectively in modelling and analysing complex spatial and structural problems. Furthermore because accurate data can be collected along the project life cycle, and stored in the Building Information Model, new

value can be added to predictive models allowing to resolve issues proactively.

Benefits

1. Improved visualization of the project, communication of design intent
2. Improved multidisciplinary collaboration
3. Reduced rework

BIM 4D-SCHEDULING

4D-BIM (four-dimensional building information modelling) is used for construction site planning related activities. The fourth dimension of BIM allows participants to extract and visualize the progress of their activities through the lifetime of the project. The utilization of 4D-BIM technology can result in improved control over conflict detection or over the complexity of changes occurring during the course of a construction project. 4D BIM provides methods for managing and visualizing site status information, change impacts as well as supporting communication in various situations such as informing site staff or warning about risks.

Benefits

Integrating BIM with 4D CAD simulation models bring benefits to participants in terms of planning optimization. Builders and manufacturers can optimize their construction activities and team coordination.

BIM 5D-ESTIMATING

5D-BIM (fifth-dimensional building information modelling) is used for budget tracking and cost analysis related activities. The fifth dimension of BIM associated with 3D and 4D (Time) allows participants to visualize the progress of their activities and related costs over time.

The utilization of 5D-BIM technology can result in a greater accuracy and predictability of project's estimates, scope changes and materials, equipment or manpower changes. 5D BIM provides methods for extracting and analysing costs, evaluating scenarios and changes impacts.

Benefits

Integrating BIM with 5D CAD simulation models enables the development of more efficient, cost-effective and sustainable constructions.

BIM 6D-SUSTAINABILITY

6D-BIM (sixth-dimensional building information modelling) helps perform energy consumption analyses. The utilization of 6D-BIM technology can result in more complete and accurate energy estimates earlier in the design process. It also allows for measurement and verification during building occupation, and improved processes for gathering lessons learned in high performance facilities.

Benefits

Integrating BIM with 6D CAD simulation models leads to an overall reduction in energy consumption.

BIM 7D-FACILITY MANAGEMENT

7D-BIM (seventh-dimensional building information modelling) is used by managers in the operation and maintenance of the facility throughout its life cycle. The seventh dimension of BIM allows participants to extract and track relevant asset data such as component status, specifications, maintenance/operation manuals, warranty data etc. The utilization of 7D-BIM technology can result in easier and quicker parts replacements, optimized compliance and a streamlined asset life cycle management over time. 7D BIM provides processes for managing subcontractor/supplier data and facility component through the entire facility life cycle.

Benefits

Integrating BIM with 7D CAD simulation models optimizes asset management from design to demolition.

OBJECTIVES

- To study about the various BIM software.
- To study about Industry Foundation Classes(IFC)

CHAPTER 2

LITERATURE REVIEW

A.H.Oti et al.(2016) studied The provision of Application Programming Interface (API) in BIM-enable tools can contribute to facilitating BIM-related research. APIs are useful links for running plug-ins and external programs but they are yet to be fully exploited in expanding the BIM scope. The modelling of n-Dimensional (nD) building performance measures can potentially benefit from BIM extension through API implementations. Sustainability is one such measure associated with buildings. For the structural engineer, recent design criteria have put great emphasis on the sustainability credentials as part of the traditional criteria of structural integrity, constructability and cost. This paper examines the utilization of API in BIM extension and presents a demonstration of an API application to embed sustainability issues into the appraisal process of structural conceptual design options in BIM. It concludes that API implementations are useful in expanding the BIM scope. Also, the approach including process modelling, algorithms and object-based instantiations demonstrated in the API implementation can be applicable to other nD building performance measures as may be relevant to the various professional platforms in the construction domain.

Miyoung Ohm et al.(2017) studied The emergence of building information modeling (BIM) has generated several BIM jobs. However, despite opinions by BIM experts, questions regarding BIM jobs and their competencies still have no clear solution. This paper addresses this question by the collection and analysis of 242 online job postings, written in English, from the US, the UK, and China. These 242 job postings comprised a total of 32,495 words, from which 35 types of job titles and 5,998 terms related to job competency were extracted. Sequentially, the 35 job types were classified into eight BIM job types by analyzing the relations between the job titles using the role and position analysis of social network analysis. The eight BIM job types were BIM project manager, director, BIM manager, BIM coordinator, BIM designer, senior architect, BIM mechanical, electrical, and plumbing (MEP) coordinator, and BIM technician. The 5,998 competency-related terms were categorized into 43 competency elements using the O*NET classification as a framework for analysis. The 43 competencies were then subcategorized into essential, common, and job-specific competencies for the eight BIM job types. The findings of this paper could contribute

to the research, industry, and academia by a) providing researchers with a scientific foundation for conducting studies related to BIM jobs and competence in the future; b) setting up guidelines for recruiting and training BIM experts in the industry; and c) allowing universities to develop BIM-related courses depending on their educational goals.

Yang Zou et al.(2016) studied Risk management in the AEC (Architecture, Engineering and Construction) industry is a global issue. Failure to adequately manage risks may not only lead to difficulties in meeting project objectives but also influence land-use planning and urban spatial design in the future growth of cities. Due to the rapid development and adoption of BIM (Building Information Modelling) and BIM-related digital technologies, the use of these technologies for risk management has become a growing research trend leading to a demand for a thorough review of the state-of-the-art of these developments. This paper presents a summary of traditional risk management, and a comprehensive and extensive review of published literature concerning the latest efforts of managing risk using technologies, such as BIM, automatic rule checking, knowledge based systems, reactive and proactive IT (information technology)-based safety systems. The findings show that BIM could not only be utilised to support the project development process as a systematic risk management tool, but it could also serve as a core data generator and platform to allow other BIM-based tools to perform further risk analysis. Most of the current efforts have concentrated on investigating technical developments, and the management of construction personnel safety has been the main interest so far. Because of existing technical limitations and the lack of "human factor" testing, BIM-based risk management has not been commonly used in real environments. In order to overcome this gap, future research is proposed that should:

- (1) have a multi-disciplinary system-thinking
- (2) investigate implementation methods and processes
- (3) integrate traditional risk management with new technologies
- (4) support the development process

Olugbenga O. Akinade et al.(2016) studied the future directions of effective Design for Deconstruction (DfD) using BIM-based approach to design coordination. After a review of extant literatures on existing DfD practices and tools, it became evident that none of the tools

is BIM compliant and that BIM implementation has been ignored for end-of-life activities. To understand how BIM could be employed for DfD and to identify essential functionalities for a BIM-based deconstruction tool, Focus Group Interviews (FGIs) were conducted with professionals who have utilised BIM on their projects. The interview transcripts of the FGIs were analysed using descriptive interpretive analysis to identify common themes based on the experiences of the participants. The themes highlight functionalities of BIM in driving effective DfD process, which include improved collaboration among stakeholders, visualisation of deconstruction process, identification of recoverable materials, deconstruction plan development, performance analysis and simulation of end-of-life alternatives, improved building lifecycle management, and interoperability with existing BIM software. The results provide the needed technological support for developing tools for BIM compliant DfD tools.

Yujie Lu et al.(2017) studied the applications of BIM for the development of green buildings, the activity of making buildings in a way that protects the natural environment. As the usefulness of BIM has been widely recognized in the building and construction industry, there is an urgent need to establish an up-to-date synthesis on the nexus between BIM and green buildings. After an in depth review of hundreds of journal articles published from 1999 to 2016 and 12 widely used types of BIM software, this study provides a holistic understanding and critical reflection on the nexus between BIM and green buildings, which is systematically illustrated by a "Green BIM Triangle" taxonomy. The proposed taxonomy indicates that the nexus between BIM and green buildings needs to be understood based on three dimensions, namely project phases, green attributes and BIM attributes. Following the proposed taxonomy, this paper systematically illustrated

- 1) The applications of BIM in supporting the design, construction, operation, and retrofitting processes of green buildings
- 2) The various functions of BIM for green building analyses such as energy, emissions, and ventilation analysis
- 3) The applications of BIM in supporting green building assessments (GBA)
- 4) Research gaps and future research directions in this area. Through critical review and synthesis of BIM and green buildings based on evidence from both academic research and industrial practices, this paper provides important guidance for building researchers and practitioners to better align BIM development with green building development in the future.

Hasan Burak Cavka et al.(2016) studied that Building information modeling (BIM) is emerging as a potential solution for facility owners to address the challenges of poor information fidelity, interoperability, and usability in project delivery to support the lifecycle of their assets' information. Despite the many benefits offered by BIM, its use for facility operations remains significantly limited. The reality is that implementing BIM in large owner organisations is a complex challenge. In particular, a significant barrier to BIM adoption for owners is the challenge of identifying and formalizing the information requirements needed to support model-based project delivery and asset management. This paper presents the results of a longitudinal research project that investigated two large owner organisations in Canada to better understand the process of developing and formulating BIM requirements to support the lifecycle of their assets. Specifically, the objectives were to formalize an iterative approach to the identification and characterization of owner requirements and to develop a conceptual framework that would relate digital and physical products to owner requirements and organisational constructs, to underpin the formalization process. As part of this research an array of requirements documentation were analysed, interviews were performed with numerous facility management personnel, and BIMs from four projects were analysed. A methodology is introduced to support a rigorous and detailed analysis of BIM requirements. The investigation of the owner requirements helped to develop an understanding of the required information content, and its alignment with BIM.

Finally the relationships between organisational constructs, owner requirements, and BIM

Shahryar Habibi(2017) studied that In order to raise awareness of the role of building information modeling (BIM) in improving energy efficiency and comfort conditions, the work introduces a strategy of combining building simulation tools and optimization methods. Furthermore, it emphasizes the fact that a combination of these strategies with BIM can improve not only the construction process but also enable exploration of alternative approaches. The work discusses the potential application of data integration methodology for an office environmental focuses on the review of the potential performance of integrated systems. It also explains how BIM can help facilitate review of results and methods for improving building performance in terms of energy efficiency and indoor environmental quality.

Shabtai Issac et al.(2017) studied that the fact that a large portion of the work in construction projects is usually carried out by different subcontractors, makes an effective work packaging process crucial for the subsequent execution planning. However, the definition of optimal work packages is currently challenging and time consuming. A method was developed to allow the work packaging process to be carried out in a more accurate and efficient way, using data from Building Information Models (BIM). This method is based on a bottom-up approach that can take into account relations between specific components, and the consequent interruptions that will occur in the construction processes. The method incorporates BIM data in Design Structure Matrices and Domrin Mapping Matrices to automatically generate a list of proposed work packages with minimal interfaces. An application of the method in a case study demonstrated that it can accelerate the work packaging process, and allow alternative solutions to be explored at an early stage in the project.

Daniele Parrone et al.(2017) studied that the seismic performance of non-structural elements is nowadays recognized to be a key issue in performance based earthquake engineering. The knowledge of construction details within a building is of paramount importance in order to reduce uncertainties and improve the quality of the analysis and design, particularly in regards to non-structural elements. The use of Building Information Modelling (BIM) could represent a new frontier in the seismic design of non-structural elements by increasing the reliability of the seismic design and/or assessment. This study discusses the effectiveness of using Building Information Models in seismic design of non- structural building elements. A simple tool has been developed to perform automatically the seismic design of sway braces for pressurized fire suppressant sprinkler piping systems based on information extracted from a Building Information Models. The effectiveness of the proposed procedure was validated via a case study were mapped. As the construction industry shifts towards model-based project delivery, this research will inform owners about how to think about handover of digital facility models, and what to require in models based on their specific needs.

Mohammed Kaseem et al.(2016) studied that the adoption of Building Information Modelling (BIM) across markets is a pertinent topic for academic discourse and industry attention. This is evidenced by the unrelenting release of national BIM initiatives, new BIM protocols, and candidate international standards. This paper is the second part of an ongoing Macro BIM Adoption study: the first paper "Macro BIM Adoption: Conceptual Structures" (Succar and Kasseem, 2015) introduced five conceptual models for assessing macro BIM adoption across

markets and informing the development of BIM adoption policies. This second paper clarifies how these models are validated through capturing the input of 99 experts from 21 countries using a survey tool, highlights the commonalities and differences between sample countries with respect to BIM adoption; and introduces sample tools and templates for either developing or calibrating BIM adoption policies. Survey data collected indicate that all five conceptual models demonstrate high levels of clarity', 'accuracy" and 'usefulness', the three metrics measured. They also indicate (1) varying rates of BIM diffusion across countries with BIM capability near the lower-end of the spectrum, (2) varying levels of BIM maturity with the mean of most macro BIM components falling below the medium level; (3) varying diffusion dynamics across countries with the prevalence of the middle-out diffusion dynamic; (4) varying policy actions across countries with a predominance of the passive policy approach; and (5) varying distribution of diffusion responsibilities among player groups with no detectable dominant pattern across countries.

Xiao Li et al.(2016) studied that Building Information Modeling (BIM) has been recognized as an emerging technological innovation which can help transform the construction industry and it has been adopted broadly in the field of built environment. Due to the rapid development of BIM research, various stakeholders require a state-of-the-art review of the BIM research and implementation. The purpose of this paper is to provide an objective and accurate summary of BIM knowledge using 1874 published BIM-related papers. The results show that 60 key research areas, such as information systems, 3D modeling, design and sustainability and 10 key research clusters, such as architecture design studio, building information and lean construction, are extremely important for the development of BIM knowledge. The results are useful for the identification of research clusters and topics in the BIM community. More importantly, these results can help highlight how BIM-related research evolves over time, thus greatly contributing to understanding the underlying structure of BIM. This study offers useful and new insights to summarize the status quo of BIM knowledge and can be used as a dynamic platform to integrate future BIM developments.

Dolli Mansuri et al.(2017) studied that the formwork systems are accountable for a significant share of the cost of reinforced concrete structures. The application of constructability principles to the design, selection and management of formwork systems in the preconstruction phase can significantly reduce the cost of reinforced concrete construction projects. Although many studies have developed tools and methodologies to automate the design and selection of

formwork systems, few studies have explored the benefits of improving the process of managing formwork. The focus of this paper is on the use of BIM along with a cascading tool to maximize the return on formwork investment and improve the management of formwork. This paper presents an approach to utilize data drawn from the building information models coupled with a cascading algorithm to efficiently manage the formwork inventory on a construction project by generating a scheduled formwork reuse plan and calculating the minimum quantity of formwork required for the project. The paper discusses the use of BIM to extract data required for the cascading tool, working of the cascading algorithm and the development of the tool. The paper ends by presenting a case study where the developed tool was applied on a construction project in Cincinnati, Ohio and 13% savings in formwork material cost was reported.

Weisheng Lu et al.(2015) studied that There is a lively debate on the application of Building Information Modelling (BIM) to construction waste management (CWM). BIM can be utilized as a less expensive, virtual, and computational environment to enable designers to ponder different design options, or contractors to evaluate different construction schemes, both with a view to minimizing construction waste generation. However, existing debate on this topic too frequently treats BIM as a cure-all silver bullet; without some major hurdles being adequately addressed, the applications of BIM will remain rhetorical. This paper aims to demystify BIM's computational application to CWM. Based on a critical literature review, a prototypical framework of a computational BIM for CWM is delineated, within which the two key prerequisites of 'information readiness' and *computational algorithms are highlighted. Then, the paper details the required information and how it can be organized in a standalone database or encapsulated in existing BIM for CWM. Learning from the historical development of data infrastructure in the field of BIM based cost management, the process to develop the required information is likely to be tortuous but is unavoidable. The paper further explores computational BIM algorithms that can manipulate the information to facilitate decision-making for CWM. Finally, the operation of computational BIM is elaborated by relating it to various prevailing procurement models within which BIM applications are contextualized. Although the framework reported here has been substantially developed for experimental application,, it is not to be taken as an immediately applicable solution but rather as an illustration of the kind of platform on which future development of computational BIM for CWM can proceed in a more efficient and effective fashion.

Pawel Nowak et al.(2015) studied that the possibilities of Building Information Modeling (BIM) techniques and relevant software for decision making optimization in construction. Some relevant description of BIM elements needed for optimization in construction investment process. Authors presents chosen tools for decision making - point of reference method. Paper consist also practical example of suggested methodology use choice of the best location of the office building.

Lieyun Ding et al.(2014) studied that the utilization of Building Information Modeling (BIM) has been growing significantly and translating into the support of various tasks within the construction industry. In relation to such a growth, many approaches that leverage dimensions of information stored in BIM model are being developed. Through this, it is possible to allow all stakeholders to retrieve and generate information from the same model, enabling them to work cohesively. To identify gaps of existing work and evaluate new studies in this area, a BIM application framework is developed and discussed in this paper. Such a framework gives an overview of BIM applications in the construction industry. A literature review, within this framework, has been conducted and the result reveals a research gap for BIM applications in the project domains of quality, safety and environmental management. A computable multi-dimensional (nD) model is difficult to establish in these areas because with continuously changing conditions, the decision making rules for evaluating whether an individual component is considered good quality, or whether a construction site is safe, also vary as the construction progresses. A process of expanding from 3D to computable nD models, specifically, a possible way to integrate safety, quality and carbon emission variables into BIM during the construction phase of a project is explained in this paper. As examples, the processes of utilizing nD models on real construction sites are described. It is believed to benefit the industry by providing a computable BIM and enabling all project participants to extract any information required for decision making. Finally, the framework is used to identify areas to extend BIM research.

CHAPTER 3

METHODOLOGY

The BIM methodology involves the coordination of different technologies for project management through a single 3D digital model that shortens the times of both the design and the production, and therefore it reduces costs. It also implies a new way of coordinating the different teams involved, improving the quality of the engineering projects, architecture and construction.

In this article we will disaggregate the changes that have taken place in the working and project development methods, optimizing the process, streamlining phases of the project and achieving a more linear and collaborative workflow.

In the early stages, where the project evolves and is being generated, the BIM methodology helps us to easily extract floors and sections from a single 3D model. This model is the germ of the project and thanks to the various displays, it allows the understanding of the different proposals by the customer and by our own team, and all that in real time.

Because of the importance of the project's implementation on its place and its adaptation to the environment, is valuable the information we can get about the energy evaluation of the building. In this way from the initial phases comparatives of different sustainable solutions are generated, allowing us to select the most appropriate from the conceptual phase. For this, we use the BIM model to study the optimum orientation of the rooms, the amount of solar radiation and lower environmental impact (6D). We may, if necessary, export the model to specific tools that complement the information obtained from the native model.

The coordination of the design team begins gaining importance in the intermediate stages. From the start, domains and the ability to modify the different design elements of each team member, should be managed. BIM model and an open environment, enables various design teams simultaneously design different parts of the project, without getting interference and expediting the process.

During the most advanced stages of the design, the use of open standards allows the use of the best specific tools in the calculation and sizing of structures and facilities. First of all, the export to IFC of the architectural model allows engineers to import into their programs modelling and

analysis for evaluation and approval by the project coordinator. Subsequently BCF file sharing enables the transmission of comments and observations as well as tracking the modifications to the project that facilitates its traceability.

The IFC files are also used to federate or integrate the different partial models of the project, and check the degree of collisions between them. By using verification tools we detect in advance possible anomalies that may cause conflict later in work, and thus avoiding more costly changes during the execution of the work.

Some unique elements require special attention during the development of the projects. To do this, we can look at design options thanks to the use of parametric tools. With them we can investigate and compare in detail alternatives that allow us to choose the most convenient solution for our client. These tools are used to develop facades, finishes and even unique elements in the interior design phase through the use of interoperable formats we can design exclusive and unique furnishing that adds value to the proposal.

When it comes to elaborate the work's documentation, the use of BIM methodology ensures the correct coordination between the three-dimensional model (3D), the two-dimensional planes exported to various formats (2D). Likewise, the use of displays of the model - with specific applications for mobile devices allows a much more completed and updated reading of the project. This information is always at the disposal of work team and the client. Finally, we can also export the IFC file model to measurement and budgeting programs (5D) and also simulation, planning and construction management (4D) to complete the information that we can extract from the model. Once the project is built, the BIM model can still be used to carry out the comprehensive management of services and building maintenance (7D).

The BIM methodology is definitely not a technological transformation by itself is not a software, it is a change of mentality. Interoperability in the complex processes has become essential in the market, traceability of actions and the inherent responsibilities of each participant in development is essential to the smooth running of the project. Customer access to all information in a simple and practical way in real time, is nowadays possible and desirable.

Softwares used for Designing G+1 Building using BIM:

- AutoCAD
- STAAD.PRO
- Revit

AutoCAD

AutoCAD is a commercial software application for 2D and 3D computer-aided design (CAD) and drafting - available since 1982 as a desktop application and since 2010 as a mobile web- and cloud-based app marketed as AutoCAD 360.

Developed and marketed by Autodesk, Inc., AutoCAD was first released in December 1982, running on microcomputers with internal graphics controllers. Prior to the introduction of AutoCAD, most commercial CAD programs ran on main-frame computers or minicomputers, with each CAD operator (user) working at a separate graphics terminal. AutoCAD is used across a wide range of industries, by architects, project managers, engineers, graphic designers, and other professionals. It is supported by 750 training centers worldwide as of 1994.

As Autodesk's flagship product, by March 1986 AutoCAD had become the most ubiquitous CAD program worldwide. As of 2014, AutoCAD is in its twenty-ninth generation, and collectively with all its variants, continues to be the most widely used CAD program throughout most of the world.

The latest version was released on 23th march 2021 as Auto- CAD 2022.



Figure No: 3.1 AutoCAD window

STAAD.PRO

STAAD.Pro is a comprehensive and integrated finite element analysis and design offering that includes a state-of-the-art user interface, visualization capabilities, and international design codes. It is capable of analyzing any structure exposed to static, dynamic, wind, earthquake, thermal, and moving loads. STAAD.Pro provides structural analysis and design for any type of project, including towers, buildings, culverts, plants, bridges, stadiums, and marine structures.

Analysis and Design

The standard STAAD.Pro analysis methods provide you with a grounding in structural and analysis requirements for an array of projects. When more advanced capabilities are required, you can extend to STAAD.Pro Advanced.

STAAD.Pro reduces the resource hours required to properly load your structure by automating the forces caused by wind, earthquakes, snow, or vehicles. No matter what material you use or in what country you design your structure, STAAD.Pro can easily accommodate your design and loading requirements, including U.S., European (including the Eurocodes), Nordic, Indian, and Asian codes. Even special codes are accommodated at no extra cost.

With an unparalleled quality-assurance program, open architecture for customization, and a 25-year track record - including such projects as the MCI Stadium in Washington D.C., Wimbledon Court No.1 in London, and the tallest transmission tower in Asia - STAAD.Pro is a suitable application for your design firm.

Features of STAAD PRO

Object-oriented 2D/3D graphical model generation with Flexible zoom, pan and multiple views.

- Built-in command file editor for text edit.
- Customizable structural templates for creating a model.
- Supports truss & beam members, plates, solids, linear and non-linear cables & curvilinear beams.
- Advanced automatic load generation facilities for area, floor & moving loads.

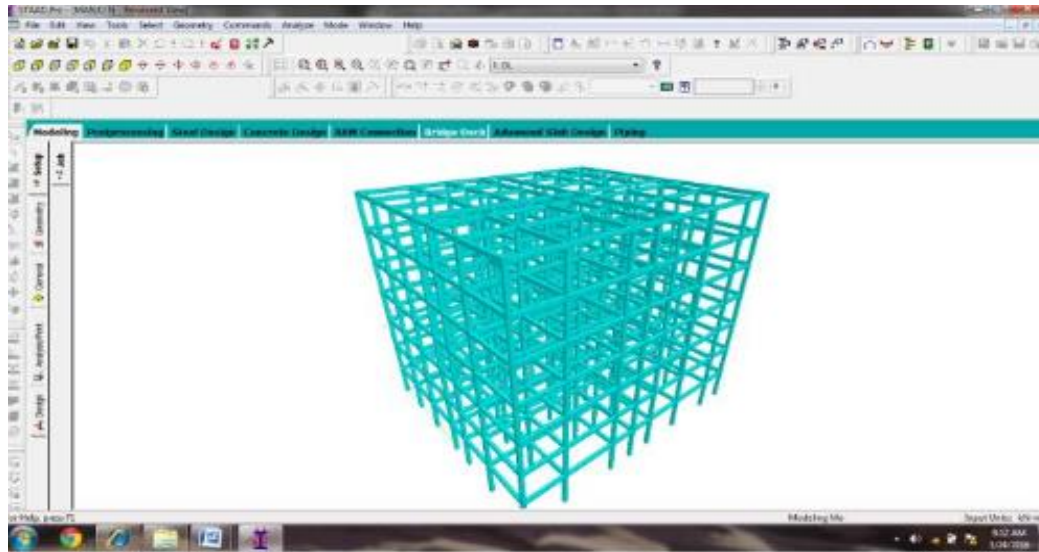


Figure 3.2 STAAD.PRO Window

Revit

Revit is building information modeling software for architects, structural engineers, MEP engineers, designers and contractors developed by Autodesk. It allows users to design a building and structure and its components in 3D, annotate the model with 2D drafting elements, and access building information from the building model's database. Revit is 4D BIM capable with tools to plan and track various stages in the building's lifecycle, from concept to construction and later maintenance and/or demolition.

Revit is Building Information Modeling (BIM) software for Microsoft Windows developed by Autodesk. It allows the user to design with parametric modelling and drafting elements with a fully integrated database. There are three software packages within the Revit BIM family of software: Revit Architecture, Revit Structure, and Revit MEP. Each package is targeted toward the architectural, structural, and mechanical/electrical/plumbing disciplines respectively. Autodesk purchased the software from Massachusetts-based Revit Technologies in 2002.

There are many categories of objects ('families' in Revit terminology), which divide into three groups:

- System Families, such as walls, floors, roofs and ceilings which are built inside a project

- Loadable Families / Components, which are built with primitives (extrusions, sweeps, etc.) separately from the project and loaded into a project for use .
- In-Place Families, which are built in-situ within a project with the same toolset as loadable components.

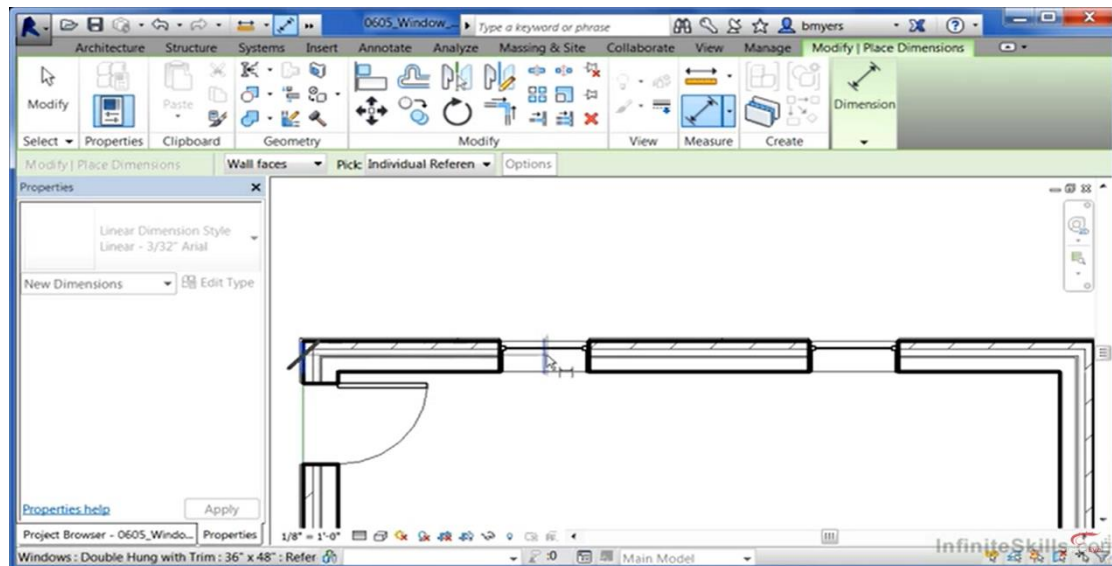


Figure 3.3 Revit window

CHAPTER 4

SOIL BEARING CAPACITY FOR G+1 BUILDING

TRIAXIAL SHEAR TEST:

REFERENCE:

IS: 2720(Part-XI)-1993

SCOPE AND PURPOSE:

- This standard lays down the method of determining the shear strength parameters of saturated soils from triaxial compression shear tests conducted under unconsolidated undrained conditions with measurements of pore water pressures.
- This test is limited to specimens in the form of right cylinders of nominal diameter 38, 50, 70, 100mm with height twice its diameter. The ratio of diameter of the sample to the maximum size of the particle in the soil shall not be less than five.
- It may be performing with or without measurement of pore pressure although for most applications the measurement of pore pressure is desirable.

APPARATUS REQUIRED:

- Loading frame
- The cell(chamber)
- Split mould- 38mm diameter and 76mm long
- Trimming knife
- Rubber membrane
- Balance
- O-Rings (2 No's)

TEST PROCEDURE:

- An undisturbed soil sample is extracted by conducting standard proctor test to the soil sample by using the OMC of soil (provided).
- Prepare the soil specimen at the desired water content and density in the large mould.
- Push the sampling tube into the large mould, remove the sampling tube filled with soil.
- For undisturbed samples, push the sampling tube into the clay sample.
- Coat the split mould lightly with a thin layer of grease. Weigh the mould.
- Extrude the sample out of the sampling tube into the split mould, using the sample extractor and the knife.
- Trim the two ends of the specimen in the split mould.
- Weigh the mould with the specimen.
- Remove the specimen from the split mould by splitting the mould into the parts.
- The sample is placed on the base of the triaxial cell. The cell must be properly set up and uniformly clamped down to prevent leakage of pressure during the test, making sure first that the sample is properly sealed with its end caps and rings (rubber) in position and that the sealing rings for the cell are also correctly placed.
- The sample is covered with rubber membrane and sealed with the help of rubber O-rings on top and bottom.
- The cell is filled with water and closed the pore pressure valve since the test is undrained.
- Note the observations recorded in digital LVDT for the readings of deformation for respective loads.
- The air pressure in the reservoir is then increased to raise the hydrostatic pressure in the required amount. The pressure gauge must be watched during the test and any necessary adjustments must be made to keep the pressure constant.
- The piston is then moved down by the handle until it is just in touch on the top of

sample, and started applying the load until specimen fails.



Figure 4.1 Extraction of Soil Sample



Figure 4.2 Doing triaxial test experiment



Figure 4.3 Triaxial test Apparatus

OBSERVATION:

Weight of sample 1 = 176.6 gm

Weight of sample 2 = 167.4 gm

1	sno	deformat	strain	PRR 1	LOAD(KG)	area	fd1	611	sno	deform	strain	PRR 2	LOAD(KG)	Area	fd2	612	
2	1	1	0.01316	10	11.56	11.4912	1.0059872	2.0059872	1	0.5	0.01316	4	12.87	11.4912	1.11999	1.61999	
3	2	2	0.02632	8	12.432	11.646486	1.0674464	2.0674464	2	1	0.02632	5	13.251	11.6465	1.13777	1.63777	
4	3	3	0.03947	9	12.584	11.806027	1.0658962	2.0658962	3	1.5	0.03947	6	13.345	11.806	1.13035	1.63035	
5	4	4	0.05263	11	12.78	11.97	1.0676692	2.0676692	4	2	0.05263	7	13.67	11.97	1.14202	1.64202	
6	5	5	0.06579	12	12.8	12.138592	1.0544881	2.0544881	5	2.5	0.06579	8	14	12.1386	1.15335	1.65335	
7	6	6	0.07895	13	13.189	12.312	1.0712313	2.0712313	6	3	0.07895	9	14.289	12.312	1.16058	1.66058	
8	7	7	0.09211	15	13.226	12.490435	1.0588903	2.0588903	7	3.5	0.09211	10	14.753	12.4904	1.18114	1.68114	
9	8	8	0.10526	17	13.25	12.674118	1.0454377	2.0454377	8	4	0.10526	11	15.288	12.6741	1.20624	1.70624	
10	9	9	0.11842	17	13.28	12.863284	1.0323958	2.0323958	9	4.5	0.11842	12	15.432	12.8633	1.19969	1.69969	
11	10	10	0.13158	18	13.3	13.058182	1.0185185	2.0185185	10	5	0.13158	12	15.758	13.0582	1.20675	1.70675	
12	11	11	0.14474	18	13.56	13.259077	1.0226956	2.0226956	11	5.5	0.14474	13	15.96	13.2591	1.2037	1.7037	
13	12	12	0.15789	18	13.678	13.46625	1.0157245	2.0157245	12	6	0.15789	13	16.87	13.4663	1.25276	1.75276	
14	13	13	0.17105	18	13.764	13.68	1.0061404	2.0061404	13	6.5	0.17105	12	16.57	13.68	1.21126	1.71126	
15	14	14	0.18421	18	13.893	13.900645	0.99945	1.99945	14	7	0.18421	11	15.98	13.9006	1.14959	1.64959	
16	15	15	0.19737	17	14	14.128525	0.9909032	1.9909032	15	7.5	0.19737		0	14.1285	0	0.5	
17	16	16	0.21053		0	14.364	0	1	16	8	0.21053		0	14.364	0		
18	17	17	0.22368		0	14.607458	0	1			0.22368		0	14.6075	0		

Table: Triaxial Shear Test Observations and Calculations

CALCULATIONS:

$$\sigma_{d1} = \frac{13.3}{12.67} = 1.05 \text{ kg/cm}^2$$

$$\sigma_{d2} = \frac{15.96}{12.67} = 1.26 \text{ kg/cm}^2$$

$$\sigma_{31} = 0.5 \text{ kg/cm}^2$$

$$\sigma_{32} = 1 \text{ kg/cm}^2$$

$$\sigma_{11} = 1.55 \text{ kg/cm}^2$$

$$\sigma_{12} = 2.26 \text{ kg/cm}^2$$

$$\sigma_{12} = \sigma_{32} \tan^2 \alpha + 2c \tan \alpha$$

$$\sigma_{11} = \sigma_{31} \tan^2 \alpha + 2c \tan \alpha$$

Solving equations,

$$2.26 = 1 \tan^2 \alpha + 2c \tan \alpha$$

$$1.55 = 0.5 \tan^2 \alpha + 2c \tan \alpha$$

By solving 2 equations,

$$0.710 = 0.5 \tan^2 \alpha$$

$$\alpha = \tan^{-1} \sqrt{\frac{0.710}{0.5}} = 49^\circ 59' \sim 50^\circ$$

$$= 2(49^\circ 59' - 50^\circ)$$

$$\Phi = 10^\circ$$

$$c = 0.35 \text{ kg/cm}^2$$

for $\Phi = 10^\circ$

$$N_c = 9.6, \quad N_q = 2.7, \quad N_\gamma = 1.2$$

From Terzaghi's Bearing Capacity,

$$q_u = c' N_c + Y D_f N_q + 0.5 Y B N_\gamma$$

For Isolated footing

$$q_u = 1.3 N_c + Y D_f N_q + 0.5 Y B N_\gamma$$

$$= 1.31.3 \times 0.35 \times 9.6 + 1.97 \times 10^{-3} \times 1.5 \times 100 \times 2.7 + 0.5 \times 1.97 \times 1.2$$

$$q_u = 4.368 + 0.8 + 0.23$$

$$q_u = 5.4 \text{ kg/cm}^2 = 530 \text{ kg/m}^2$$

$$q_{nu}=5.4 - (1.97 \times 10^{-3} \times 1 \cdot 5 \times 100)$$

$$q_{nu}= 5.10 \text{ kg/cm}^2$$

$$q_{ns} = \frac{5.10 \times 98.1}{F.S} = \frac{500.31}{2.5} = 200.12 \text{ kg/m}^2$$

$$q_s = 200.12 + (19.7 \text{ kg/m}^2 \times 1.5 \text{ m})$$

$$q_s = 230.5 \text{ KN/m}^2$$

SHEAR STRENGTH OF SOIL SAMPLE CALCULATIONS:

$$\text{Sample 1} \quad \sigma_{31} = 0.5 \text{ kg/cm}^2$$

$$\sigma_n = \left(\frac{\sigma_{11} + \sigma_{31}}{2} \right) + \left(\frac{\sigma_{11} - \sigma_{31}}{2} \right) \cos 2\alpha$$

$$\begin{aligned} \sigma_n &= \left(\frac{1.55 + 0.5}{2} \right) + \left(\frac{1.55 - 0.5}{2} \right) \cos 2(50^\circ) \\ &= 0.933 \text{ kg/cm}^2 \end{aligned}$$

$$S_1 = c + \sigma_n \tan \Phi$$

$$S_1 = 0.35 + 0.933 \times \tan \Phi$$

$$S_1 = 0.524 \text{ kg/cm}^2$$

Likewise for Sample 2

$$S_2 = 0.35 + 2.3 \times \tan(10^\circ)$$

$$S_2 = 0.755$$

$$\text{Average Shear Strength of Soil from Triaxial test} = \frac{0.524 + 0.755}{2}$$

$$S = 0.639 \text{ kg/cm}^2$$

CHAPTER 5

SOFTWARES USED IN BIM FOR PLANNING AND DESIGNING G+1

Softwares used for Designing G+1 Building using BIM:

- AutoCAD
- STAAD.PRO
- Revit

5.1 AUTOCAD 2D

Introduction:

AutoCAD 2D, a powerful computer-aided design (CAD) software developed by Autodesk, is a cornerstone in the field of engineering and design. It is renowned for its versatility, precision, and efficiency in creating 2-dimensional drawings. Below are key aspects and applications of AutoCAD 2D, highlighting its significance in various industries:

Key Features of AutoCAD 2D:

- User-Friendly Interface:** AutoCAD 2D boasts an intuitive interface, making it accessible to both beginners and seasoned professionals.
- Precision and Accuracy:** The software enables users to create drawings with unparalleled precision, ensuring accuracy in design and documentation.
- Extensive Drawing Tools:** AutoCAD 2D provides a comprehensive set of drawing tools, including lines, circles, arcs, and polygons, facilitating the creation of intricate designs.
- Dimensioning and Annotation:** Precise dimensioning tools allow for the addition of measurements and annotations, enhancing the clarity and communicative value of drawings.
- Layer Management:** The layering system in AutoCAD 2D allows users to organize elements efficiently, making it easier to manipulate, edit, and control the visibility of different components within a drawing.

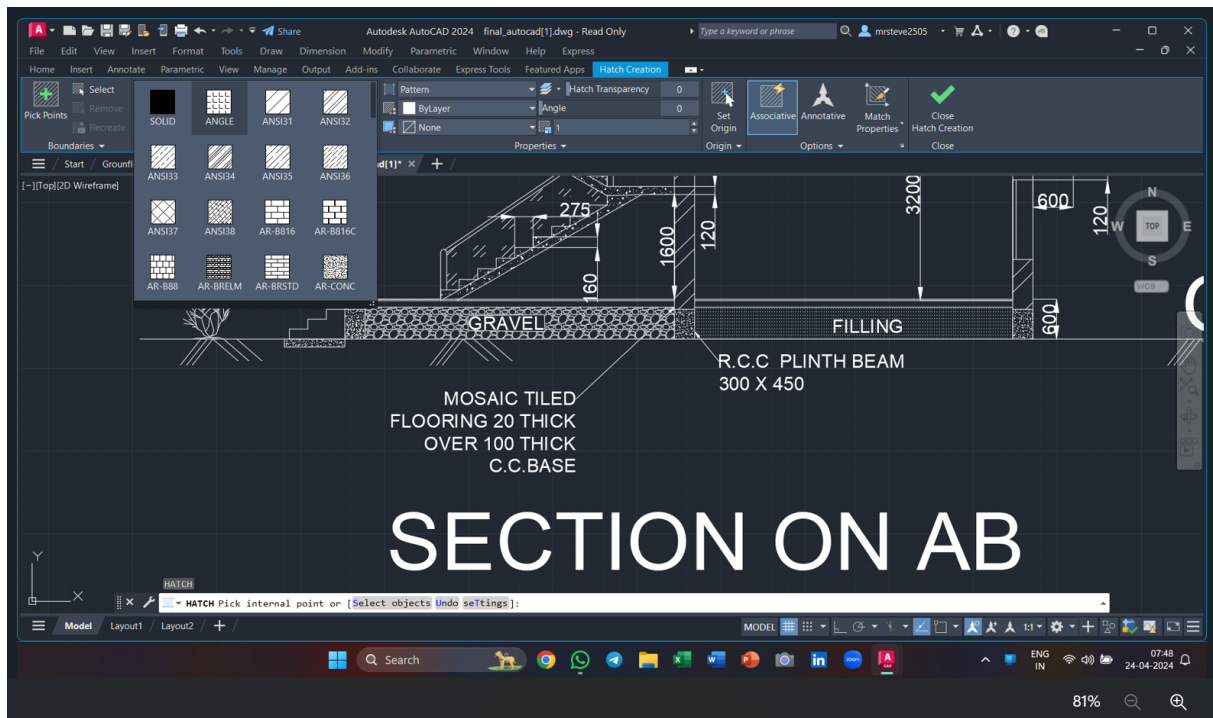
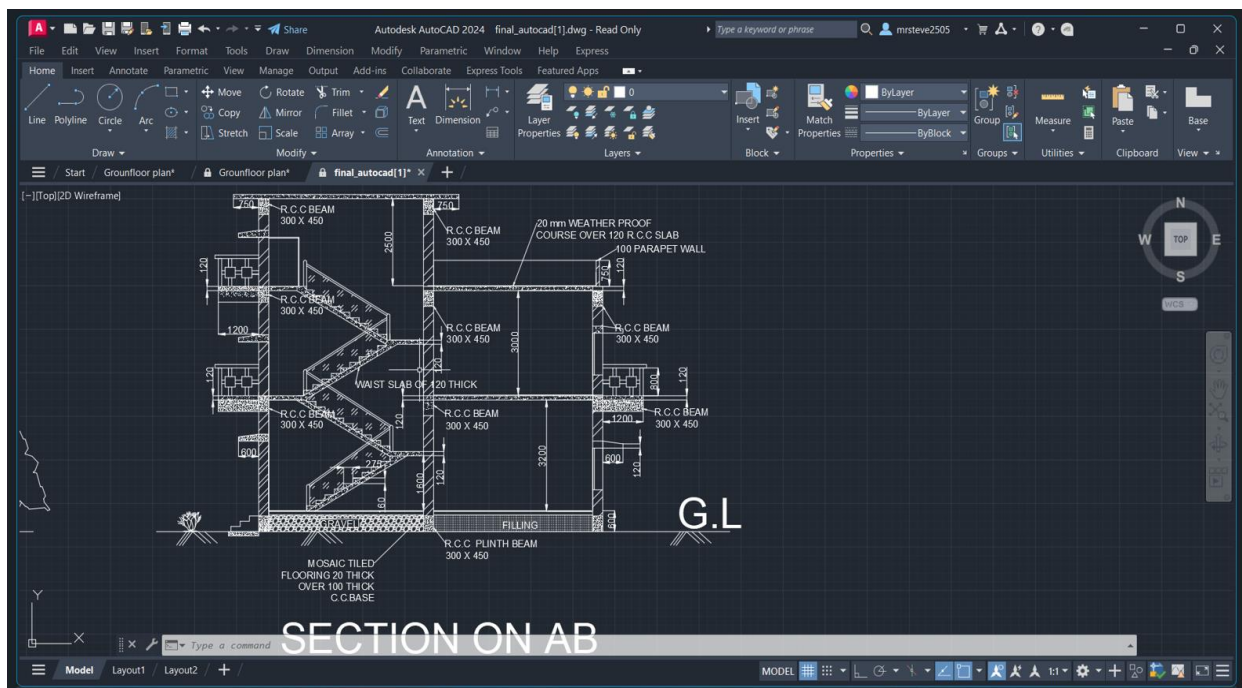
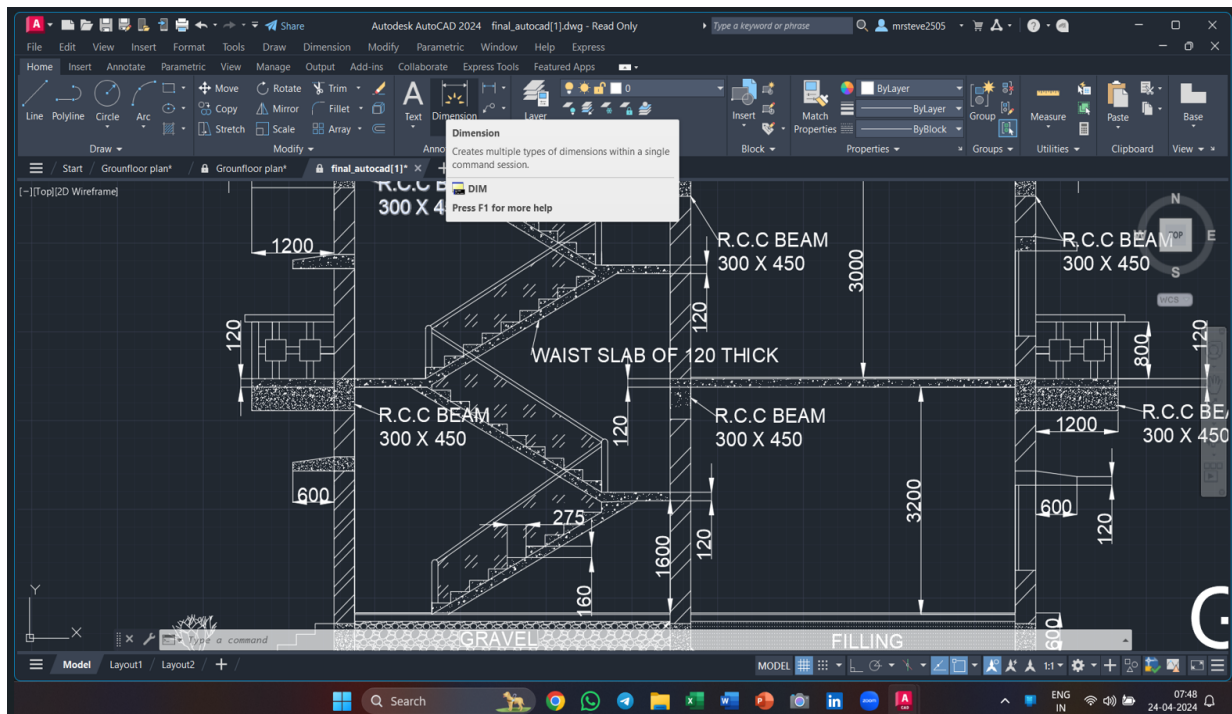


Figure No5.1 Planning Building using Hatch Tool in AutoCAD



FigureNo5.2 Section of the Building in AutoCAD



FigureNo.5.3 Planning Building using Dimensioning tool in AutoCAD

Drafting of sectional view of house

- First to draw the walls of the house select line command and draw a line and the take offset of 0.3m using offset command .
- Draw all the walls required using offset as per given measurements.
- Take line command to give slab of house and draw as per required measurements.
- And then for the footings, first select some colour to the footings and then select the line command and draw the footings according to the measurements.
- And then copy those footings for all four walls using copy command.
- Draw the doors and windows of the house and copy them using copy command wherever they are required.
- Use rectangle and line command to draw windows and doors of the house .draw them as per required measurement.
- Draw the lines to connect the walls using line command and trim the extras using tr double enter.
- Draw lines for sand bed ,rock bed etc.
- Now we have to provide hatch to the house . For the dado first draw a line using line command and then click on hatch select the suitable hatch and click on it.
- For slab and base of the footings concrete hatch must be given. Click on the hatch and select the concrete hatch and click on the selected objects like slab and base of the footings and for concrete bed.

- Now provide the hatch for the walls using hatch option . select the hatch with double lines for the walls .And for the stone bed use the appropriate hatch.
- Now we have to enter the dimensions and text for the required.
- Click on the dimension layer, click on linear. select the first point and then second point for giving the dimensions.
- For continuous measurements, click annotate, click continuous for the continuous measurements.
- Now enter the text for the required.
- And the sectional view of house is drafted using auto cad.

Steps required to draw plan of Ground floor and First floor of Residential building using Autocad 2024

Step 1: Setting Up the Drawing Environment

The first step in creating a plan for a residential building in AutoCAD 2024 is to set up the drawing environment. Start by launching AutoCAD and selecting the appropriate template for your drawing. Choose a template that matches the units (e.g., feet or meters) and scale you want to use for your plan. Once the template is loaded, set up the drawing units and grid settings according to your preferences.

Step 2: Drawing the Exterior Walls

Begin by drawing the outline of the building's ground floor. Use the Line or Polyline tool to create the exterior walls, ensuring they are accurately aligned and dimensioned according to the building's design. You can input specific dimensions using the keyboard or rely on the dynamic input feature to draw walls to scale. Pay attention to corners and intersections to ensure that the walls connect seamlessly.

Step 3: Adding Interior Walls

Once the exterior walls are in place, add interior walls to define the layout of rooms and other spaces on the ground floor. Again, use the Line or Polyline tool to draw the walls accurately, making sure they are perpendicular to the exterior walls and aligned with the overall design. Consider the placement of doors, windows, and other openings as you draw the interior walls, leaving appropriate spaces for them.

Step 4: Inserting Doors and Windows

After the walls are in place, insert doors and windows into the drawing. AutoCAD 2024 provides a variety of door and window blocks that you can use to add these elements quickly. Access the block library through the Insert tab or by using the Insert command, and select the desired door and window types. Place them accurately within the walls, ensuring they align with openings and follow the building's design specifications.

Step 5: Adding Dimension and Annotation

Once the walls, doors, and windows are in place, add dimensions and annotations to the drawing to provide context and information about the building's layout. Use the Dimension and Text tools to add measurements, labels, and other annotations to the plan. Ensure that dimensions are clear, legible, and accurately represent the distances between walls, openings, and other features.

Step 6: Creating Layers and Assigning Properties

To maintain organization and manageability in your drawing, create layers for different elements such as walls, doors, windows, dimensions, and annotations. Assign properties such as color, line type, and line weight to each layer to distinguish between different elements and make editing easier. Utilize the Layer Properties Manager to create and manage layers efficiently.

Step 7: Drawing the First Floor Plan

After completing the ground floor plan, repeat the same steps to draw the first-floor plan. Start by creating a new drawing for the first floor, or copy the ground floor plan and modify it as needed. Adjust the walls, doors, windows, and other elements to reflect the layout and design of the first floor, taking into account any differences in configuration or features.

Step 8: Reviewing and Finalizing the Plans

Once both the ground floor and first floor plans are drawn, review the drawings carefully to ensure accuracy and completeness. Check for any inconsistencies, errors, or omissions in the design, and make any necessary adjustments. Verify that the plans adhere to building codes, regulations, and design standards applicable to residential buildings in your area. Once you're satisfied with the plans, save the drawings in the desired format and prepare them for sharing or further development.

By following these steps, you can create detailed plans for the ground floor and first floor of a residential building using AutoCAD 2024. From setting up the drawing environment to adding dimensions and annotations, each step plays a crucial role in producing accurate and comprehensive drawings for construction or presentation purposes. With practice and familiarity with AutoCAD's tools and features, you can efficiently create professional-quality plans for residential buildings.

Step To Draw the Elevation of a Residential Building Using AutoCAD 2024

Before starting to draw the elevation, it's crucial to set up your drawing environment properly in AutoCAD 2024. This involves configuring your units, layers, and drawing settings to ensure accuracy and efficiency throughout the process.

Step 1: Firstly, launch AutoCAD and select the appropriate template or start a new drawing with the necessary units (typically architectural units for building design). You can set the units by typing "UNITS" in the command line and selecting the desired unit type.

Next, organize your drawing by creating layers for different elements of the elevation. For instance, you might have layers for walls, doors, windows, etc. Use the "LAYER" command to create and manage layers effectively.

Finally, adjust your drawing settings according to your preferences. This includes things like grid settings, snap settings, and object snaps. These settings will help you maintain precision while drawing the elevation.

Step 2: Drawing the Basic Structure

With your drawing environment set up, you can begin drawing the basic structure of the building elevation. Start by drawing the outline of the building using the "LINE" or "POLYLINE" command. Ensure that you accurately represent the shape and dimensions of the building.

Once the outline is complete, add details such as walls, windows, and doors. Use the appropriate commands like "RECTANGLE" for walls and "RECTANG" for doors and windows. Pay close attention to dimensions and alignments to maintain accuracy. Consider using object snaps like "ENDPOINT," "MIDPOINT," and "PERPENDICULAR" to ensure precise placement of elements relative to each other.

Step 3: Adding Detail and Dimension

With the basic structure in place, it's time to add detail and dimension to the elevation drawing. This involves adding features such as window panes, door swings, and architectural details.

Utilize commands like "OFFSET" to create offsets for window and door frames. You can also use "ARRAY" to duplicate elements such as windows along a wall.

For dimensioning, use the "DIM" command to add linear dimensions to the elevation drawing. Ensure that dimensions are placed clearly and accurately to convey the size and scale of the building.

Consider using annotation scales to control the appearance of dimensions based on the viewport scale.

Step 4: Applying Text and Annotations

To provide additional information and context to your elevation drawing, include text and annotations. Use the "TEXT" command to add labels for elements such as rooms, materials, and special features.

Ensure that text is legible and appropriately sized for the drawing scale. You can adjust text properties such as font, size, and justification using the text style manager.

Consider adding leader lines and arrows to point to specific elements or dimensions within the drawing.

Step 5: Reviewing and Finalizing the Drawing

Before finalizing your elevation drawing, take the time to review it thoroughly for accuracy and completeness. Check for any missing elements, incorrect dimensions, or inconsistencies in the drawing.

Make any necessary adjustments or corrections to ensure that the drawing accurately represents the intended design.

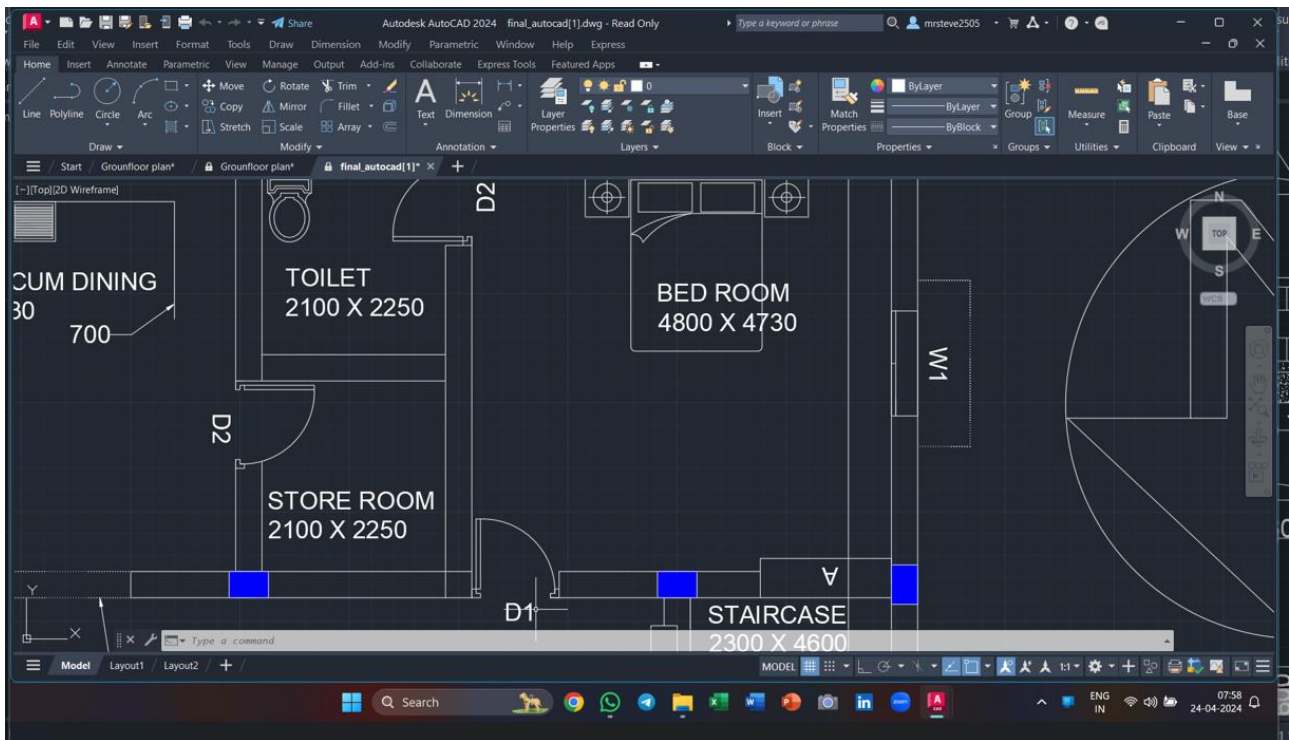


Figure5.3 Dimensions of plan in AutoCAD

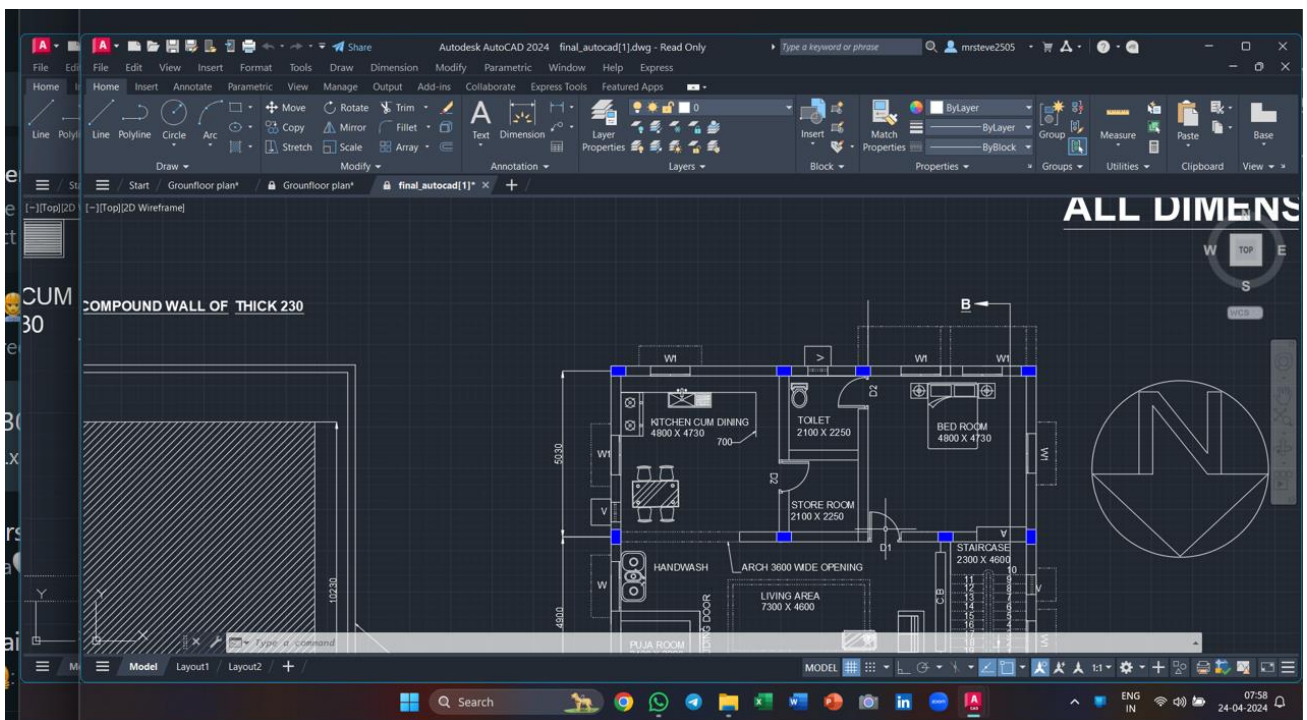


Figure 5.4 Representing columns line plan

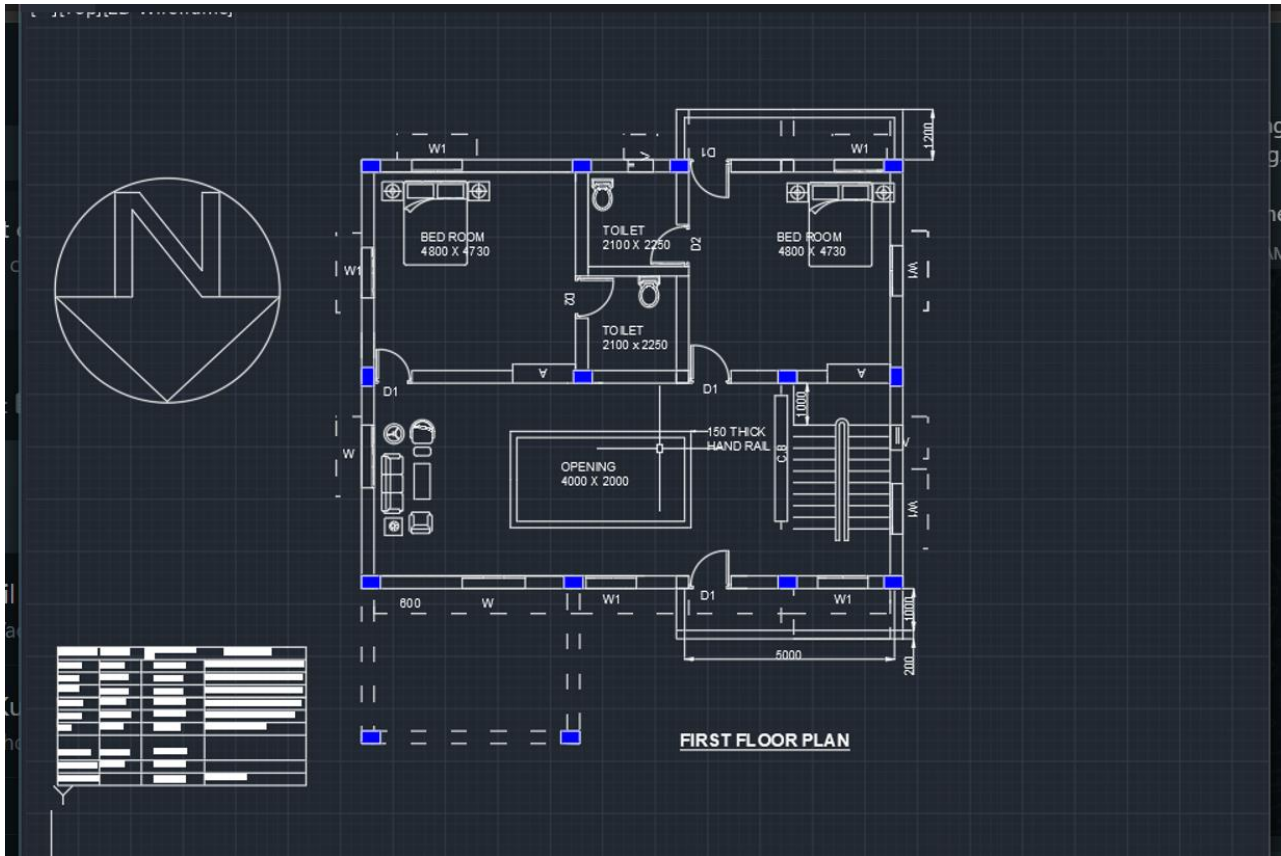


Figure 5.5 Line plan of Buliding

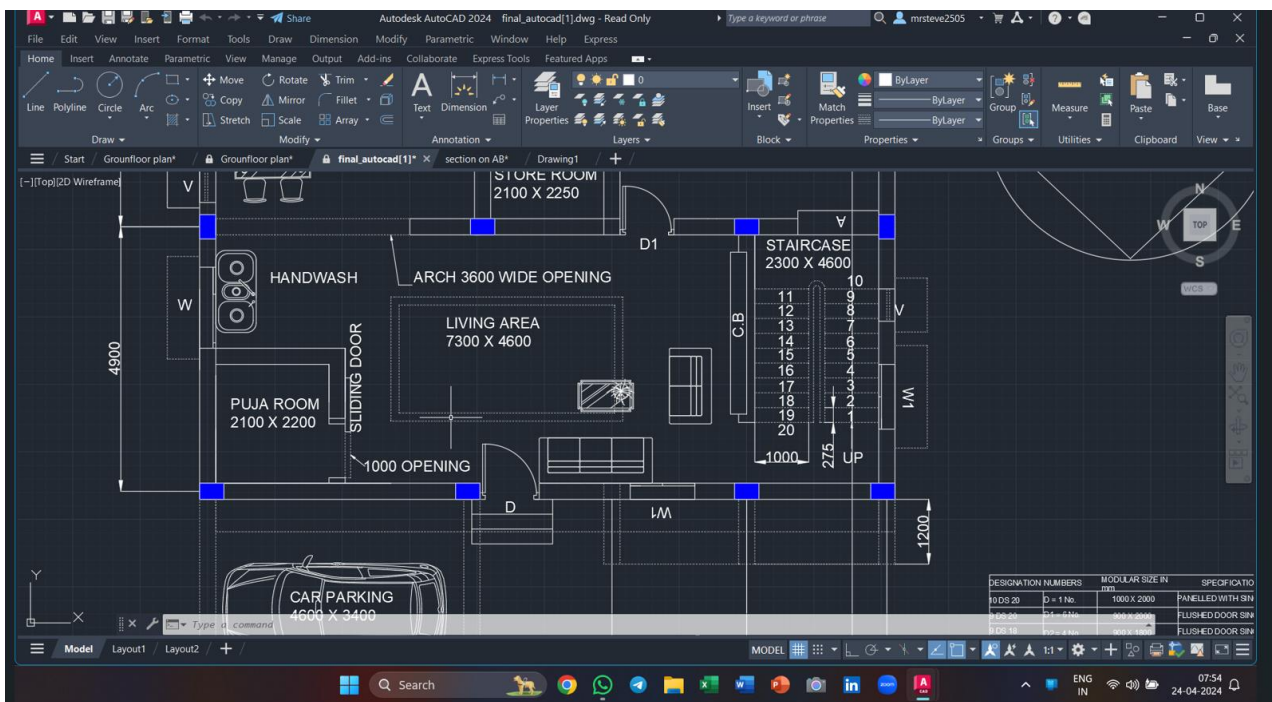


Figure5.6 Showing floor plans in AutoCAD

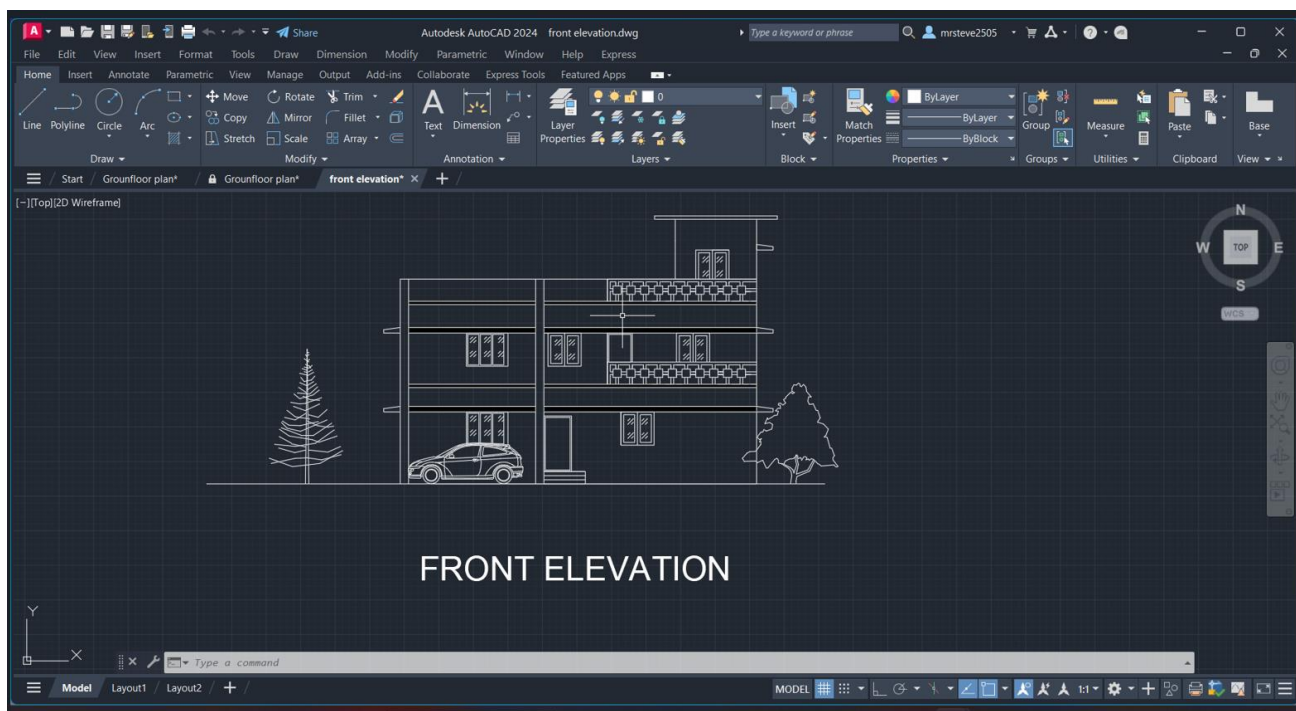
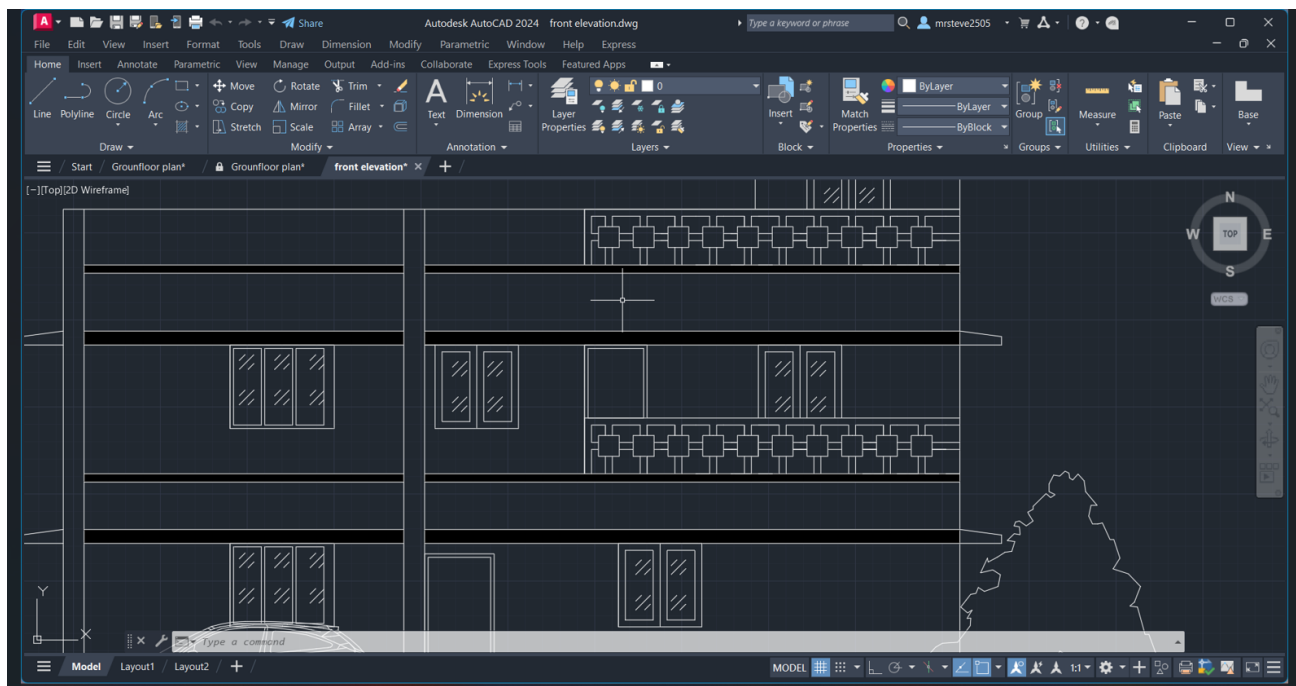


Figure5.7 Showing Elevation View in AutoCAD

SIZE OF THE PLOT
50 FT X 60 FT(15m x 16m)

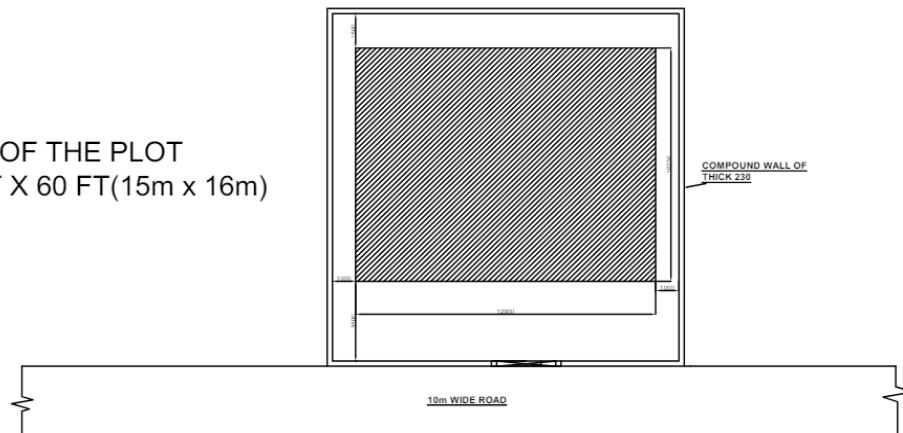


Figure5.8 Site Plot View

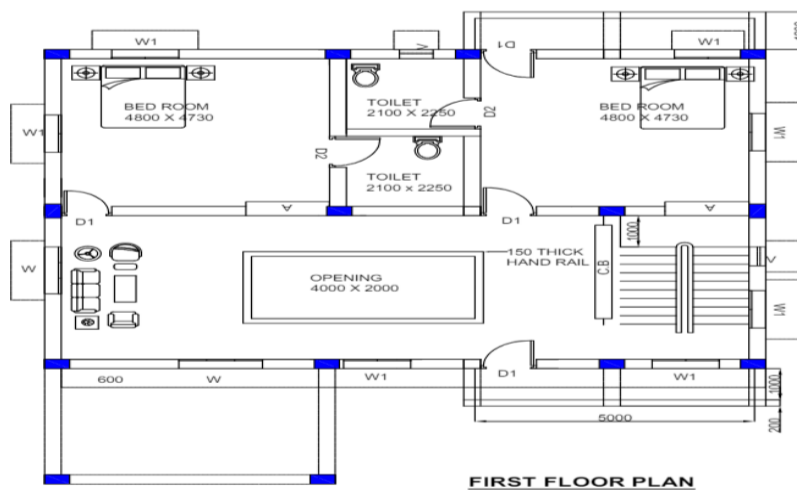


Figure5.9 First floor plan

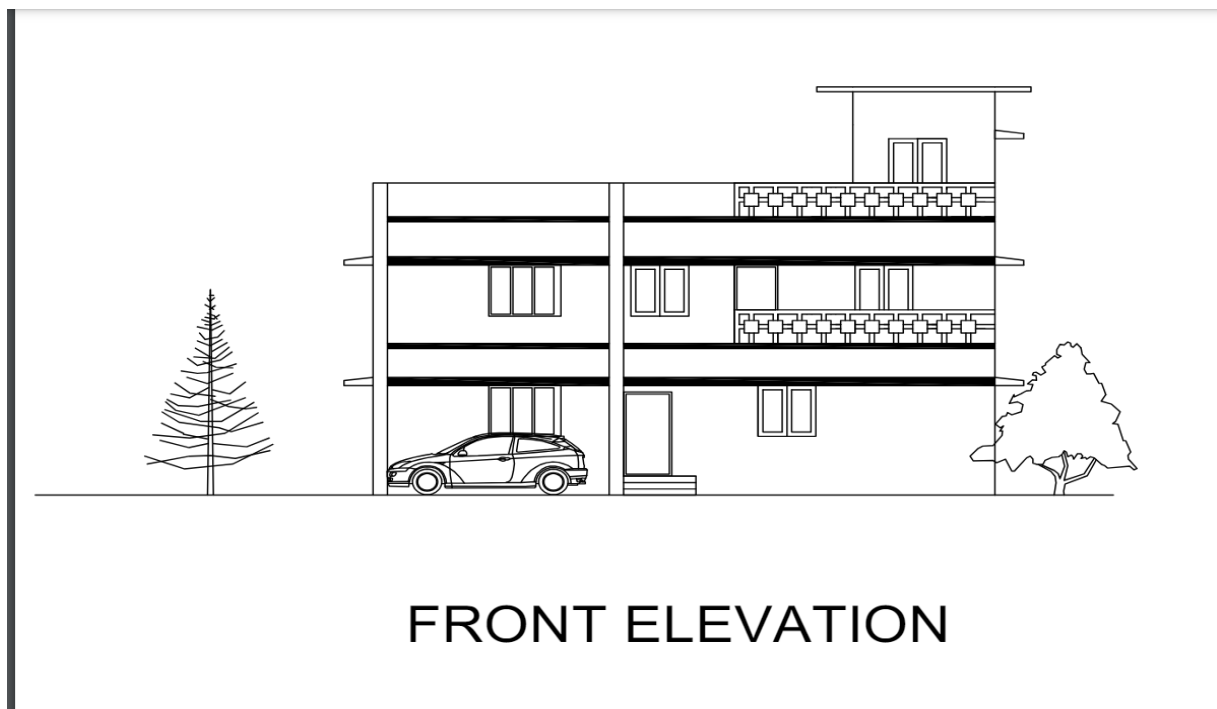


Figure 5.10 Front Elevation plan

5.2 STAAD.PRO

Design process involves the selection and detailing of the components that make up the structural system. The main object of reinforced concrete design is to achieve a structure that will result in a safe economical solution.

The objective of the design is

1. Foundation design
2. Column design
3. Beam design
4. Slab design

These all are designed under limit state method

LIMIT STATE METHOD

The object of design based on the limit state concept is to achieve an acceptability that a structure will not become unserviceable in its life for the use for which it is intended. i.e. it will not reach a limit state. In this limit state method all relevant states must be considered in design to ensure a degree of safety and serviceability.

LIMIT STATE :

The acceptable limit for the safety and serviceability requirements before failure occurs is called limit state.

LIMIT STATE OF COLLAPSE:

This is corresponding to the maximum load carrying capacity. Violation of collapse limit state implies failures in the source that a clearly defined limit state of structural usefulness has been exceeded. However, it does not mean complete collapse.

This limit state corresponds to

- a) Flexural
- b) Compression
- c) Shear
- d) Torsion

Limit state of Serviceability:

This state corresponds to development of excessive deformation and is used for checking member in which, magnitude of deformations may limit the rise of the structure of its components.

- a) Deflection
- b) Cracking
- c) Vibration

FEATURES OF THE STRUCTURE

The Duplex house (i.e., Ground floor + 1st floor) residential building consists construction area of 147 sq.m

Provided dimensions are as follows:

- * Width of staircase : 2.3 m
- * Clear height of room : 3.3 m
- * Thickness of walls : 300 mm
- * Thickness of parapet wall: 100 mm
- * Height of parapet wall : 1.2 m

The plan of building is prepared with the help of AutoCAD 2020 and is shown in FIG.1. The frames of building are analyzed using Staad pro software.

Figure 5.11 Plinth beam layout

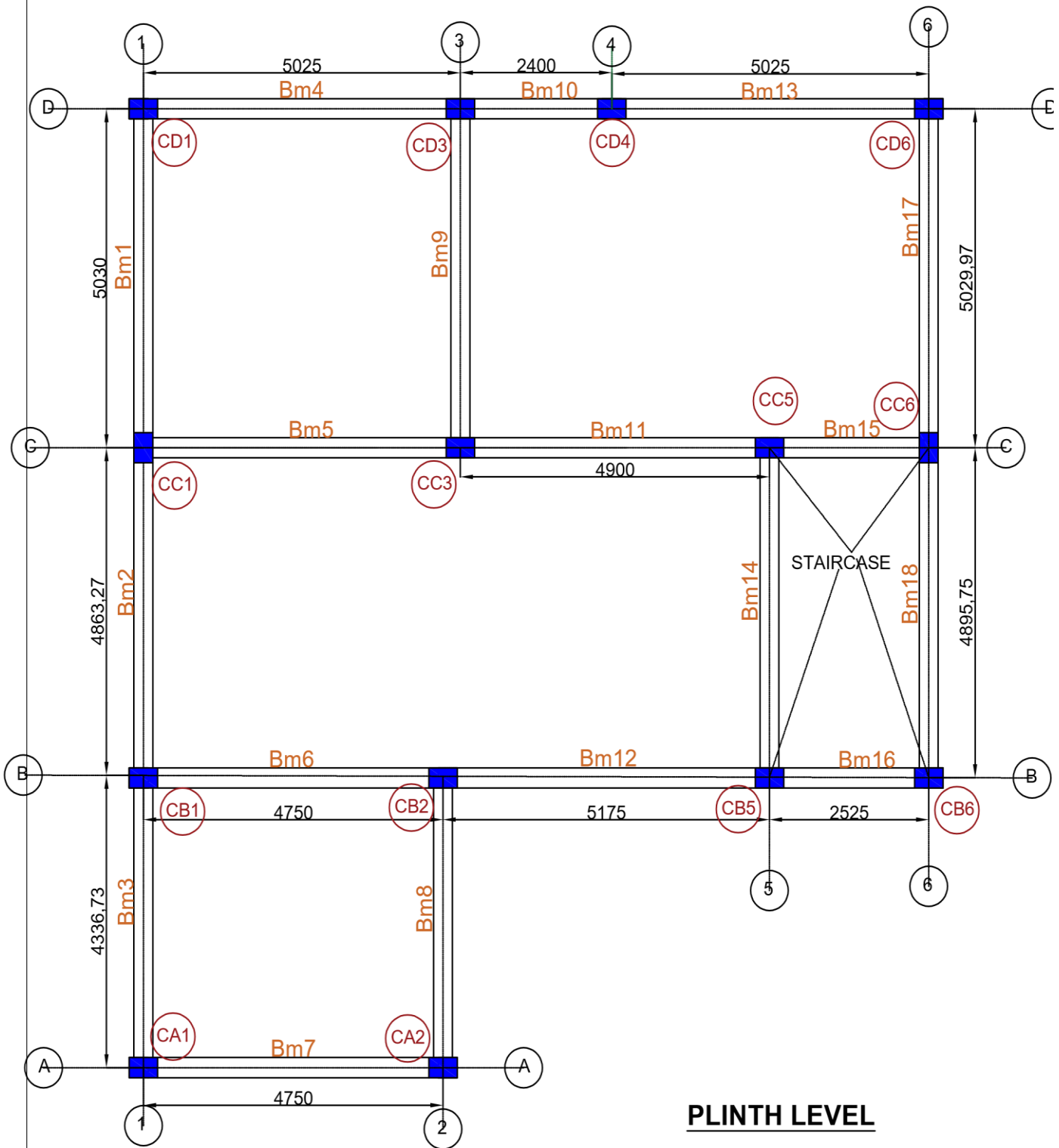


Figure 5.12 Beam and Column layout for ground floor

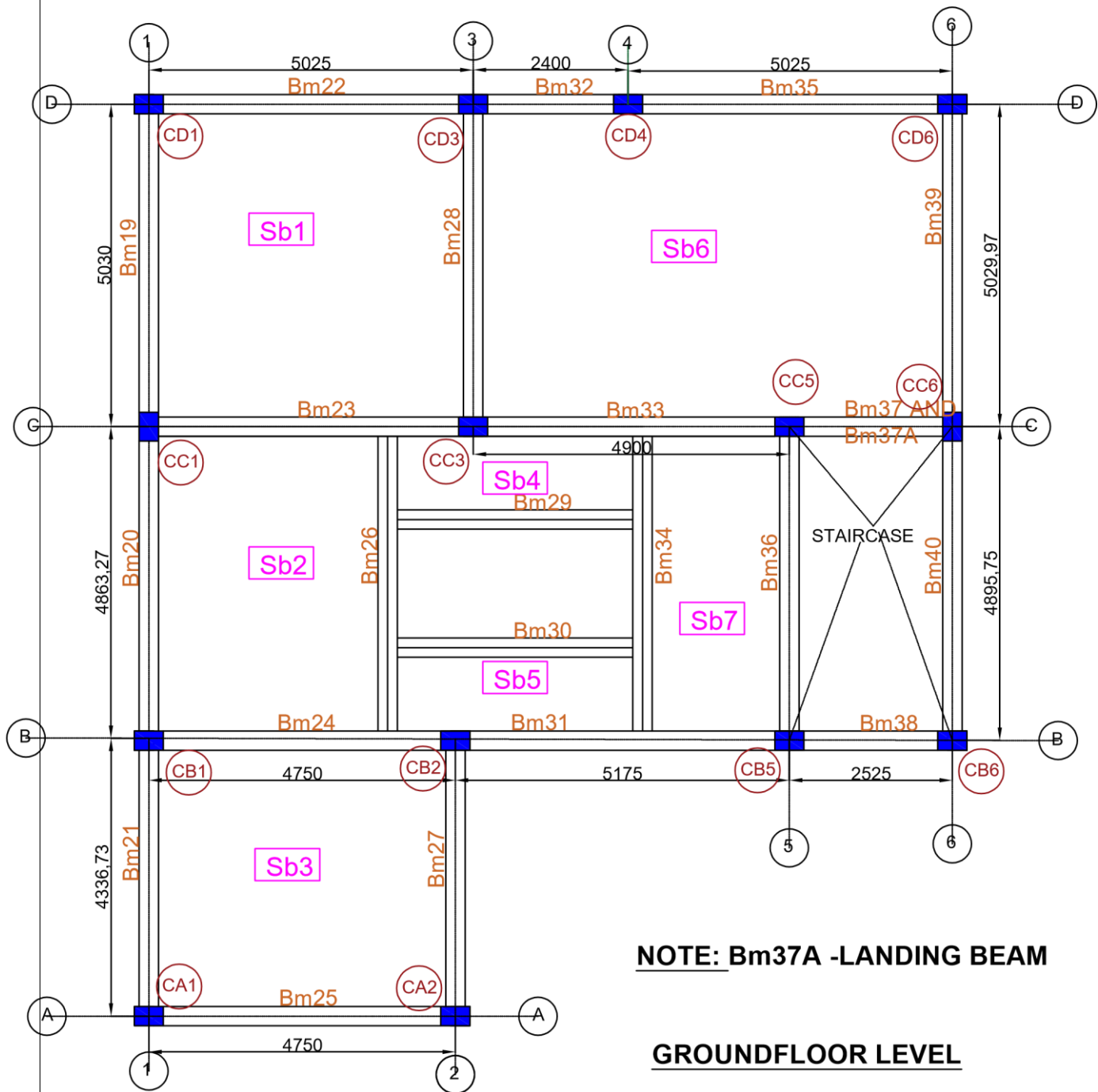
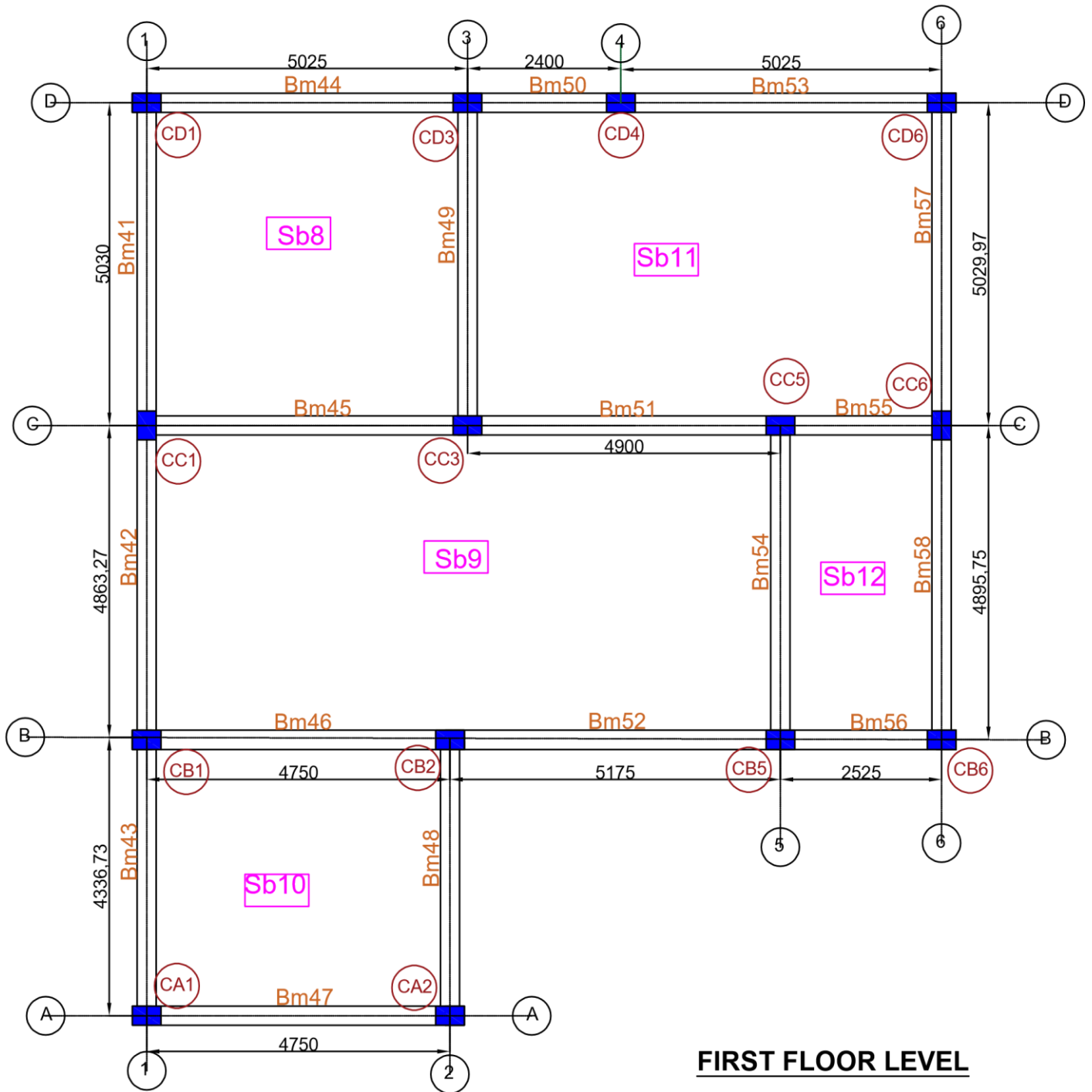


Figure 5.13 Beam and Column layout for first floor



FIRST FLOOR LEVEL

SEPCIFICATIONS:

DESIGNS ACCORDING TO

IS: 456 – 2000

IS: 875 – 1987 (part 1)

IS: 875 – 1987 (part 2)

MATERIALS

Concrete Mix: M20 grade

Steel grade: Corrosive resistant Fe415

LOADS

Live Loads : As per IS: 875 (part-2) – 1987 -
2kN/m²

Dead Loads : Self weight of structural components
and Floor finishes of
1 kN/m² and self weight walls

SBC of Soil : 200 kN/m²

UNIT WEIGHT OF MATERIALS

RCC: 25 kN/m³

Brick Masonry: 20 kN/m³

ANALYSIS OF STRUCTURE

Analysis of structure:

Staad pro is an effective software tool for the analysis and design of structural members. The steps for analyzing structure using staad is as follows.

- Modelling of structure
- Application of loads
- Analysis

STAAD PRO Model of the structure

The structure is to be modelled as per the plan using Staad pro with fixed supports. The 3-D model of the structure is shown in fig

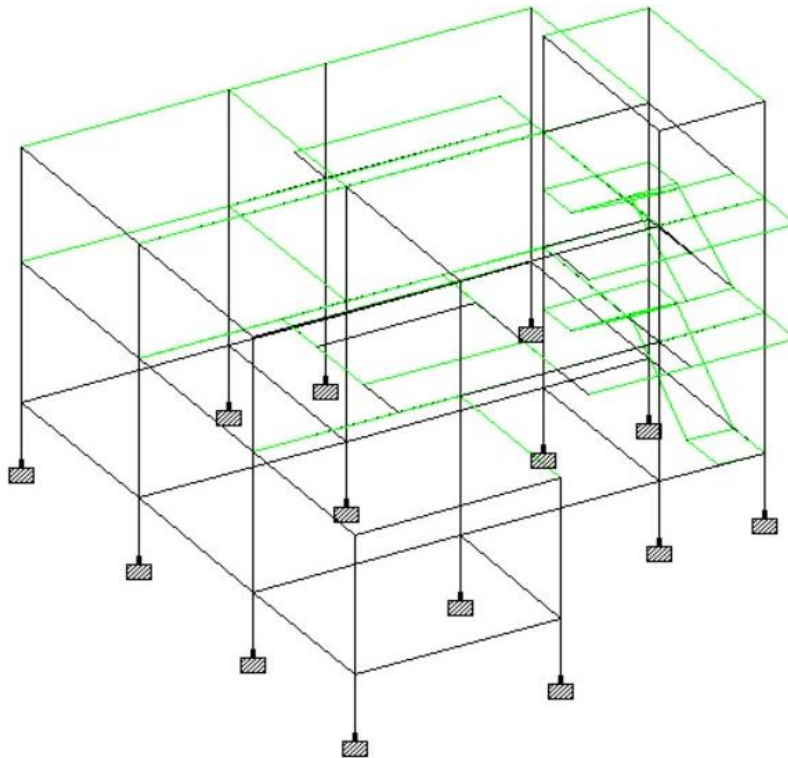


Figure 5.15 STAAD PRO Model of the Structure

Application of loads:

There are various types of loads acting on the structure are

- Dead Load
- Live

Dead Load:

In contrast, dead loads (also known as permanent or static loads, or permanent actions) remain relatively constant over time and comprise, for example, the weight of a building's structural elements, such as beams, walls, roof and flooring components. Dead loads result from the weight of the structure or other fixed elements before any live loads are taken into consideration. They may include permanent non-structural elements such as partitions, immovable fixtures and even built-in cupboards. As per IS 456:2000, the unit weights of plain concrete and reinforced concrete are 24 kN/m^2 and 25 kN/m^2 respectively.

Live Load:

Live loads (also known as applied or imposed loads, or variable actions) may vary over time and often result from the occupancy of a structure. Typical live loads may include; people, the action of wind on an elevation, furniture, vehicles, the weight of the books in a library and so on. A live load can be expressed either as a uniformly distributed load (UDL) or as one acting on a concentrated area (point load). The live or imposed on the slabs for a

residential building is considered as 2 kN/m^2 .

DISTRIBUTION OF LOADS TO FRAME MEMBERS:

- Dead and live loads on the two-way slabs are distributed on to frame members according to triangular or trapezoidal distribution. If slab is square, then distribution is triangular and if slab is rectangular, then distribution is trapezoidal.
- The load distribution is mainly based on the YIELD LINE THEORY which is defined by JOHASEN.
- The load distribution in the structure is shown below



LOAD COMBINATIONS:

According to partial safety factor clause 36.4 of IS 456-2000, the load combinations are as follows

S.NO	Load Combinations
1.	1.5 (DL+LL)
2.	1.0 (DL+LL)

NOTE:

1. While considering earthquake effects, substitute EL or WL.
2. 0.9 value is to be considered when stability against overturning or stress reversal is critical.
3. The earthquake load is considered in both directions along the frames and the members are designed for critical load combinations.

CRITICAL FRAMES:

Based on the loads coming on to the frames from the structure, the critical frames are selected. The various layouts of the structure are shown below. The location of columns and beams are shown in below fig.

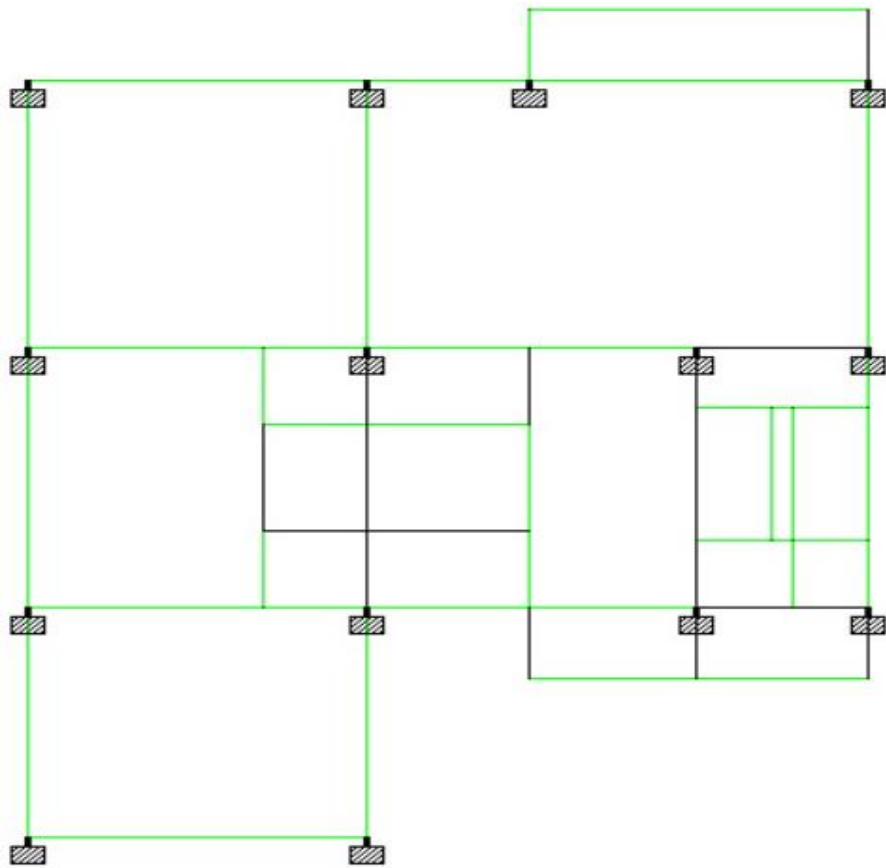


Figure 5.16 Beam and column layout (Top View)

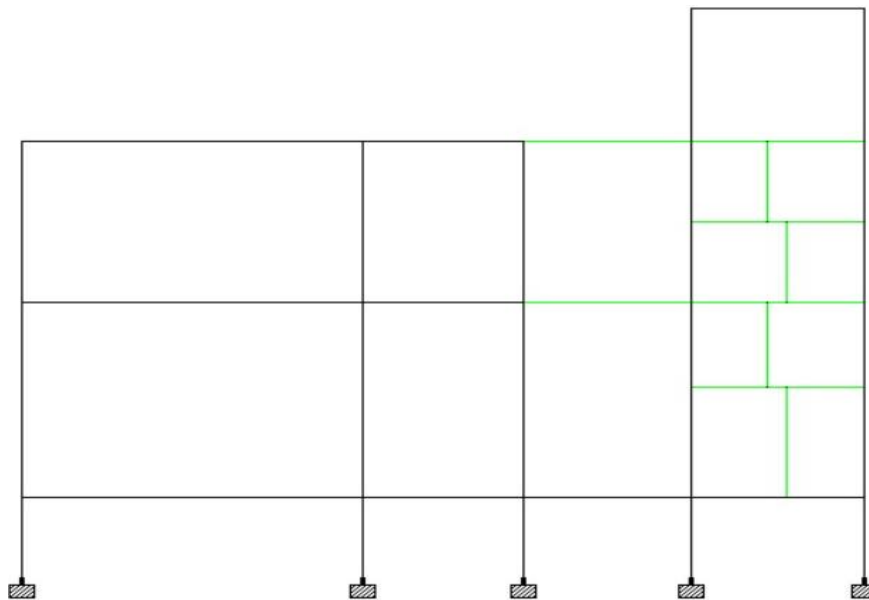


Figure 5.17 Typical Frame Layout (Front View)

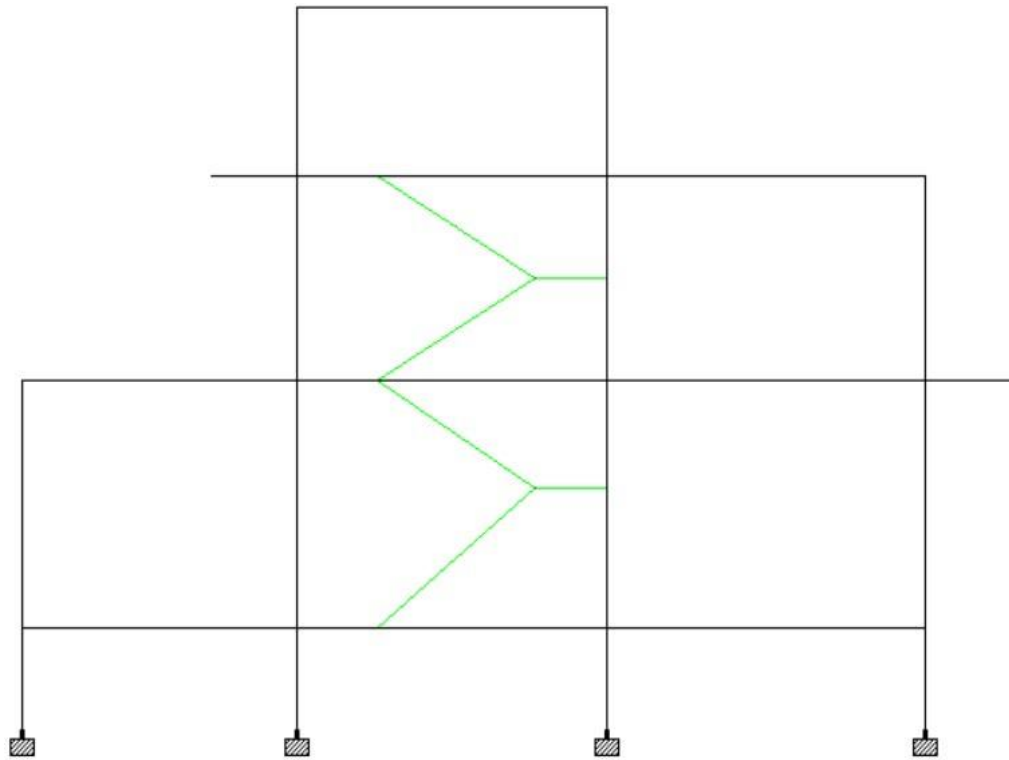


Figure 5.18 Typical Frame Layout (Side View)

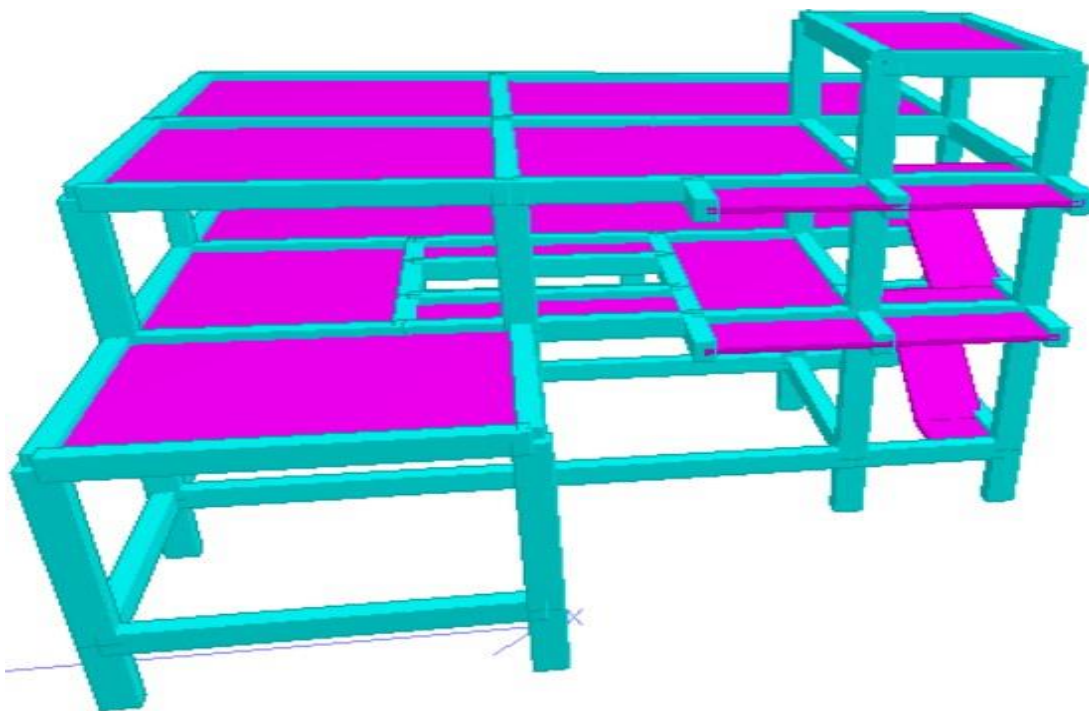


Figure 5.19 Rendered View of the Structure

ANALYSIS AND DESIGN RESULTS:

The structure is modelled in STAAD PRO with material property as concrete. The frames are assigned with fixed supports and the combination of dead loads and live loads are applied to each member.

The staad output of following elements is presented.

- Slabs
- Beams
- Columns
- Footings

SLAB DESIGN

Design of Slabs: -

Assume M₂₀ grade Concrete and Fe-415 Steel

Provide an overall depth D = 120 mm

Provide an effective depth d = 100 mm
(i.e 20 mm cover)

a) Slab Design (Two Way Slab)

Slab no 1: - (Living Room 5.10 m × 5.030 m)

➤ Loads Considered: -

$$\text{Self-weight of slab} = 25 \times 0.12 = 3.00 \text{ kN/m}^2$$

$$\text{floor finish} = 1 \text{ kN/m}^2$$

$$\text{Live load on slab} = 2 \text{ kN/m}^2$$

$$\text{Total working load} = 6 \text{ kN/m}^2$$

$$\text{Design or factored load} = 6 \times 1.5 = 9 \text{ kN/m}^2$$

$$\text{Effective shorter span } L_x = 5.030 \text{ m}$$

$$\text{Effective longer span } L_y = 5.10 \text{ m}$$

$$\text{Aspect Ratio} = \frac{L_y}{L_x} = \frac{5.1}{5.030} = 1.01 \leq 2 \text{ (Design as Two-way slab)}$$

➤ Design Moments & Shear Forces: -

This slab corresponds to Case-4 of Table 26 of IS:456-2000 as Two adjacent Edges discontinuous.

Moment Coefficients: -

$$\alpha_{x(-ve)} = 0.0475$$

$$\alpha_{x(+ve)} = 0.0355$$

$$\alpha_{y(-ve)} = 0.047$$

$$\alpha_{y(+ve)} = 0.035$$

Moments: -

$$\begin{aligned} M_{ux(-ve)} &= \alpha_{x(-ve)} W_u l_x^2 \\ &= 0.0475 \times 9 \times 5.030^2 = 10.816 \text{ kN-m} \end{aligned}$$

$$\begin{aligned} M_{ux(+ve)} &= \alpha_{x(+ve)} W_u l_x^2 \\ &= 0.0355 \times 9 \times 5.030^2 = 8.08 \text{ kN-m} \end{aligned}$$

$$\begin{aligned} M_{uy(-ve)} &= \alpha_{y(-ve)} W_u l_y^2 \\ &= 0.047 \times 9 \times 5.030^2 = 10.7 \text{ kN-m} \end{aligned}$$

$$\begin{aligned} M_{uy(+ve)} &= \alpha_{y(+ve)} W_u l_y^2 \\ &= 0.035 \times 9 \times 5.030^2 = 7.97 \text{ kN-m} \end{aligned}$$

Shear force: -

$$V_u = \frac{W_u l}{2} = \frac{9 \times 5.030}{2} = 22.635 \text{ kN}$$

➤ Minimum Depth Required: -

The minimum depth required to resist bending moment.

$$\begin{aligned} M_u &= 0.138 f_{ck} b d^2 \\ 10.816 \times 10^6 &= 0.138 \times 20 \times 1000 \times d^2 \\ d_{\text{required}} &= 62.6 \text{ mm} < d_{\text{provided}} \end{aligned}$$

Hence provided depth is adequate.

➤ Reinforcement: -

Along short span (-ve) @ support.

$$\begin{aligned} M_{ux} &= 0.87 f_y A_{st} d \left(1 - \frac{f_y \times A_{st}}{f_{ck} \times b \times d}\right) \\ 10.816 \times 10^6 &= 0.87 \times 415 \times A_{st} \times 100 \times \left(1 - \frac{415 \times A_{st}}{20 \times 1000 \times 100}\right) \end{aligned}$$

$$A_{st} = 320.994 \text{ mm}^2$$

Using 10mm diameter bars, spacing of bars

$$S = \frac{ast}{A_{st}} \times 1000 = \frac{\frac{\pi}{4} \times 10^2}{320.994} \times 1000 = 244.67 \text{ mm}$$

Maximum spacing is i) $3d = 3 \times 100 = 300\text{mm}$

ii) 300mm whichever is less Hence provide 10mm bars at 240 mm c/c.

Along short span (+ve) @ midspan.

$$M_{ux} = 0.87 f_y A_{st} d \left(1 - \frac{f_y \times A_{st}}{f_{ck} \times b \times d}\right)$$

$$8.08 \times 10^6 = 0.87 \times 415 \times A_{st} \times 100 \times \left(1 - \frac{415 \times A_{st}}{20 \times 1000 \times 100}\right)$$

$$A_{st} = 235 \text{ mm}^2$$

Minimum Reinforcement $A_{st,min} = 0.12\%$ of gross area

$$= \frac{0.12}{100} \times 1000 \times 120 = 144 \text{ mm}^2$$

Using 10mm diameter bars, spacing of bars

$$S = \frac{ast}{Ast} \times 1000 = \frac{\frac{\pi}{4} \times 10^2}{235} \times 1000 = 334 \text{ mm}$$

Maximum spacing is i) $3d = 3 \times 100 = 300\text{mm}$

ii) 300mm

whichever is less Hence

provide 10mm bars at 300mm c/c.

Along longer span (-ve) @ support.

These bars will be placed above the bars in X-direction

Hence $d = 100 - 10 = 90 \text{ mm}$

$$M_{uy} = 0.87 f_y A_{st} d \left(1 - \frac{f_y \times A_{st}}{f_{ck} \times b \times d}\right)$$

$$10.7 \times 10^6 = 0.87 \times 415 \times A_{st} \times 90 \times \left(1 - \frac{415 \times A_{st}}{20 \times 1000 \times 90}\right)$$

$$A_{st} = 360 \text{ mm}^2$$

Using 10mm diameter bars, spacing of bars

$$S = \frac{ast}{Ast} \times 1000 = \frac{\frac{\pi}{4} \times 10^2}{360} \times 1000 = 218\text{mm}$$

Maximum spacing is i) $3d = 3 \times 120 = 360\text{mm}$

ii) 300mm whichever is less Hence provide

10mm bars at 210 mm c/c.

Along longer span (+ve) @ midspan.

$$M_{uy} = 0.87 f_y A_{st} d \left(1 - \frac{f_y \times A_{st}}{f_{ck} \times b \times d}\right)$$

$$7.97 \times 10^6 = 0.87 \times 415 \times A_{st} \times 90 \times \left(1 - \frac{415 \times A_{st}}{20 \times 1000 \times 90}\right)$$

$$A_{st} = 262 \text{ mm}^2$$

Using 10mm diameter bars, spacing of bars

$$S = \frac{a_{st}}{A_{st}} \times 1000 = \frac{\frac{\pi}{4} \times 10^2}{260} \times 1000 = 306\text{mm}$$

Maximum spacing is i) $3d = 3 \times 120 = 360\text{mm}$

ii) 300mm

whichever is less hence provide 10mm bars at 300mm c/c.

b) Slab Design: (One Way Slab): -

Slab no : 5 (utility area 3.95×1.45)

➤ Loads Considered: -

$$\text{Self-weight of slab} = 25 \times 0.120 = 3.00 \text{ kN/m}^2$$

$$\text{floore finish} = 1 \text{ kN/m}^2$$

$$\text{Live load on slab} = 2 \text{ kN/m}^2$$

$$\text{Total working load} = 6.00 \text{ kN/m}^2$$

$$\text{Design or factored load} = 6 \times 1.5 = 9 \text{ kN/m}^2$$

$$\text{Effective shorter span } L_x = 1.45 \text{ m}$$

$$\text{Effective longer span } L_y = 3.95 \text{ m}$$

$$\text{Aspect ratio} = \frac{L_y}{L_x} = \frac{3.95}{1.45} = 2.72 > 2 \text{ (Design of one way slab)}$$

➤ Design Moments & Shear Forces: -

Moments: -

$$M_u = \frac{W_u \times l^2}{8} = \frac{9 \times 1.45^2}{8} = 2.37 \text{ kN-m}$$

Shear force: -

$$V_n = \frac{W_n l}{2} = \frac{9 \times 1.45}{2} = 2.625 \text{ kN}$$

➤ Minimum Depth Required: -

The minimum depth required to resist bending moment.

$$M_u = 0.138 f_{ck} b d^2$$

$$2.37 \times 10^6 = 0.138 \times 20 \times 1000 \times d^2$$

$$d_{\text{required}} = 30 \text{ mm} < d_{\text{provided}}$$

Hence provided depth is adequate.

➤ Main Reinforcement: -

$$M_u = 0.87 f_y A_{st} d \left(1 - \frac{f_y \times A_{st}}{f_{ck} \times b d}\right)$$

$$2.37 \times 10^6 = 0.87 \times 415 \times A_{st} \times 100 \times \left(1 - \frac{415 \times A_{st}}{20 \times 1000 \times 100}\right)$$

$$A_{st} = 67.00 \text{ mm}^2$$

Minimum Reinforcement $A_{st, \min} = 0.12\%$ of gross area

$$= \frac{0.12}{100} \times 1000 \times 120 = 144 \text{ mm}^2$$

$$A_{st \text{ required}} < A_{st \text{ min}}$$

So provide minimum A_{st}

Using 10mm diameter bars, spacing of bars

$$S = \frac{a_{st}}{A_{st}} \times 1000 = \frac{\frac{\pi}{4} \times 10^2}{144} \times 1000 = 545.5 \text{ mm}$$

Maximum spacing is i) $3d = 3 \times 100 = 300 \text{ mm}$

ii) 300mm

whichever is less Hence provide 10mm bars at 300mm c/c.

➤ Distribution Reinforcement: -

Minimum Reinforcement $A_{st, \min} = 0.12\%$ of gross area

$$= \frac{0.12}{100} \times 1000 \times 120 = 144 \text{ mm}^2$$

Using 8mm diameter bars, spacing of bars

$$S = \frac{ast}{Ast} \times 1000 = \frac{\frac{\pi}{4} \times 8^2}{144} \times 1000 = 350 \text{ mm}$$

Maximum spacing is i) $5d = 5 \times 100 = 500\text{mm}$

ii) 450mm

whichever is less Hence provide 8mm bars at 350mm c/c.

Design of Staircase: -

Staircase dimensions = 2.3m × 3.675m

Height of the floor = 3.2m

$$\text{Height of one flight} = \frac{3.2}{2} = 1.6\text{m}$$

Let us assume rise = 160mm

Thread = 275mm

$$\text{No of rises} = \frac{3200}{160} = 20$$

No of rises for each flight = 10 No's

Length required = $275 \times 9 = 2475\text{mm} = 2.475\text{m}$

Width of landing = $3.675 - 2.475 = 1.2\text{m}$

Adopt width of staircase = 1.2 m

➤ Effective Span: -

$$\begin{aligned} \text{Effective span} &= \text{centre to centre distance of walls} \\ &= 2.3 + 0.23 = 2.53\text{m} \end{aligned}$$

➤ Thickness of slab: -

$$\text{Assume effective depth } d = \frac{\text{span}}{25} = \frac{2530}{25} = 100.64 \text{ mm}$$

Adopt $d = 100 \text{ mm}$, $D = 125 \text{ mm}$

➤ Loads: -

Loads per meter width as shown below

$$\begin{aligned} \text{Weight of Slab} &= D \sqrt{1 + \left[\frac{R}{T} \right]^2} \times 25 \\ &= 0.125 \times \sqrt{1 + \left[\frac{0.16}{0.275} \right]^2} \times 25 = 3.49\text{kN/m} \end{aligned}$$

Live load = 2

kN/m²Floor Finish

$$= 1 \text{ kN/m}^2$$

Total load = 8.365 kN/m

Factored load $W_u = 8.365 \times 1.5 = 12.54 \text{ kN/m}$

$$\text{Factored Bending Moment} = \frac{W_u}{8} \times l^2 = \frac{12.54}{8} \times 2.51^2$$

➤ Minimum depth required: -

The minimum depth required to resist maximum bending moment.

$$M_u = 0.138 f_{ck} b d^2$$

$$9.9 \times 10^6 = 0.138 \times 30 \times 1000 \times d^2$$

$$d_{\text{required}} = 48.90 \text{ mm} < d_{\text{provided}}$$

Hence provided depth is adequate.

STRUCTURAL DRAWINGS AND DETAILING

Reinforcement Detailing of Slabs:

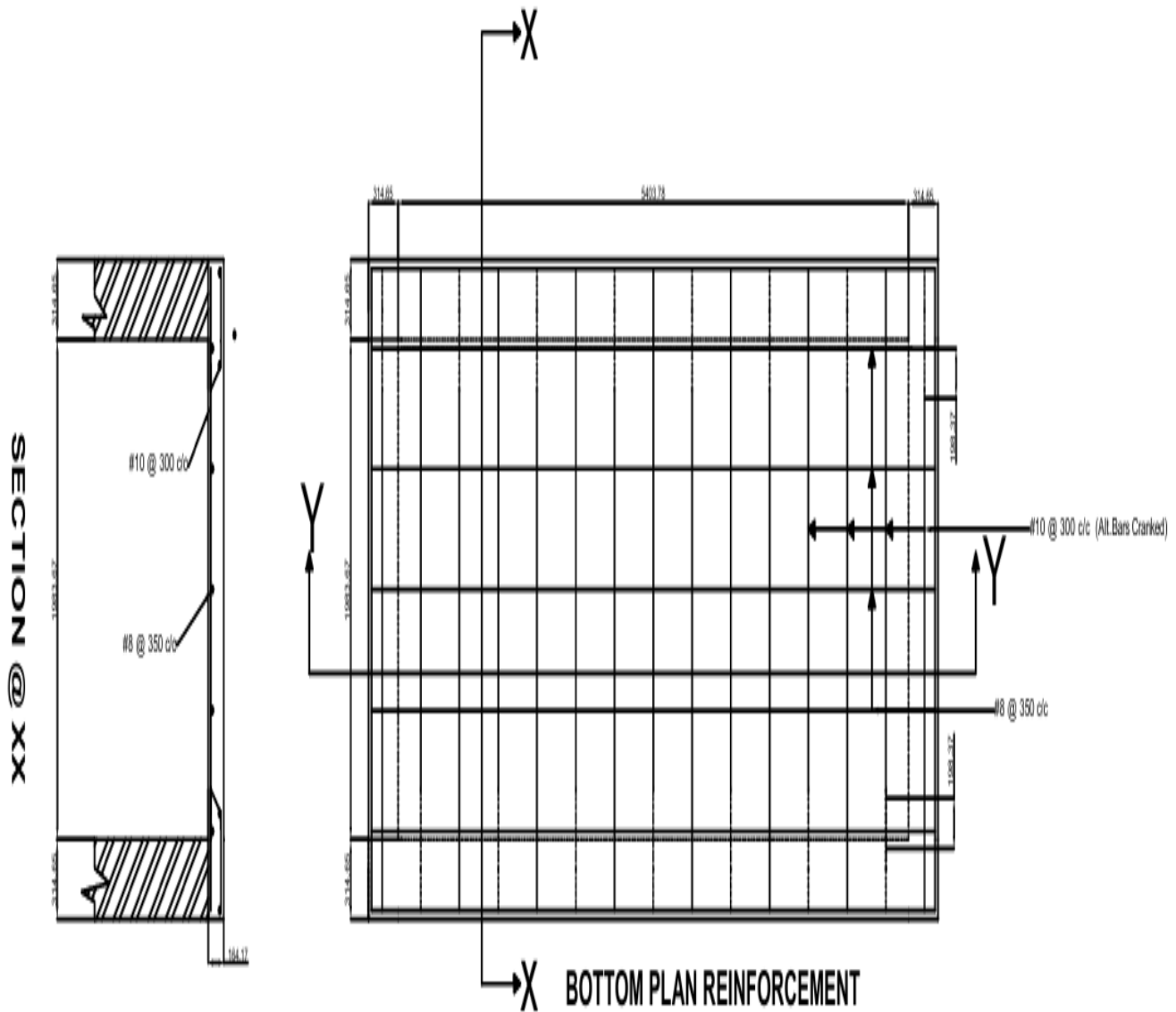
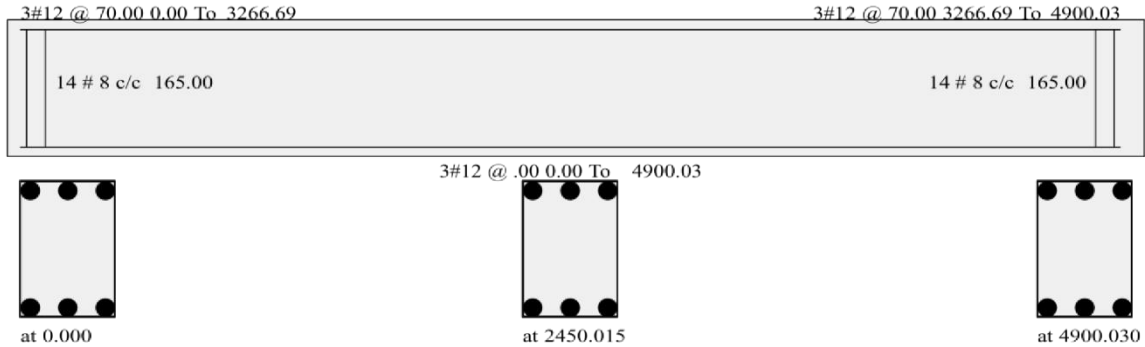


Figure 5.20 Reinforcement Detailing of Slab

STAAD.Pro Query Concrete Design

Beam no. Bm 2 @ plinth level

Design Code: IS-456



IS-456 LIMIT STATE DESIGN
BEAM NO. Bm 2 DESIGN RESULTS

M20 Fe415 (Main) Fe415 (Sec.)

LENGTH: 4900.0 mm SIZE: 230.0 mm X 375.0 mm COVER: 25.0 mm

TOP 257.46
REINF. (Sq. mm)

BOTTOM 257.46
REINF. (Sq. mm)

TOP 3-12i
REINF. 1 layer(s)

BOTTOM 3-12i
REINF. 1 layer(s)

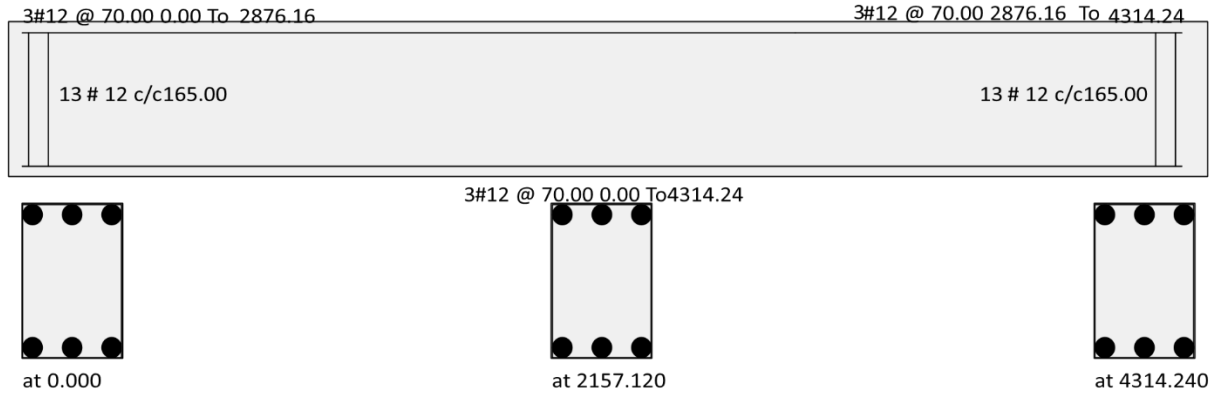
SHEAR 2 legged 8i
REINF. @ 165 mm c/c

FigureNo.21 Reinforcement Detailing of Beam

STAAD.Pro Query Concrete Design

Beam no. Bm 21

Design Code: IS-456



IS-456 LIMIT STATE DESIGN
BEAM NO. Bm 2 DESIGN RESULTS

M20 Fe415 (Main) Fe415 (Sec.)

LENGTH: 4314.0 mm SIZE: 230.0 mm X 375.0 mm COVER: 25.0 mm

TOP 257.46
REINF. (Sq. mm)

BOTTOM 257.46
REINF. (Sq. mm)

TOP 3-12i
REINF. 1 layer(s)

BOTTOM 3-12i
REINF. 1 layer(s)

SHEAR 2 legged 8i
REINF. @ 165 mm c/c

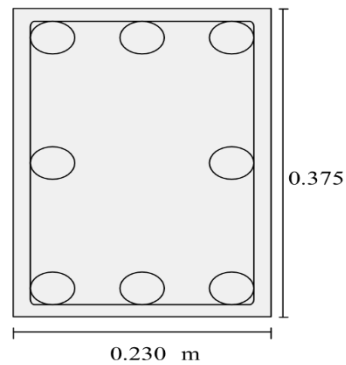
Figure No. 22 Reinforcement Detailing of beams

Figure5.23 Reinforcement Detailing of Columns:

STAAD.Pro Query Concrete Design

Beam no. CB 1

Design Code: IS-456



Design Results

Fy(Mpa)	415
Fc(Mpa)	20
As Reqd(mm ²)	617.000000
As (%)	1.049000
Bar Size	12
Bar No	8

IS-456 LIMIT STATE DESIGN

COLUMN NO. CB1 DESIGN RESULTS

M20 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3650.0 mm CROSS SECTION: 230.0 mm X 375.0 mm COVER: 40.0 mm

REQD. STEEL AREA : 616.96 Sq.mm.

REQD. CONCRETE AREA: 85633.05 Sq.mm.

MAIN REINFORCEMENT : Provide 8 - 12 dia. (1.05%, 904.78 Sq.mm.)
(Equally distributed)

TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

Figure5.23 Reinforcement Detailing of Footings:

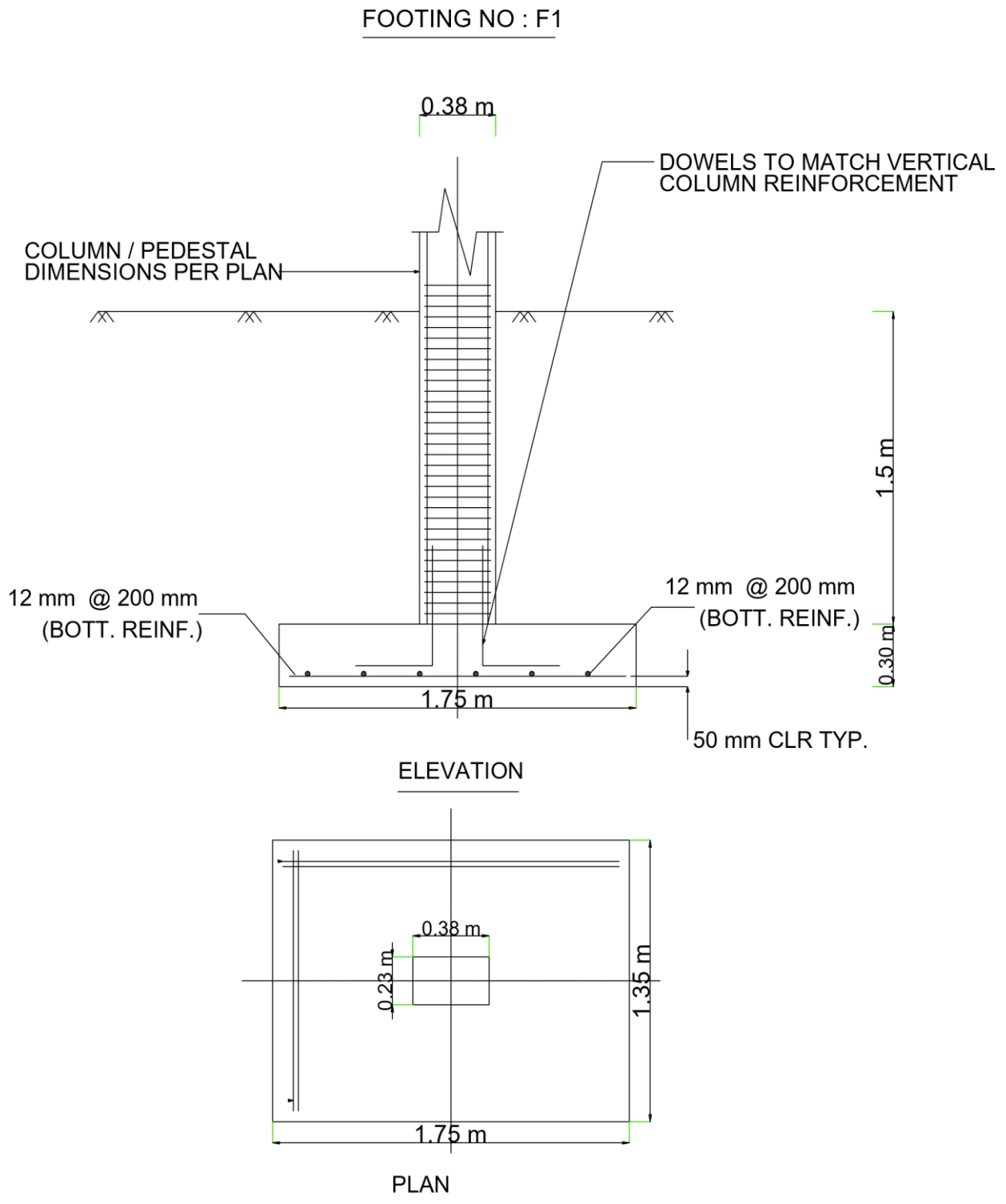


Figure 5.24 Reinforcement Detailing of Staircase

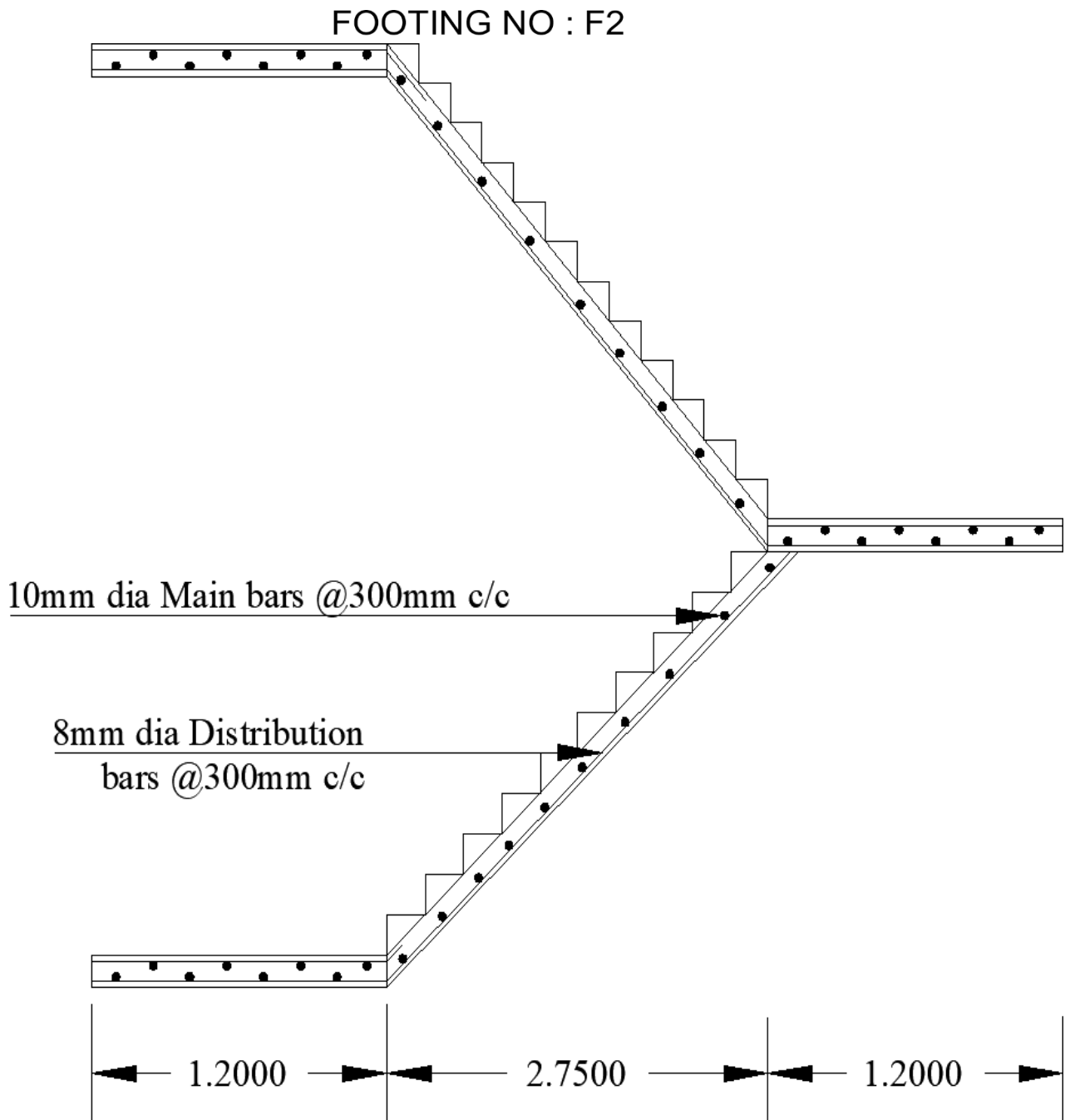
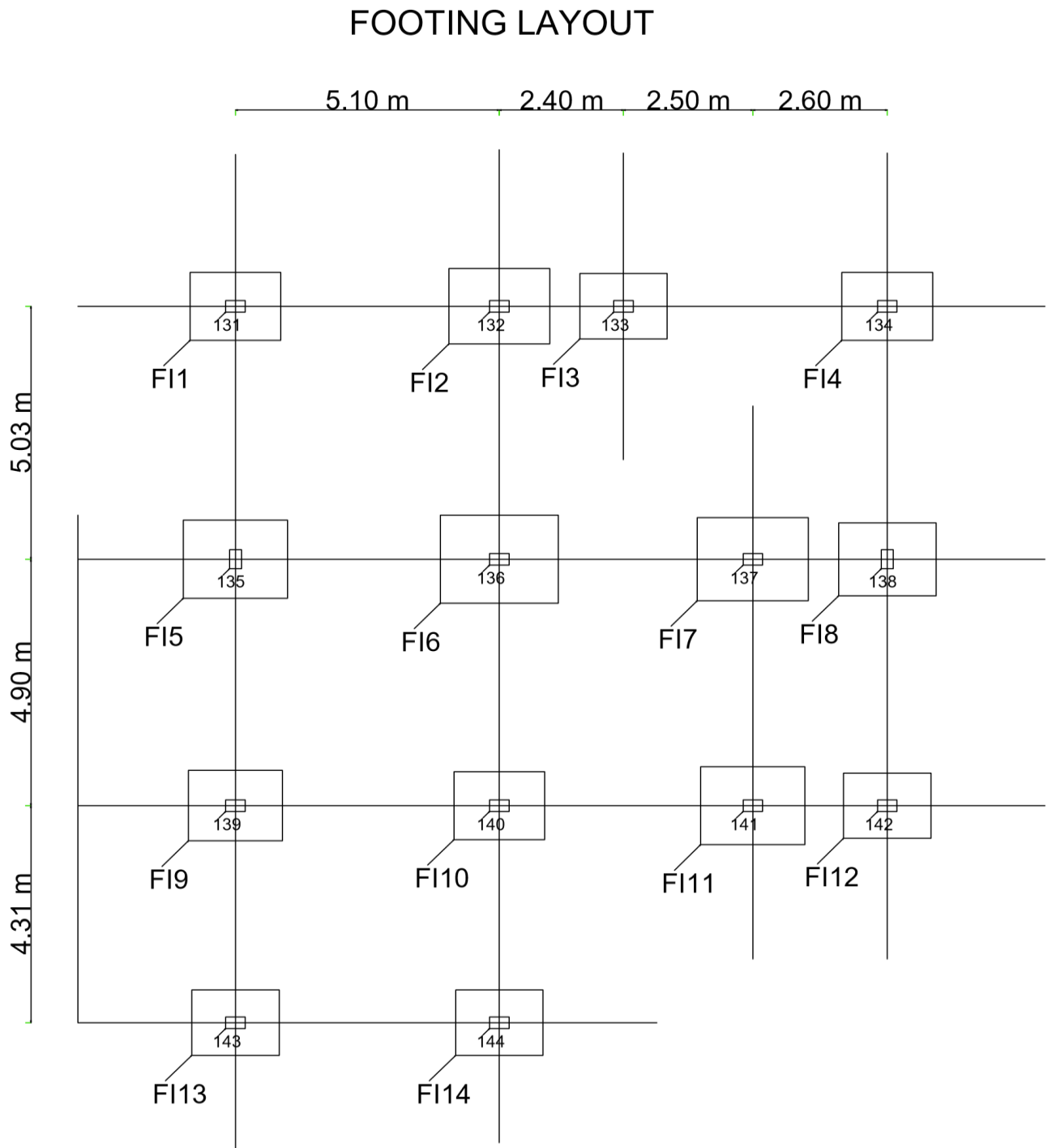


Figure5.25 FOOTING LAYOUT



5.3 REVIT ARCHITECTURE

BUILDING THE MODEL OF G+1 BUILDING

When a user makes a building, model, or any other kind of object in Revit, they may use Revit's rendering engine to make a more realistic image of what is otherwise a very diagrammatic model. This is accomplished by either using the premade model, wall, floor, etc., tools, or making her or his own models, walls, materials, etc. Revit 2010 comes with a plethora of predefined materials, each of which can be modified to the user's desires. The user can also begin with a "Generic" material. With this, the user can set the rotation, size, brightness, and intensity of textures, gloss maps (also known as shine maps), transparency maps, reflection maps, oblique reflection maps, hole maps, and bump maps, as well as leaving the map part out and just using the sliders for any one (or all or none) of the aforementioned features of textures.

An experienced user can create realistic and accurate families ranging from furniture to lighting fixtures, as well as import existing models from other programs. Revit families can be created as parametric models with dimensions and properties. This lets users modify a given component by changing predefined parameters such as height, width or number in the case of an array. In this way a family defines a geometry which is controlled by parameters, each combination of parameters can be saved as a type, and each occurrence (instance in Revit) of a type can also contain further variations. For example, a swing door may be a Family. It may have types describing different sizes, and the actual building model will have instances of those types placed in walls where instance-based parameters could specify the door hardware uniquely for each occurrence of the door.

Although Revit software comes with a range of families out of the box (OOTB), they are limited, so users can find a need to build their own families or buy them from online stores such as Plan-sort, Bimbandit or Andekan. A number of websites offer families for free including Revitcity, Augi and National Bimlibrary.

PARAMETRIC COMPONENTS

Parametric Components, also known as families, are the basis for all building components designed in Revit Architecture. They offer an open, graphical system for design thinking and form making as well as an opportunity to express design intent at increasingly detailed levels.

Use Parametric Components for the most elaborate assemblies, such as cabinetry and equipment, as well as for the most elementary building parts, such as walls and columns. Best of all, no programming language or coding is required.

SCHEDULES

Schedules are just another view of the Revit Architecture model. A change to a schedule view is automatically reflected in every other view and vice versa. Functionality includes associative split-schedule sections and selectable design elements via schedule views, formulas, and filtering.

LINE DIAGRAM OF ALL FLOORS

HOW I CREATED LINE DIAGRAM

- Setting the units to meters
- Then set the dimensions
- Creation of ground, first, and terrace
- The floor height is given for ground floor= 3200 mm, first floor= 3000 mm , head room= 2500 mm
- Using the wall tool, line diagram is created for ground, first floor.



Figure 5.26 Ground Floor line plan

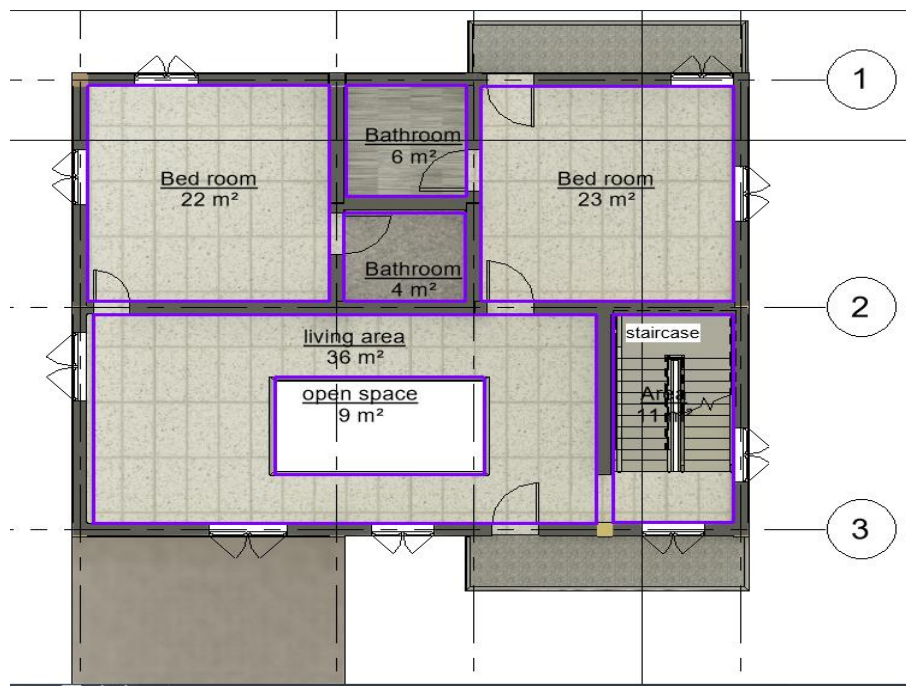


Figure 5.27 First Floor line Plan

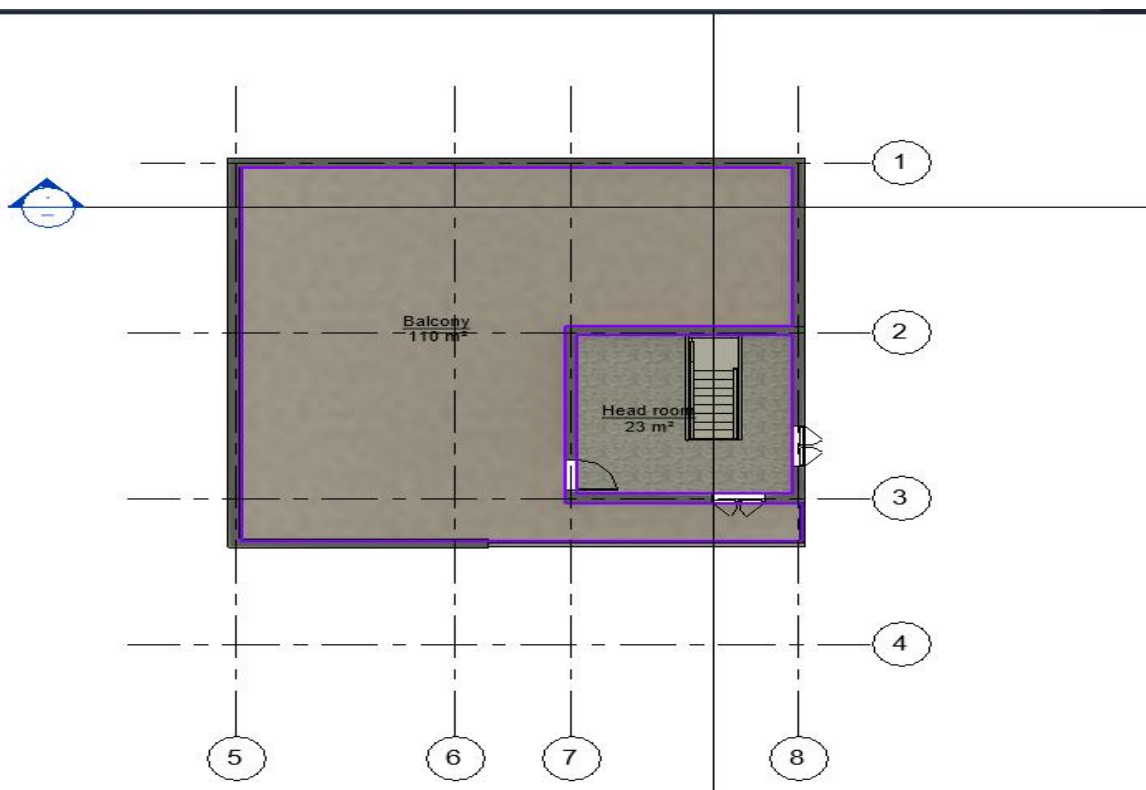


Figure 5.27 Roof line plan

STEPS TO CREATE FOUNDATION IN REVIT

1. **Activate the Structural Tab:** Switch to the "Structure" tab on the ribbon at the top of the screen. This tab contains tools for creating structural elements like footings.
2. **Create Isolated Footing:**
 - Click on the "Foundation" tool in the Structure panel. Alternatively, you can type "FO" and press Enter.
 - Choose "Isolated" from the Foundation Type dropdown menu.
 - Select the desired footing type and size from the Type Selector.
 - Place the isolated footing by clicking on the desired location in the plan view. You may need to adjust the footing's size and position according to your design requirements.
3. **Adjust Footing Properties:**
 - Once the isolated footing is created, you can select it and modify its properties using the properties palette. You can adjust dimensions, materials, and other parameters as needed.

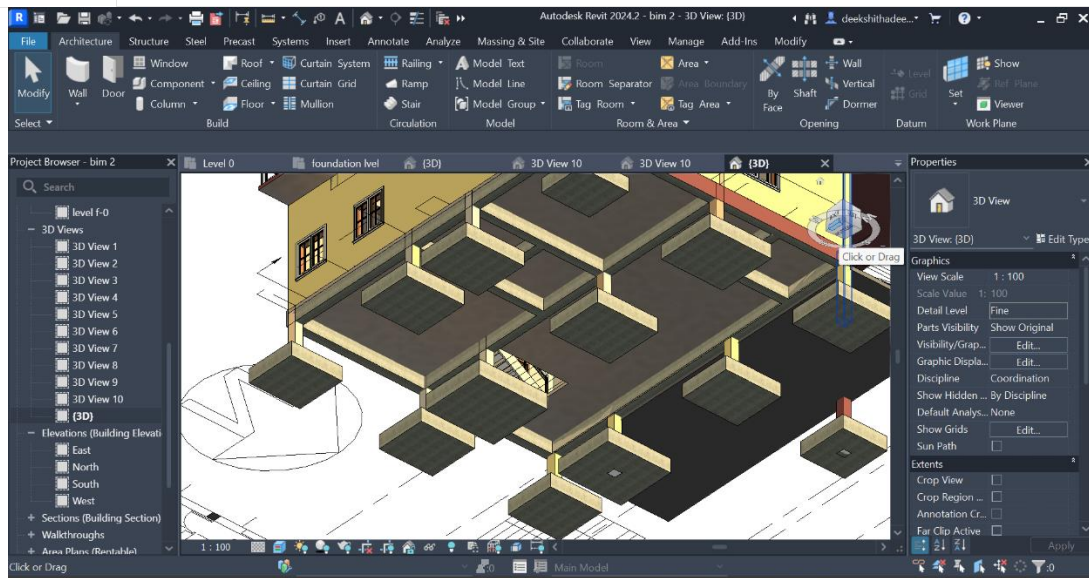


Figure5.28 Footing 3D Diagram



Figure 5.29 Wall, door and Window Tool

- Doors, windows and staircases to 3 floors are added using door, window and staircase tools
- The glass windows for living room and bed room have been created using Stire front wall
- The is added all three floors using Model text tool.

STEPS TO CREATE MASONRY GENERIC WALL IN REVIT

1. **Open Revit and Start a New Project:** Open Revit and start a new project or open an existing one where you want to create the masonry wall.
2. **Select Wall Tool:** In the ribbon menu, go to the "Architecture" tab, and click on the "Wall" tool.
3. **Choose Wall Type:** In the Properties palette, under "Type Selector," select a wall type that's close to what you want. You can choose a generic wall type and modify it later.
4. **Edit Wall Type:** If the selected wall type is not suitable, you can duplicate and edit it. Click on the "Edit Type" button in the Properties palette to modify the properties of the wall type.
5. **Change Wall Structure:** In the "Edit Type" dialog, you can modify the wall's structure. Add layers to represent the masonry construction. Typically, a masonry wall consists of layers such as brick or block, mortar, insulation, and plaster. Adjust the thickness and material of each layer accordingly.
6. **Edit Materials:** Assign appropriate materials to each layer. You can either choose from the existing material library in Revit or create custom materials to represent the appearance of masonry.
7. **Adjust Wall Height and Length:** Click on the wall in the drawing area to select it. You can then adjust the wall's height and length by dragging the grips or by entering precise dimensions in the Properties palette.
8. **Finish Editing:** Once you're satisfied with the wall's properties, click "OK" to finish editing the wall type.

9. **Place Wall in the Project:** With the wall type configured, you can now place masonry walls in your project by clicking in the drawing area.

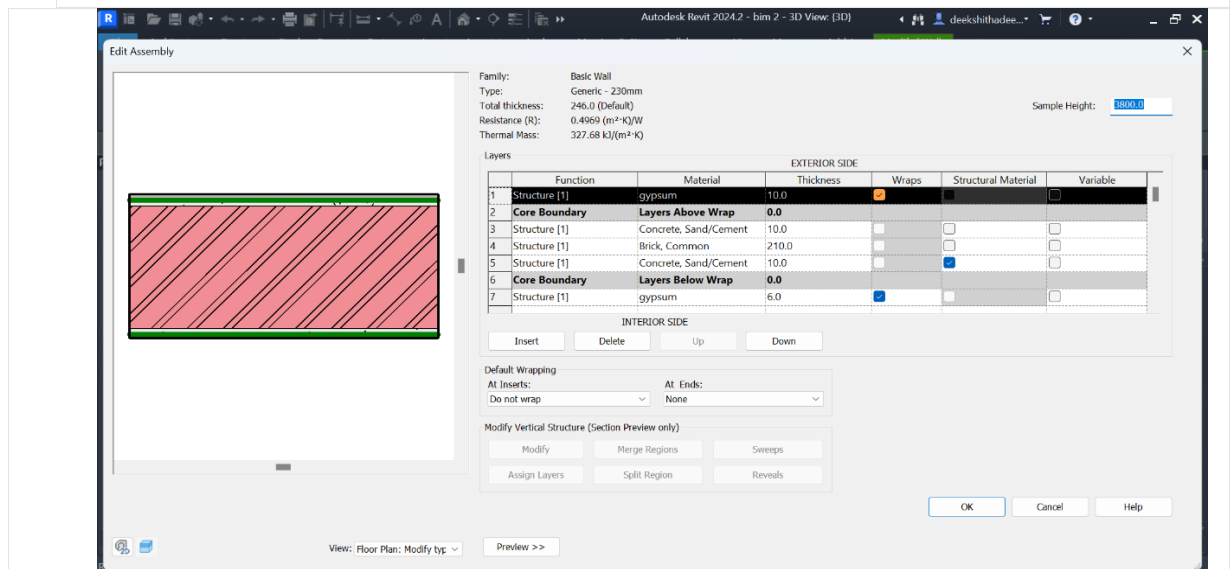


Figure 5.30 Generic Brick Wall Properties

STEPS TO CREATE COLUMNS AND BEAMS IN REVIT

1. **Activate the Structural Tab:** Go to the "Structure" tab on the ribbon at the top of the screen. This tab contains tools for creating structural elements like columns and beams.
2. **Create Columns:**
 - Click on the "Column" tool in the Structure panel. Alternatively, you can type "CO" and press Enter.
 - Choose the column type from the dropdown menu, such as architectural or structural columns.
 - Place the column by clicking on the desired location in the plan view. You can specify the height and other properties in the properties palette.
3. **Create Beams:**
 - Click on the "Beam" tool in the Structure panel. Alternatively, you can type "BE" and press Enter.
 - Choose the beam type from the dropdown menu, such as steel, concrete, or wood.

- Sketch the path of the beam by clicking to define its start and end points. You can create straight or curved beams.
- Adjust the beam properties such as height, width, and elevation in the properties palette.

4. **Modify Columns and Beams:**

- Once columns and beams are created, you can select them and modify their properties using the properties palette. You can adjust dimensions, materials, and other parameters as needed.
- You can also use editing tools like Move, Rotate, or Mirror to modify the position or orientation of columns and beams.

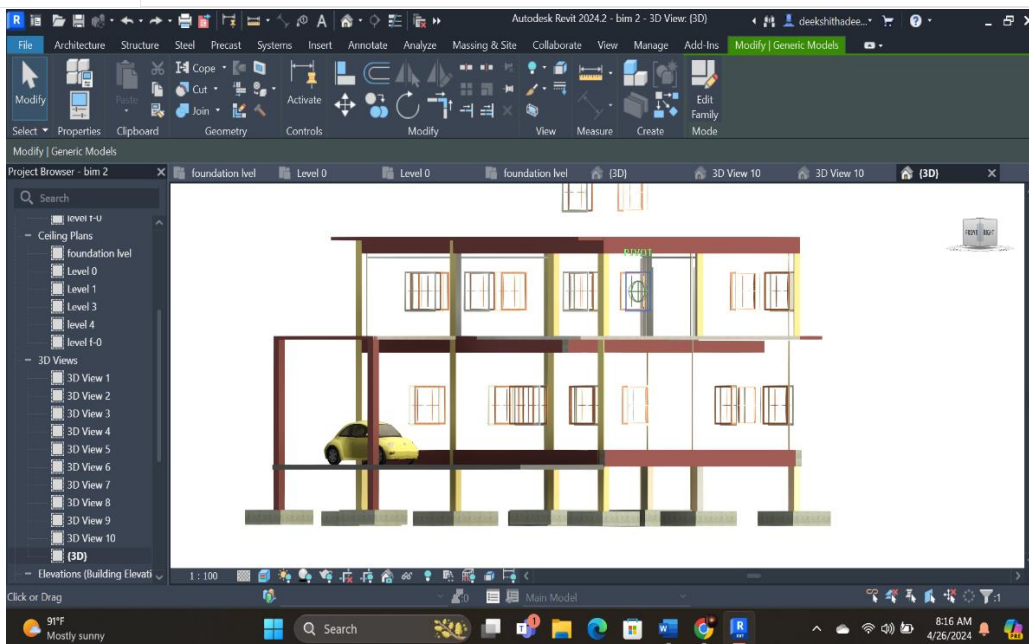


Figure 5.31 Beams and columns of structure

HOW I TO CREATED FLOORS

1. **Open Revit:** Launch Revit software on your computer.
2. **Start a New Project or Open an Existing One:** You can either start a new project or open an existing project where you want to add floors.

3. **Select the Level:** In the project browser, navigate to the level where you want to create the floor. Levels represent the different heights or stories in your building.
4. **Activate the Floor Tool:** Go to the "Architecture" tab on the ribbon at the top of the screen. Then, click on the "Floor" button in the Build panel. Alternatively, you can type "FL" and press Enter.
5. **Choose the Floor Type:** Revit offers different types of floors such as Slab, Structural, or Architectural. Select the appropriate floor type for your project from the dropdown menu.
6. **Sketch the Floor Boundary:** Click to place points to define the perimeter of your floor. You can create straight lines by clicking, or create curved lines by clicking and dragging. Double-click to finish the sketch.
7. **Adjust Floor Properties:** Once the floor is created, you can adjust its properties such as height, thickness, material, etc., by selecting the floor and modifying its properties in the Properties palette.
8. **Modify the Floor Shape:** You can further modify the floor shape by selecting it and using the Modify tools like Move, Copy, or Trim/Extend.
9. **Add Openings:** If your floor needs openings for stairs, ramps, or other elements, you can use the "Floor Opening" tool to create them.

Finish Editing: Once you're satisfied with your floor, click the "Finish Edit Mode" button or press the Esc key to exit the sketch mode.

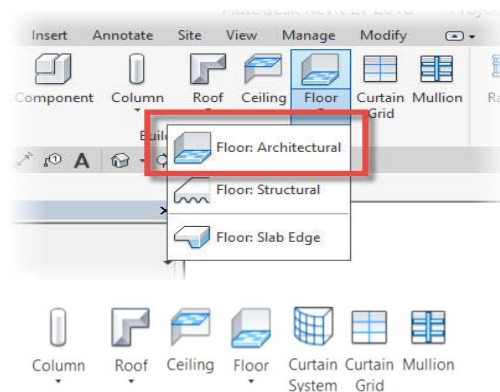


Figure 5.32 Architectural tool bar

CREATION OF CEILING IN REVIT

1. **Activate the Ceiling Tool:** Go to the "Architecture" tab on the ribbon at the top of the screen.
2. **Place Ceilings:**
 - Click on the "Ceiling" tool in the Build panel. Alternatively, you can type "CE" and press Enter.
 - Choose the ceiling type you want to create (e.g., suspended ceiling, plaster ceiling).
 - Click to place ceilings on the desired rooms or areas of your building. Revit will automatically detect the boundaries of rooms.
3. **Adjust Ceiling Properties:**
 - Once the ceilings are placed, you can select them and modify their properties using the Properties palette. You can adjust parameters such as height, material, thickness, and appearance.

STEPS TO CREATE STAIRCASE ALONG WITH RAILING

1. **Create Staircase:** Use the Stair tool to draw the outline of the staircase. Adjust parameters such as number of risers, tread width, and overall dimensions.
2. **Edit Stair Properties:** Modify the properties of the staircase as needed, such as the type of stair (straight, L-shaped, U-shaped, etc.), materials, and dimensions.
3. **Add Railing:** Once the staircase is in place, use the Railing tool to draw the railing along the edges of the staircase. You can choose different railing types and adjust their properties.
4. **Edit Railing Properties:** Customize the railing properties, such as height, handrail type, baluster spacing, and materials.

5. Adjust Connections: Ensure that the railing connects properly to the staircase and any surrounding elements. Use the Modify tools to adjust connections and alignments if necessary.

6. Review in 3D: Switch to 3D view to review the staircase and railing assembly. Make any final adjustments to ensure everything looks correct and functions properly.

STEPS TO APPLICATION OF PAINTS TO BUILDING

1. Access the Material Browser:

- Go to the Manage tab on the ribbon at the top of the screen.
- Click on the "Materials" button in the Settings panel. This will open the Material Browser where you can manage and create materials.

2. Create or Choose a Material:

- In the Material Browser, you can either choose an existing material from the library or create a new material.
- To create a new material, click on the "Create Material" button and specify the properties such as name, appearance, color, texture, reflectivity, transparency, etc.

3. Apply Material to Elements:

- Once you have the desired material created or selected, you can apply it to the building elements in your project.
- Select the element (e.g., wall, floor, column) to which you want to apply the material.
- Go to the Properties palette and find the Material parameter.
- Click on the button next to the Material parameter to open the Material Browser.
- Choose the material you want to apply from the Material Browser and click OK.

4. Adjust Material Properties (Optional):

- After applying the material to an element, you can further adjust its properties if needed.
- Double-click on the element to select it and then click on the "Edit Type" button in the Properties palette.

- This will open the Material Editor where you can modify the material's properties such as color, texture, reflectivity, etc.

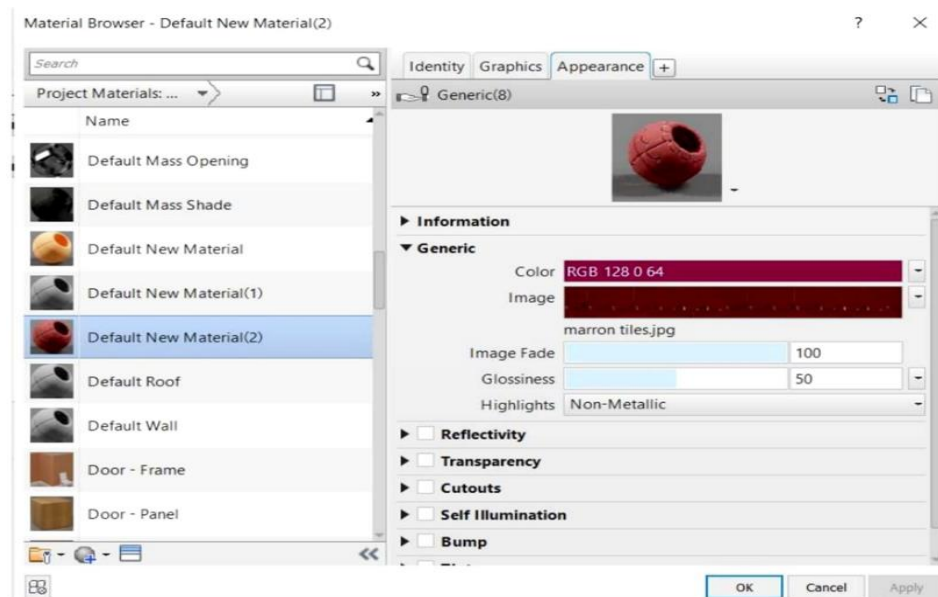


Figure 5.33 Painting Material Browser

CREATION OF TOPO SURFACE AND SITE COMPONENTS

Creating a Topography Surface:

1. **Activate the Site Tab:** Switch to the "Site" tab on the ribbon at the top of the screen. This tab contains tools for creating and editing site elements.
2. **Create a Toposurface:**
 - Click on the "Toposurface" tool in the Model Site panel. Alternatively, you can type "TO" and press Enter.
 - Choose the method for creating the toposurface (e.g., by points, from import, by sketching).
 - Follow the prompts to define the shape and elevation of the toposurface. You can specify elevation points manually, import data from a file, or sketch the boundary.
3. **Edit Topo-surface:**
 - After creating the topo-surface, you can further edit it to refine the terrain.
 - Use tools like Add Points, Delete Points, Move Points, or Smooth Surface to adjust the shape and elevation of the toposurface as needed.

1. **Adding Site Components: Activate the Site Tab:** Switch back to the "Site" tab on the ribbon at the top of the screen.
2. **Choose Site Components:**
 - Revit offers various site components such as roads, sidewalks, parking lots, trees, shrubs, and more.
 - Click on the desired site component from the Site Components panel. If needed, you can access additional components by clicking on the drop-down arrow next to each category.
3. **Place Site Components:**
 - Click to place the selected site components on the toposurface or adjacent to it. You can adjust their size, orientation, and position as needed.
4. **Adjust Site Component Properties:**
 - Once site components are placed, you can select them and modify their properties using the Properties palette. You can adjust parameters such as material, color, height, density, and other settings.



Topo-surface and Site component tool

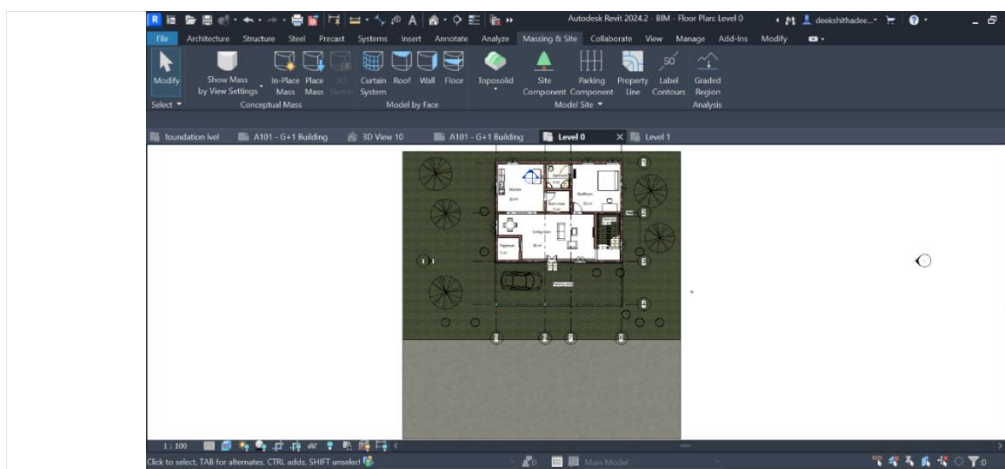


Figure 5.34 Topo-surface and site components in line plan

Figure 5.35 North Section view:



Figure 5.36 South Section View

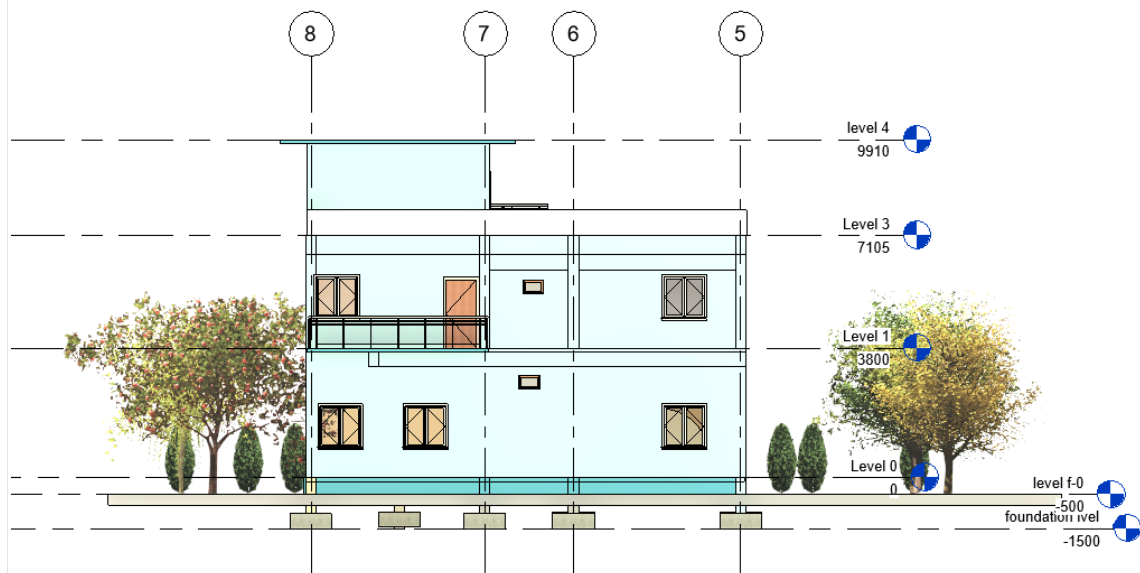


Figure 5.37 East Section View

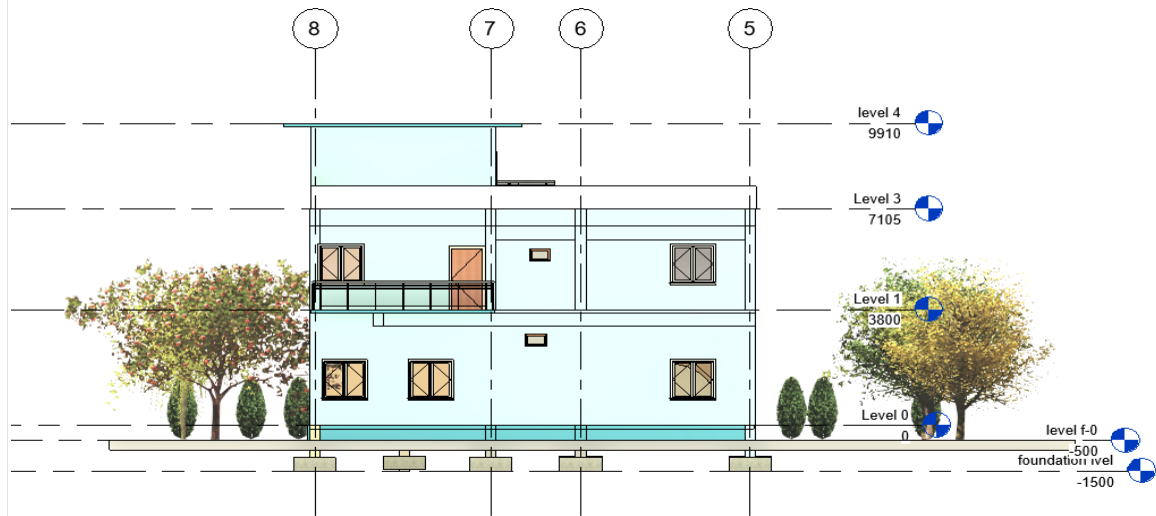


Figure 5.38 West Section view

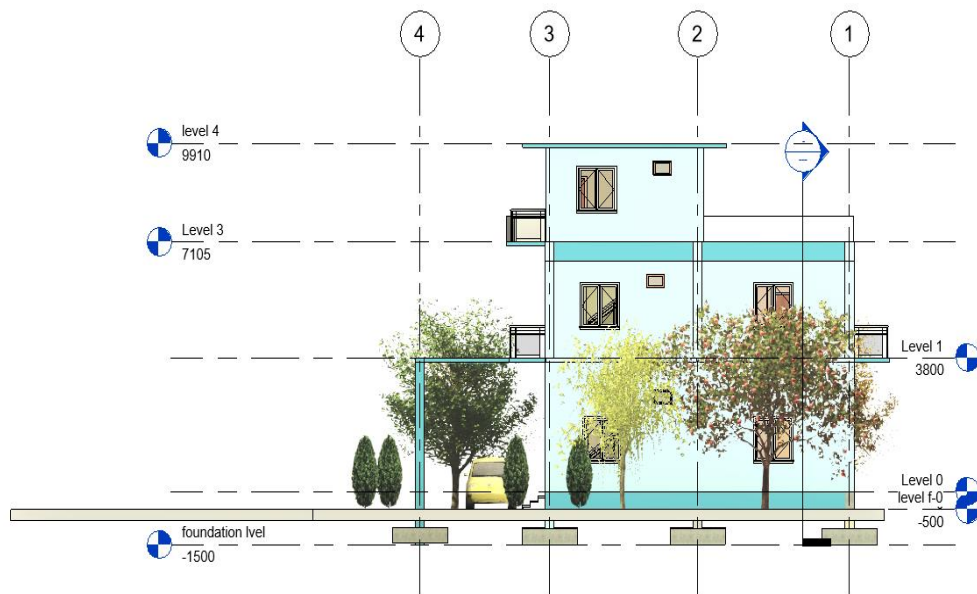


Figure 5.39 Ground floor 3D view:



Figure 5.40 First floor 3D view



Figure 5.41 Top floor view of Building

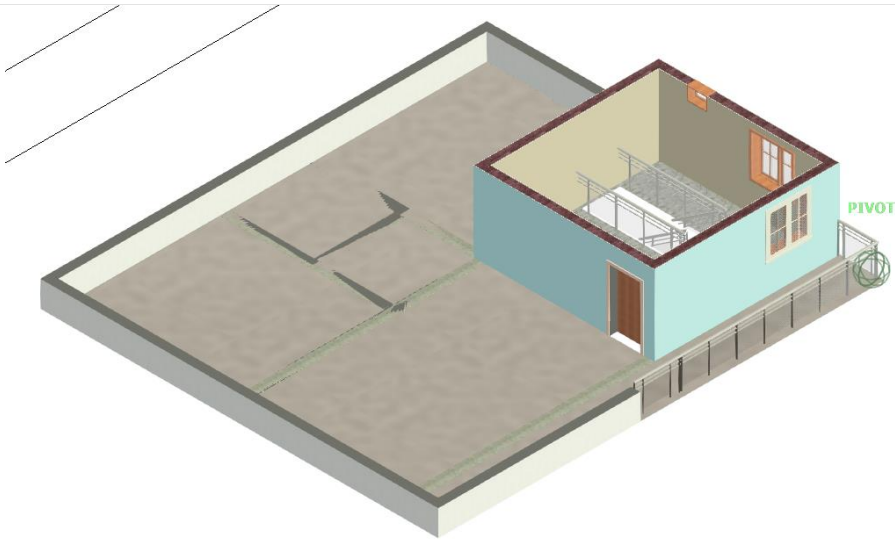


Figure 5.42 3D model of G+1 Building



CHAPTER 6

PLUMBING WORK USING REVIT FOR G+1

Plumbing work using Revit

Revit facilitates an integrated approach to building design by allowing architects, structural engineers, and MEP (mechanical, electrical, and plumbing) professionals to work within a single model. This Integration ensures that all elements of a building's design are coordinated, reducing conflicts and streamlining the design process.

Plumbing systems often need to be carefully coordinated with mechanical and electrical systems as well as structural components like beams and walls. Revit allows for precise 3D modeling, which helps in identifying and resolving spatial conflicts before construction begins. This can prevent costly on-site errors and redesigns.

Changes are a constant in construction projects, and Revit makes it easier to manage these changes. When modifications are made to a model, Revit automatically updates all related views and sheets, including floor plans, sections, and elevations. This ensures consistency and accuracy across all project documentation.

Revit supports BIM (Building Information Modeling) processes, which are essential for modern construction projects involving multiple stakeholders. With features like worksharing, different team members can work on the same model simultaneously, regardless of their location. This facilitates better communication and decision-making among teams.

Revit automatically generates detailed, customizable schedules for plumbing fixtures, fittings, piping, and other components. These schedules provide comprehensive information that can be used for installation, procurement, and maintenance purposes.

Steps to Create Plumbing work in Revit

1. Setting Up Your Project

Create a New Project: Start by setting up a new project using a template that suits your needs, whether it's a commercial or residential building.

Define Levels and Grids: Establish levels and grids which will help in placing and routing your plumbing components accurately.

2. Importing Architectural and Structural Models

Link Other Disciplines: If you are working as part of a larger team, link architectural and structural models to coordinate your work with other disciplines.

Use Copy/Monitor: Utilize the Copy/Monitor features for elements that are crucial for plumbing, like walls and floors, to ensure any changes in other models are updated in your project.

3. Creating Plumbing Systems

Piping Systems: Use the 'Piping' workspace to create domestic hot and cold water systems, and waste systems. Revit includes tools for automatically generating piping layouts based on the fixtures you select.

Plumbing Fixtures: Insert plumbing fixtures like sinks, toilets, and showers from the Revit library or import from manufacturer-specific models.

System Browser: Use the system browser for managing and reviewing the systems in your project to ensure there are no clashes or issues.

4. Routing Pipes and Adding Accessories

Automatic Routing: Utilize Revit's automatic routing solutions to easily connect fixtures with appropriate pipes and fittings.

Manual Adjustments: Make manual adjustments as necessary to optimize the system layout and to accommodate specific design constraints.

Accessories: Don't forget to add in all necessary accessories and equipment, such as pumps, valves, and water heaters.

5. Sloping of Piping

Gravity Systems: Pay particular attention to waste and vent systems that require precise sloping to function correctly. Revit enables setting slopes for pipes, ensuring that gravity systems work efficiently.

6. Analyzing and Simulating

Flow and Pressure Calculations: Use Revit's calculation tools to check the flow, velocity, and pressure drop across your piping systems.

System Analysis: Leverage tools like Autodesk Insight for more in-depth analysis, including energy analysis if needed.

7. Coordination and Clash Detection

Interdisciplinary Review: Regularly coordinate with other disciplines to avoid clashes. Use Revit's 'Interference Check' tool to detect and resolve clashes between plumbing and other building systems.

Revit Worksharing: Utilize Revit worksharing features to allow multiple users to work on the same project file simultaneously.

8. Documentation and Detailing

Drawing Sheets: Create detailed drawing sheets that include plan views, sections, and details specifically for the plumbing systems.

Tagging and Annotations: Add tags, dimensions, and annotations to provide clarity and ensure that the plumbing systems are well documented.

Schedules: Generate schedules for fixtures, fittings, piping, and equipment to facilitate installation and procurement.

9. Exporting and Sharing

Export Formats: Prepare your plumbing model for export to various formats as required by other stakeholders, such as contractors or other engineering disciplines.

BIM Collaboration: Share the Revit model using BIM collaboration tools to ensure all stakeholders have the latest information.

10. Revisions and Updates

Manage Revisions: Keep track of any changes and updates in your model with Revit's revision management tools.

By following these steps, you can ensure that your Revit plumbing project is detailed, accurate, and well-coordinated with other building systems and disciplines. Always keep learning and updating your skills as Revit and BIM methodologies evolve.

CHAPTER 7

INSTALLATION OF ELECTRICAL LIGHTENING SYSTEM USING USING REVIT FOR G+1

Importance of Electrical Lightening System using Revit

Revit allows architects and designers to create detailed 3D models of buildings, including lighting fixtures and their placement. This level of accuracy ensures that lighting designs are appropriately Integrated with other building elements, leading to efficient and effective lighting solutions.

Revit enables simulation and analysis of lighting designs, including daylighting analysis, energy consumption calculations, and lighting levels. By using these tools, designers can optimize lighting designs for energy efficiency and occupant comfort.

BIM platforms like Revit facilitate coordination among different disciplines involved in building design, such as architecture, MEP (Mechanical, Electrical, and Plumbing), and structural engineering. Integrating lighting design within the BIM environment ensures that lighting systems are coordinated seamlessly with other building systems, reducing conflicts and errors during construction.

Revit's visualization capabilities allow designers to create realistic renderings and walkthroughs, providing clients and stakeholders with a clear understanding of the proposed lighting design. This enhances communication and decision-making throughout the design process.

Revit generates detailed documentation and specifications for lighting fixtures, including schedules, luminaire data, and specifications. This streamlines the procurement process and ensures that contractors have accurate information for installation.

Steps to create Electrical Lightening System using Revit

1.Setting Up the Project:

- **Open Revit Software:** Launch the Revit software on your computer.
- **Open Existing Project:** Navigate to the location of your existing project file (.rvt) using the "Open" option in Revit's startup dialog or the "File" menu. Select the project file and click "Open."
- **Project Units:** Check and adjust the project units if necessary. Go to the "Manage" tab and click on "Project Units." Ensure that the units for length, area, volume, and other parameters are set correctly based on your project requirements.
- **Save Changes:** Once you've completed the setup and configuration, save any changes made to the project file by clicking on the "Save" or "Save As" option in the "File" menu.

2. Creating Electrical Systems:

- **Accessing the Electrical System Tool:** To create electrical systems in Revit, first, ensure you're in the appropriate view (such as a floor plan or a reflected ceiling plan) where you want to place the electrical components. Then, navigate to the "Systems" tab on the ribbon.
- **Selecting the Electrical System Type:** In the "Systems" tab, locate the "Electrical" panel. Here, you'll find various electrical system types, such as Lighting, Power, Data, and Fire Alarm. Choose the type of electrical system you want to create based on your project requirements.
- **Placing System Components:** After selecting the desired electrical system type, you can start placing system components in the model. For example, if you're creating a lighting system, you would place lighting fixtures such as ceiling lights, wall sconces, or pendant lights in the appropriate locations on the floor plan.
- **Defining Circuiting:** As you place electrical components, Revit automatically assigns them to electrical circuits based on the system type. You can define circuiting parameters such as circuit number, panel assignment, and load classification for each component. Revit also provides tools to manually adjust circuit assignments if needed.

3. Placing Fixtures and Devices:

Once you've selected the electrical system type, you can start placing fixtures and devices in the model. Choose the desired fixture or device from the Revit library, which includes a wide range of pre-built families for lighting fixtures, outlets, switches, panels, and other electrical components.

4. Performing Electrical Analysis (Optional):

Revit offers tools for performing electrical analysis, such as voltage drop calculations and circuit sizing. This helps ensure that your electrical design meets performance requirements and standards.

5. Generating Documentation:

Produce construction documents, drawings, and reports from your Revit model. This includes plans, elevations, schedules, and specifications related to the electrical systems in your project. Such as autocad, dwg, ... files.

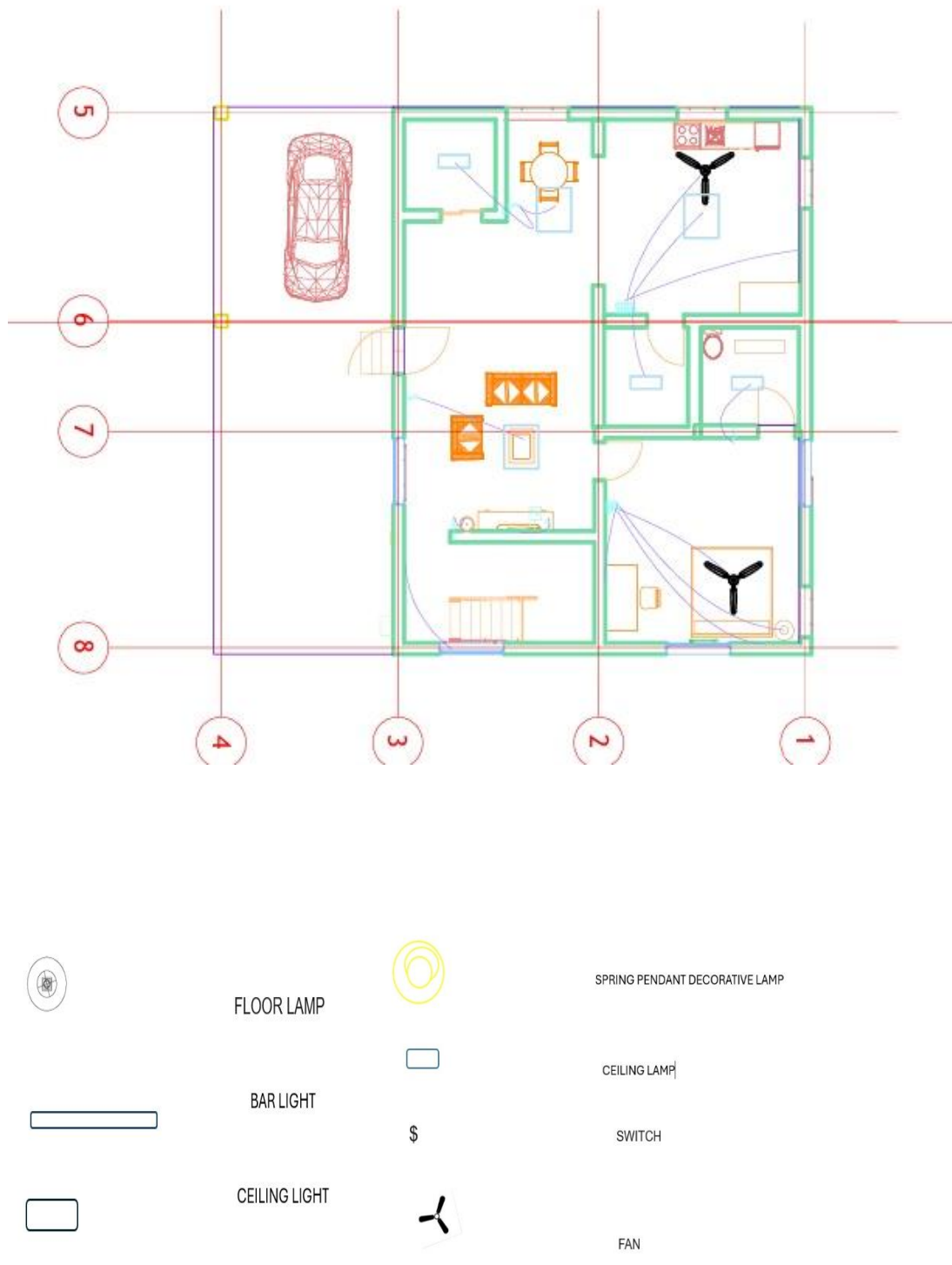


Figure 7.1 Ground floor electrical and lightening planning

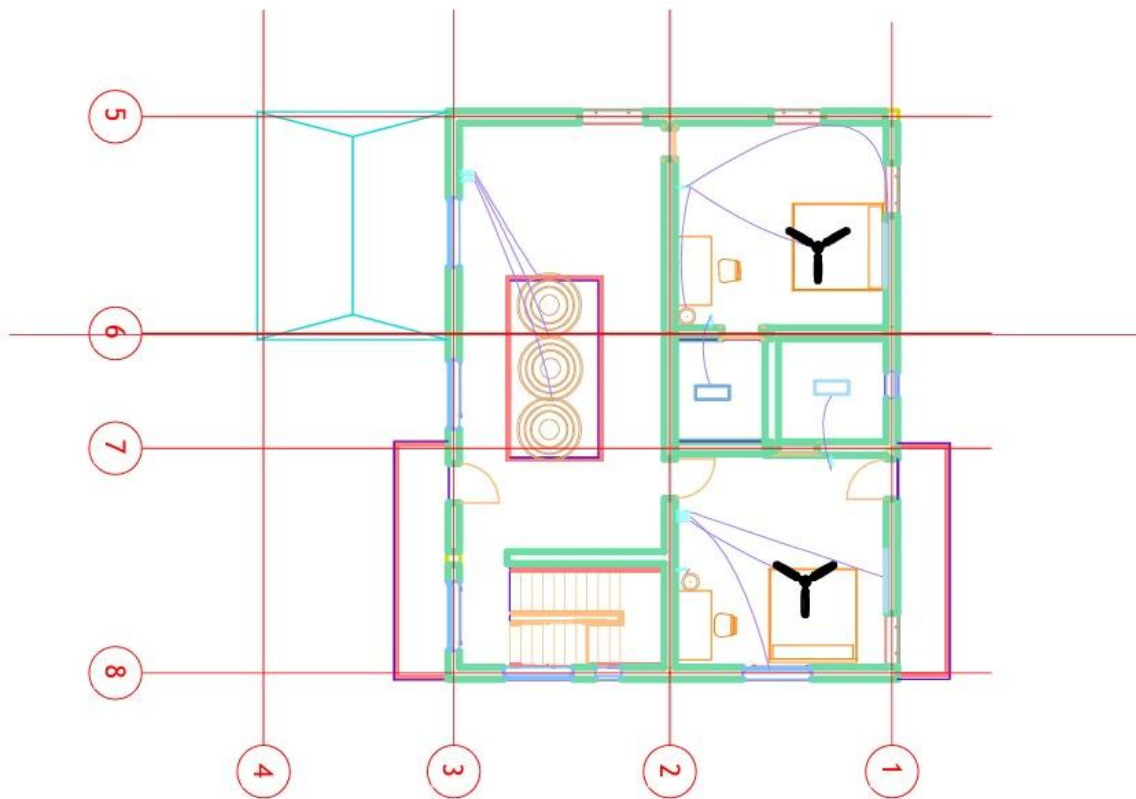
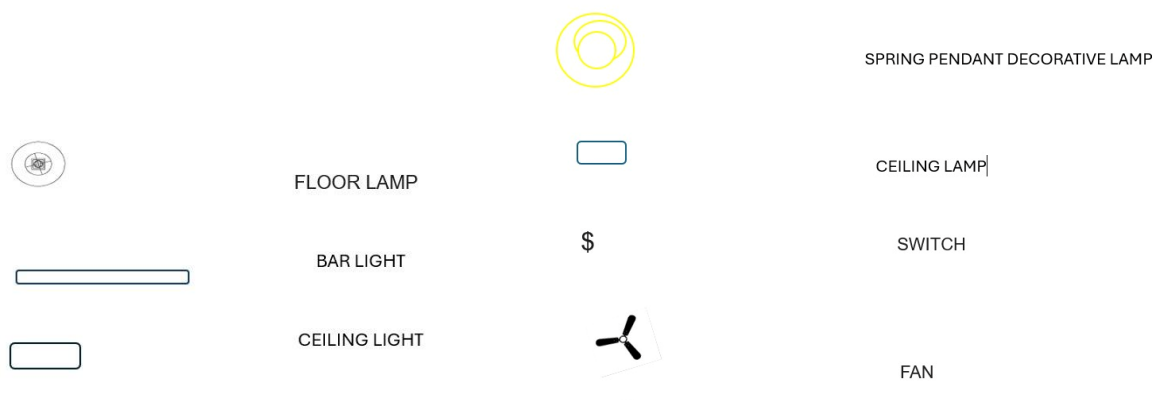


Figure No.7.2 First floor Electrical and Lightening Planning



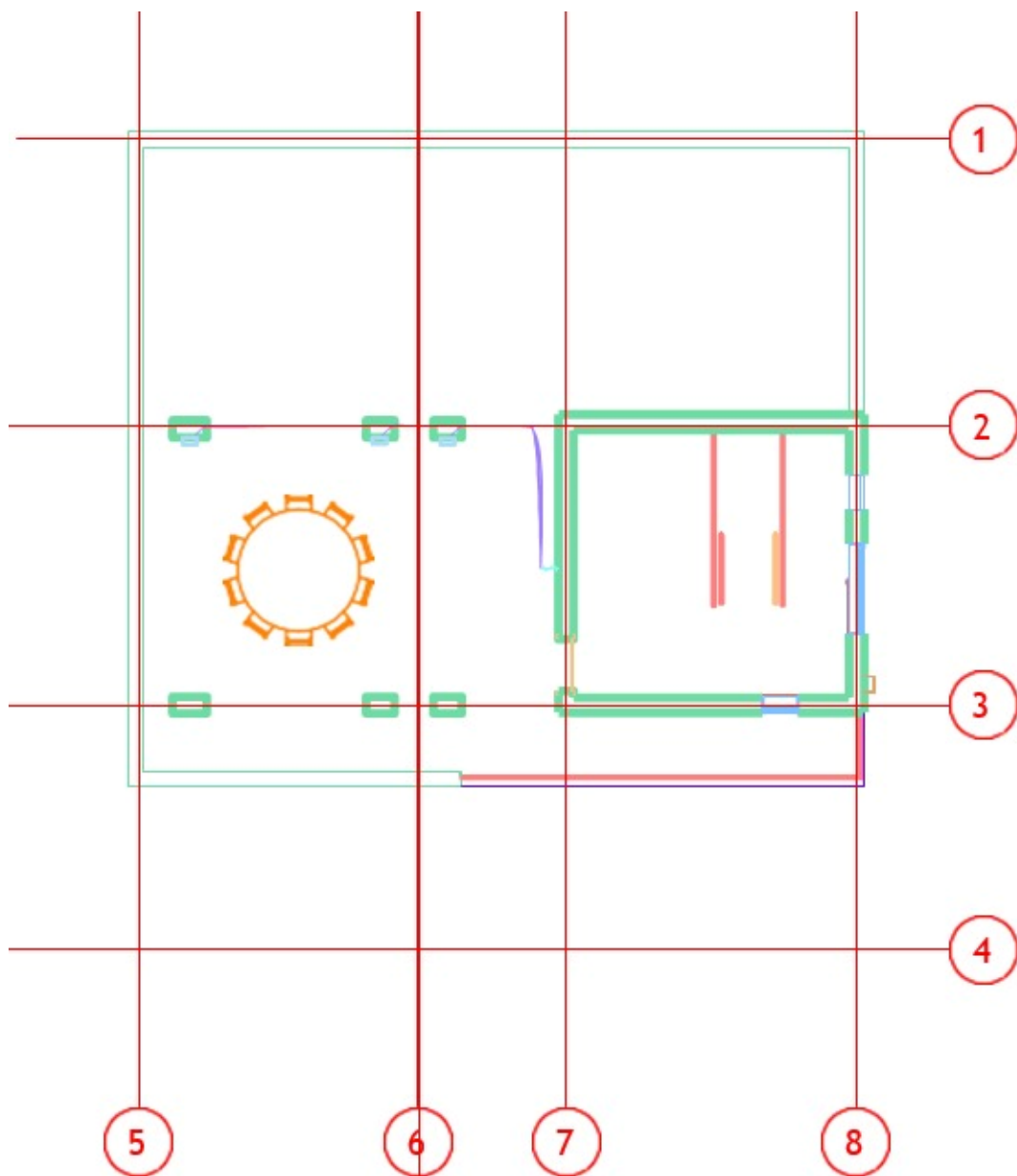


Figure No.7.3 Head Room electrical and lightening Plan

CHAPTER 8

QUANTITY ESTIMATION AND COSTING

7.1 Estimation and Costing

Estimation and costing refer to the processes involved in predicting the expenses associated with a construction project. Estimation involves calculating the quantities of materials, labor, and equipment needed for the project, while costing involves determining the monetary value of these resources and activities. It's crucial for budgeting, planning, and ensuring the financial feasibility of construction projects.

7.2 Components used for Estimation and Costing

- 1. Budgeting:** They help in establishing accurate budgets for projects, ensuring that resources are allocated efficiently.
- 2. Planning:** Estimation and costing aid in the planning phase by providing a clear understanding of the resources required and their associated costs, allowing for better project scheduling.
- 3. Resource Allocation:** They assist in allocating materials, labor, and equipment effectively, preventing shortages or overages that can lead to delays and cost overruns.
- 4. Bid Preparation:** For contractors, accurate estimation and costing are crucial for preparing competitive bids and securing contracts.
- 5. Risk Management:** By identifying potential cost implications early in the project, estimation and costing help in managing and mitigating financial risks.
- 6. Client Satisfaction:** Clients rely on accurate cost estimates to make informed decisions about project feasibility and investment.

Overall, estimation and costing play a vital role in ensuring the success, efficiency, and

profitability of construction projects.

In estimation and costing for construction projects:

1. **Excavation:** Refers to the process of digging and removing earth or soil to prepare the site for construction. It involves calculating the quantity of soil to be excavated, the cost of excavation equipment and labor, and any disposal or backfilling expenses.

2. **RCC Work (Reinforced Concrete Cement):** Involves estimating the materials and labor required for constructing structural elements using reinforced concrete, such as columns, beams, slabs, and foundations. This includes the cost of concrete, reinforcement bars, formwork, labor, and any additives or admixtures.

3. **Brickwork:** Involves estimating the materials and labor needed for constructing walls, partitions, and other masonry structures using bricks and mortar. This includes calculating the quantity of bricks, mortar, scaffolding, labor, and any additional materials like sand and cement.

4. **Plastering:** Involves estimating the materials and labor required for applying plaster or other finishing materials to walls and ceilings to create a smooth surface. This includes calculating the quantity of plaster, labor for application, scaffolding, and any additives or finishing touches.

5. **Painting:** Involves estimating the materials and labor needed for painting interior and exterior surfaces of buildings. This includes calculating the quantity of paint, primer, brushes, rollers, labor for preparation and application, scaffolding, and any additional materials like masking tape and drop cloths.

6. **Concrete:** Refers to estimating the materials and labor required for casting concrete in various construction elements such as foundations, slabs, walls, and columns. This includes calculating the quantity of concrete, reinforcement bars, formwork, labor for mixing and pouring, curing materials, and any additives or admixtures.

QUANTITY ESTIMATION OF GROUND FLOOR AND FIRST FLOOR

GROUND FLOOR

S No	Particulars of items and details of work	No	Length (m)	Breadth (m)	Depth(m)	Quantity (cum)
Earth work--						
1 Earth work in excavation in foundation						
	footing-1	1	1.755	1.350	0.300	0.711
	footing-2	1	1.950	1.500	0.350	1.024
	footing-3	1	1.690	1.300	0.300	0.684
	footing-4	1	1.755	1.350	0.300	0.816
	footing-5	1	2.015	1.550	0.350	1.234
	footing-6	1	2.275	1.750	0.370	1.389
	footing-7	1	2.145	1.650	0.400	1.244
	footing-8	1	1.885	1.450	0.360	0.950
	footing-9	1	1.820	1.400	0.350	0.860
	footing-10	1	1.755	1.350	0.350	0.952
	footing-11	1	2.015	1.550	0.400	1.048
	footing-12	1	1.690	1.300	0.300	0.659
	footing-13	1	1.690	1.300	0.300	0.659
	footing-14	1	1.690	1.300	0.300	0.659
total						12.889 m ^{cu}
2 Earthwork in filling in plinth						
	bedroom	1	4.800	4.730	0.480	10.898
	living room	1	7.300	4.600	0.480	16.118
	kitchen & dining	1	4.800	4.730	0.480	10.898
	toilet	1	2.100	2.250	0.480	2.268
	store room	1	2.100	2.250	0.480	2.268
	stair case	1	2.300	4.600	0.480	5.078
total						47.529 m ^{cu}
3 concrete						
cement concrete in foundation						
	footing-1	1	1.755	1.350	0.500	1.185
	footing-2	1	1.950	1.500	0.500	1.463
	footing-3	1	1.690	1.300	0.500	1.099
	footing-4	1	1.755	1.350	0.500	1.185
	footing-5	1	2.015	1.550	0.500	1.562
	footing-6	1	2.275	1.750	0.500	1.991

footing-7	1	2.145	1.650	0.500	1.770
footing-8	1	1.885	1.450	0.500	1.367
footing-9	1	1.820	1.400	0.500	1.274
footing-10	1	1.755	1.350	0.500	1.185
footing-11	1	2.015	1.550	0.500	1.562
footing-12	1	1.690	1.300	0.500	1.099
footing-13	1	1.690	1.300	0.500	1.099
footing-14	1	1.690	1.300	0.500	1.099

	cu
total	18.935 m

4 R.C.C work 1:2:4 excluding steel reinforcement
bars and its bending including centering and
shuttering and binding steel

R.C.C. slab --

Bedroom	1	4.800	4.730	0.120	2.724
Balcony	2	5.000	1.000	0.120	1.200
kitchen cum dining	1	4.800	4.730	0.120	
living room(deducting the opening of 4*2*0.12=0.96cum	1	7.300	4.600	0.120	3.070
car parking	1	4.600	3.400	0.120	1.877
Toilet	1	2.100	2.250	0.120	0.567
store room	1	2.100	2.250	0.120	0.567

	cu
total of slab	10.005 m

lintels--

over door

D	1	1.400	0.230	0.100	0.032
D1	1	1.300	0.230	0.100	0.030
D2	2	1.300	0.230	0.100	0.060

over windows

W	1	1.900	0.230	0.100	0.044
W1	7	1.600	0.230	0.100	0.258

total	0.4232
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**5 First class brick work in cement mortar in superstructure
in 30 cm walls of main rooms--**

Bed room,Kitchen cum dining,Toilet,store room

	11.10			16.849
long wall(12.900-4*0.45=11.100)	2	0	0.230	3.300 8
short wall	4	4.730	0.230	3.300 14.360
partition wall between toilet and store	1	2.250	0.230	3.300 1.708
Living Area				

long wall	1	7.900	0.230	3.300	5.9961
short wall	2	4.600	0.230	3.300	6.9828
Puja room					
long wall	1	2.400	0.230	3.300	1.8216
short wall	1	2.200	0.230	3.300	1.6698
stair case				3.300	
long wall	1	4.600	0.230	3.300	3.4914
short wall	1	2.300	0.230	3.300	1.7457
				total	52.917 m ^{cu}

Deduct--

Door openings

D	1	1.000	0.230	2.000	0.460
D1	1	0.900	0.230	2.000	0.414
D2	2	0.900	0.230	1.800	0.745
Opening for puja room	1	1.000	0.230	1.800	0.414
Windows opening					
W	1	1.500	0.230	1.200	0.414
W1	7	1.200	0.230	1.200	2.318
Ventilator V	3	0.600	0.230	0.400	0.166

Total deduction 4.9312

net
total 47.986 cu m

Wood work-Door and Window---

6 sal wood work in chaukhats in doors and windows--

Door

D	1	5.100	0.100	0.075	0.038
D1	1	5.000	0.100	0.075	0.038
D2	2	4.600	0.100	0.075	0.069
Sliding door	1	4.700	0.100	0.075	0.035

Window

W	1	5.400	0.100	0.075	0.041
W1	7	4.800	0.100	0.075	0.252

total 0.4725 m^{cu}

73 cm thick pannelled shutters of deadar wood in door and windows

Door

D	1	0.870	1.870	1.627
D1	1	0.770	1.870	1.440
D2	2	0.770	1.670	2.572
Sliding door	1	0.870	1.670	1.453
Window				
W	1	1.380	0.880	1.214
W1	7	1.080	0.880	6.653
			total	14.958 sq 7 m

8 Doors and windows fittings of oxidized iron	same as item(14)	total	14.95sq m
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Steel and Iron work----

9 Iron work in hold fasts and window bars---

	6@1K		
Hold fasts in doors	4 g	each	24 Kg
	4@1K		
Hold fasts in windows	8 g	each	32 Kg
20mm dia window bars @ 2.47 Kg per m---			
1 window (W)	1	8.000	1.200
7 Windows(W1)	7	7.000	1.000
			68.2 kg

total	224.45 kg
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Plastering and pointing

10 12mm plastering with 1:6 cement local sand mortor in walls

Inside plastering

	19.06	
Bed rooms	1	0
	23.80	
Living Room	1	0
	13.80	
staircase	1	0
	18.86	
Kitchen and dining	1	0
		3.300 62.898
		3.300 78.540
		3.300 45.540
		3.300 62.238

store room	1	8.700	3.300	28.710
Toilet	1	8.700	3.300	28.710
Shelves-cup board	1	9.000	3.300	29.700

	290.79
total	6 sq m

Deduct--

Door openings				
D	1	1.000	1.700	1.700
D1	1	0.900	1.700	1.530
D2	2	0.900	1.500	2.700
Sliding door	1	1.000	1.500	1.500
Stair case entrance openings	2	2.300	1.500	6.900
Window openings				
W	1	1.500	1.200	1.800
W1	7	1.200	1.200	10.080

	sq
Total of deductions	26.21 m

Net total of Inside plaster	264.58 sq 6 m
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11 Outside plastering

	22.93	85.987
Outer side (3.300+0.45)	1 0	3.750 5

Deduct--

Window openings				
W	1	1.500	1.200	1.800
W1	7	1.200	1.200	10.080
Door openings				0.000
D	1	1.000	1.700	1.700
D1	1	0.900	1.700	1.530
D2	2	0.900	1.500	2.700

	sq.
Total of deductions	17.810 m

Net total of Outside plaster =	68.177 cu 5 m
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Grand total of Inside and Outside plasterings =	sq m
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12 Skirting 20 cm high with 12mm thick 1:3

cement mortor neat cement finished---

	19.60	
Bed rooms	1 0	19.600
	23.80	
Living Room	1 0	23.800
	18.86	
Kitchen cum dining	1 0	18.860
toilet	1 8.700	8.700
	13.80	
staircase	1 0	13.800
store room	1 8.700	8.700
Doors	5 0.300	1.500
Jambs at entrance	2 0.300	0.600
deduct door openings	5 1.000	5.000

Net total	100.56
	0 m

14 Dado 12mm thick 1:3 cement coarse sand mortar
neat cement finished(in kitchen,bath)

	18.66	
Kitchen 50cm high	1 0	0.500 9.33
Bathroom 1m high	1 8.700	1.000 8.7
Jambs of bathroom	1 0.200	1.000 0.2

total	sq.
	18.23 m

Deduct--

Bathdoor D2	1 0.900	1.800 1.62
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total of deductions	sq.
	1.62 m

Net total	sq.
	16.61 m

15 Flooring

2.5 cm C.C 1:2:4 floor over and including
7.5cm lime concrete--

Bed rooms	1 4.800	4.730	22.704
Staircase	1 2.300	4.600	10.58
Living Room	1 7.300	4.600	33.58
sill of entrance	1 2.300	0.300	0.69
Kitchen	1 4.800	4.730	22.704
Toilet	1 2.100	2.250	4.725
store room	1 2.100	2.250	4.725

	sq.
total	89.128 m

16 2.5 cm C.C 1:2:4 floor (Without lime concrete)

sills of door D	1	1.000	0.200	0.2
sills of door D1	1	0.900	0.200	0.18
sills of door D2	2	0.900	0.200	0.18
stair case landing	1	2.300	0.900	2.07

	sq.
total	2.63 m

17 Painting

painting two coats over one coat of priming-

Panelled door D	2.25	1.000	2.000	4.5
Panelled door D1	2.25	0.900	2.000	4.05
Panelled door D2	4.5	0.900	1.800	7.29
panelled window W	2.25	1.500	1.200	4.05
panelled window W1	15.8	1.200	1.200	22.68
				3.1893
Painting window bars W	2.25	1.350	1.050	75
				17.364
Painting window bars W1	15.8	1.050	1.050	38

	63.123 sq
total	75 m

18 coal tar painting two coats

on back of chaukhats--

Door D	1	5.100	0.100	0.51
Door D1	1	5.000	0.100	0.5
Door D2	2	4.600	0.100	0.92
Window W	1	5.400	0.100	0.54
Window w1	7	4.800	0.100	3.36

	sq.
total	5.83 m

19 White washing and colour washing

white washing 3 coats walls (inside)

	19.06		
Bed rooms	1	0	3.300 62.898
	13.80		
staircase	1	0	3.300 45.540
	18.66		
Kitchen and dining	1	0	3.300 61.578

store room	1	8.700	3.300	28.710
Toilet	1	8.700	3.300	28.710
Shelves-cup board	1	9.000	3.300	29.700
			257.13	sq
total			6	m

Deduct--

Door openings				
D	1	1.000	1.700	1.700
D1	1	0.900	1.700	1.530
D2	2	0.900	1.500	2.700
Sliding door	1	1.000	1.500	1.500
Stair case entrance openings	2	2.300	1.500	6.900
Window openings				
W	1	1.500	1.200	1.800
W1	7	1.200	1.200	10.080
				sq
Total of deductions			26.210	m

Net total	230.92	sq.
	6	m

19 colour washing two coats over one coat of white washing outside walls

	22.93			sq
Outer side	1	0	3.750	85.988
				m

Deduct--

Window openings				
W	1	1.500	1.200	1.800
W1	7	1.200	1.200	10.080
Door openings				
D	1	1.000	1.700	1.700
D1	1	0.900	1.700	1.530
D2	2	0.900	1.500	2.700
				sq.
Net Total=			17.810	m

Misc. items

20 10 mm dia A.C. rain water pipes(6 nos) 6 3.600

FIRST FLOOR

S	No	Particulars of items and details of work	No	Length (m)	Breadth (m)	Depth(m)	Quantity (cum)
		1 R.C.C work 1:2:4 excluding steel reinforcement bars and its bending including centering and shuttering and binding steel					
		R.C.C. slab --					
		bedroom	2	4.800	4.730	0.120	5.449
		living room	1	7.300	4.600	0.120	4.030
		Mumty Room including projections of 750mm on four sides	1	3.800	6.100	0.120	2.782
		toilets	2	2.100	2.250	0.120	1.134
							cu
						total of slab	10.613 m
		Staircase					
		Steps	40	1.000	0.275	0.160	1.76
		Hand rail Glazed	4	3.939	0.100	0.590	0.9296
		landing	4	1.000	2.600	0.120	1.248
		Waist slab	4	3.939	1.000	0.120	1.89072
							cu
						Total	5.82832 m
		lintels--					
		over door					
		D1	3	1.300	0.230	0.100	0.090
		D2	3	1.300	0.230	0.100	0.090
		over windows					
		W	2	1.900	0.230	0.100	0.087
		W1	8	1.600	0.230	0.100	0.294
							cu
						total	0.561 m

First class brick work in cement mortar in 2 superstructure in 30 cm walls of main rooms--

Bed rooms ,Toilets

long wall(12.900-4*0.45=11.100)	2	11.100	0.230	3.000	15.318
short wall	4	4.730	0.230	3.000	13.055
partition wall between toilets	1	2.250	0.230	3.000	1.553
Living Area				3.000	
long wall	1	7.900	0.230	3.000	5.451
short wall	2	4.600	0.230	3.000	6.348
Mumty room				3.000	

long wall	1	4.600	0.230	2.500	2.645
short wall	1	2.300	0.230	2.500	1.3225
total					44.139 m ^{cu}

Deduct--

Door openings

D1	3	0.900	0.230	2.000	1.242
D2	3	0.900	0.230	1.800	1.118

Windows opening

W	2	1.500	0.230	1.200	0.828
W1	8	1.200	0.230	1.200	2.650
Ventilator V	1	0.600	0.200	0.400	0.048

Total deduction 5.885

net
total 38.254 cu m

Wood work-Door and Window---

sal wood work in chaukhats in doors and
3 windows--

Door

D1	3	5.000	0.100	0.075	0.113
D2	3	4.600	0.100	0.075	0.104

Window

W	2	5.400	0.100	0.075	0.081
W1	8	4.800	0.100	0.075	0.288

total 0.585 m^{cu}

43 cm thick pannelled shutters of deadar
wood in door and windows

Door

D1	3	0.770		1.870	4.320
D2	3	0.770		1.670	3.858

Window

W	2	1.380		0.880	2.429
W1	8	1.080		0.880	7.603

total 18.209 m^{sq}

5 Doors and windows fittings

of oxidized iron	same as item(14)	total	14.95sq m
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Steel and Iron work----

6 Iron work in hold fasts and window bars---

Hold fasts in doors	46@1Kg each	24 Kg
Hold fasts in windows	84@1Kg each	32 Kg

20mm dia window bars @ 2.47 Kg per m---

1 window (W)	2	8.000	1.200	19.2 m
7 Windows(W1)	8	7.000	1.000	49 m
				68.2 kg

total	224.45 kg
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Plastering and pointing

7 12mm plastering with 1:6 cement local sand mortar in walls

Inside plastering

Bed rooms	2	19.060	3.000	114.360
Living Room	1	23.800	3.000	71.400
Mumty room	1	13.800	3.000	41.400
Toilets	2	8.700	3.000	52.200
Shelves-cup board	1	9.000	3.000	27.000

total	264.960 m	sq
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Deduct--

Door openings

D1	3	0.900	1.700	4.590
D2	3	0.900	1.500	4.050

Window openings

W	2	1.500	1.200	3.600
W1	8	1.200	1.200	11.520

Total of deductions	23.760 m	sq
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Net total of Inside plaster	241.2 m	sq
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8 Outside plastering

Outer side (3.300+0.45)	1	22.930	3.750	85.9875 m ^{sq.}
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Deduct--

Window openings

W	2	1.500	1.200	1.800
W1	8	1.200	1.200	11.520
Door openings				0.000
D1	3	0.900	1.700	1.530
D2	3	0.900	1.500	4.050

Total of deductions	18.900 m ^{sq.}
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Net total of Outside plaster =	67.0875 m ^{cu}
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9 Skirting 20 cm high with 12mm thick 1:3 cement mortar neat cement finished---

Bed rooms	2	19.600		39.200
Living Room	1	23.800		23.800
toilets	2	8.700		17.400
Mumty Room	1	13.800		13.800
Doors	6	0.300		1.800
Jambs at entrance	2	0.300		0.600
deduct door openings	6	1.000		6.000

Net total	102.600 m
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Dado 12mm thick 1:3 cement coarse sand
10 mortar

neat cement finished(in kitchen,bath)

Bedrooms 50cm high	2	19.660	0.500	19.66
toilets 1m high	2	8.700	1.000	17.4
Jambs of bathroom	1	0.200	1.000	0.2

total	37.26 m ^{sq.}
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Deduct--

Bathdoor D2	2	0.900	1.800	3.24
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total of deductions	3.24 m ^{sq.}
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Net total	34.02 m ^{sq.}
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11 Flooring

2.5 cm C.C 1:2:4 floor over and including

7.5cm lime concrete--

Bed rooms	2	4.800	4.730	45.408
Living Room(deducting opening of area 4*2=8sq.m)	1	7.300	4.600	25.58
sill of entrance	1	2.300	0.300	0.69
Toilets	2	2.100	2.250	9.45

	sq.
total	81.128 m

12 2.5 cm C.C 1:2:4 floor (Without lime concrete)

sills of door D1	3	0.900	0.200	0.54
sills of door D2	3	0.900	0.200	0.54
stair case landing	1	2.300	0.900	2.07

	sq.
total	3.15 m

13 Painting

painting two coats over one coat of
priming-

Panelled door D1(3*2+0.25=6.75)	6.75	0.900	2.000	12.15
Panelled door D2	6.75	0.900	1.800	10.935
panelled window W	4.5	1.500	1.200	8.1
panelled window W1	18	1.200	1.200	25.92
Painting window bars W	4.5	1.350	1.050	6.37875
Painting window bars W1	18	1.050	1.050	19.845

	sq
total	83.3288 m

14 coal tar painting two coats

on back of chaukhats--

Door D1	3	5.000	0.100	1.5
Door D2	3	4.600	0.100	1.38
Window W	2	5.400	0.100	1.08
Window w1	8	4.800	0.100	3.84

	sq.
total	7.8 m

15 White washing and colour washing

white washing 3 coats walls (inside)

Bed rooms	2	19.060	3.300	125.796
Mumty room	1	13.800	3.300	45.540

Toilets	2	8.700	3.300	57.420
Shelves-cup board	1	9.000	3.300	29.700

total	258.456 m	sq
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Deduct--

Door openings

D1	3	0.900	1.700	4.590
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D2	3	0.900	1.500	4.050
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Stair case entrance openings	2	2.300	1.500	6.900
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Window openings

W	2	1.500	1.200	3.600
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W1	8	1.200	1.200	11.520
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Total of deductions	30.660 m	sq
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Net total	227.796 m	sq.
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16 colour washing two coats over one coat of
white washing outside walls

Outer side	1	22.930	3.750	85.988 m	sq
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Deduct--

Window openings

W	2	1.500	1.200	1.800
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W1	8	1.200	1.200	11.520
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Door openings

D1	3	0.900	1.700	1.530
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D2	3	0.900	1.500	4.050
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Net Total=	17.370 m	sq.
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COSTING FOR GROUND FLOOR AND FIRST FLOOR

			GROUND FLOOR			
			Abstract of Estimated cost			
sno	particulars of items of work	quantity	unit	rate Rs	per	amount Rs.
Earthwork-						
1	Earthwork in excavation in foundation	12.89	cu m	750	/cu m	96.675
2	Earthwork in filling plinth	47.529	cu m	714	/cu m	339.35706
Concrete-						
3	concrete in foundation	18.935	cu m	680	/cu m	12875.8
4	R.C.C work 1:2:4 excluding steel reinforcement bars and its bending including centering and shuttering and binding steel	10.005	cu m	9173.5	/cu m	91780.8675
Brick work						
5	First class brick work in lime mortar in superstructure in 30 cm walls of main rooms--	47.986	cu m	8288	/cu m	397707.968
Wood work-Door and Window---						
7	sal wood work in chaukhats in doors and window	0.4725	cu m	8720.85	/cu m	4120.601625
8	3 cm thick pannelled shutters of deadend wood in door and windows	14.9587	sq m	3259.05	/sq m	48751.15124
9	Doors and windows fittings of oxidized iron	14.9587	sq m	282.95	/sq m	4232.564165
Steel and Iron work----						
10	Iron work in hold fasts and window bars---	2.2245	quintal	870	/q	1935.315
Plastering and pointing						
11	12mm plastering with 1:6 cement local sand mortar in walls	332.763	sq m	1280.9	/sq m	426236.1267

12 Skirting 20 cm high with 12mm thick 1:3 cement mortar neat cement finished---	100.56	m	1637.7	/m	164687.112
13 Dado 12mm thick 1:3 cement coarse sand mortar neat cement finished(in kitchen,bath)	16.61	sq m	1298.8	/sq m	21573.068
Flooring					
14 2.5 cm C.C 1:2:4 floor over and including 7.5cm lime concrete--	89.128	sq m	1515	/sq m	135028.92
15 2.5 cm C.C 1:2:4 floor (Without lime concrete)	2.63	sq m	1222.75	/sq m	3215.8325
Painting					
16 painting two coats over one coat of priming-	63.123	sq m	1145.95	/sq m	72335.80185
17 coal tar painting two coats on back of chaukhats--	5.83	sq m	404	/sq m	2355.32
White washing and colour washing					
18 white washing 3 coats walls (inside)	230.926	sq m	720	/sq m	166266.72
19 colour washing two coats over one coat of white washing outside walls	17.81	sq m	1020	/sq m	18166.2
Miscellaneous items					
20 10 mm dia A.C. rain water pipes(6 nos)	21.6	m	270	m	583.2
					1550715.5

work	Labour requir	no	per	quantity	rate(per day)	amount Rs
Earthwork-						
	Mazdoor	3.25	10 cu m	60.419	600	11781.705
Concrete-						
	mazdoor	10	10 cu m	28.935	600	17361
	bhisti	2	10 cu m	28.935	450	2604.15
	head mason	0.25	10 cu m	28.935	450	325.51875
	mason	2.5	10cu m	28.935	600	4340.25
Brick work						
	mazdoor	15	10 cu m	78.429	600	70586.1
	bhisti	2	10 cu m	78.429	450	7058.61
	head mason	0.5	10 cu m	78.429	450	1764.6525
	mason	8	10 cu m	78.429	600	37645.92
Plastering						
	mazdoor	1.5	10 sq m	332.763	600	29948.67
	bhisti	0.1	10 sq m	332.763	450	1497.4335
	head mason	0.2	10 sq m	332.763	450	2994.867
	mason	1	10 sq m	332.763	600	19965.78
Painting						
	painter	0.5	10 sq m	68.953	750	2585.7375
flooring						
	mazdoor	1	10 sq m	91.758	600	5505.48
	bhisti	0.25	10 sq m	91.758	450	1032.2775
	head mason	1.5	10 sq m	91.758	450	6193.665
	mason	6	10 sq m	91.758	600	33032.88

wood work	carpenter	0.75	10 sq m	30.37	750	1708.3125
						257933.01
					total	1808648.542
					Add 8% for water supply and sanitary work	144691.8834
					Add 8% for Electrification work	144691.8834
					total	2098032.309
					Add 3% for contingencies	62940.96926
					add 2% for work charged establishment	41960.64617
					Total	2202933.924
					total	4011582.466

Plinth area rate of ground floor

Plinth area = 12.7 x 10.23

129.921

plinth area rate inclusive of water supply and sanitary work and electrif 520227/129.921

30877.09043 per sq m

First floor

Abstract of Estimated cost

particulars of items of work	quantity	unit	rate Rs	per	amount Rs.
Concrete-					
R.C.C work 1:2:4 excluding steel reinforcement bars and its bending including centering and shuttering and binding steel	16.43	cu m	9173.5	/cu m	150720.61
Brick work					
First class brick work in lime mortar in superstructure in 30 cm walls of main rooms--	38.254	cu m	8288	/cu m	317049.15
Wood work-Door and Window---					
sal wood work in chaukhats in doors and windows--	0.585	cu m	8720.85	/cu m	1906.5443
3 cm thick pannelled shutters of deadar wood in door and windows	18.209	sq m	3259.05	/sq m	5152.2366
Doors and windows fittings of oxidized iron	18.209	sq m	282.95	/sq m	15841.83
Steel and Iron work----					
Iron work in hold fasts and window bars---	2.2245	quintal	870	/q	2849.3621

Plastering and pointing

12mm plastering with 1:6 cement local sand mortar in walls	308.288	sq m	1280.9	/sq m	400403.81
Skirting 20 cm high with 12mm thick 1:3 cement mortar neat cement finished---	102.6	m	1637.7	/m	168028.02
Dado 12mm thick 1:3 cement coarse sand mortar neat cement finished(in kitchen,bath)	34.02	sq m	1298.8	/sq m	41597.955

Flooring

2.5 cm C.C 1:2:4 floor over and including 7.5cm lime concrete--	81.128	sq m	1515	/sq m	32775.712
2.5 cm C.C 1:2:4 floor (Without lime concrete)	3.15	sq m	1222.75	/sq m	3851.6625

Painting

painting two coats over one coat of priming-coal tar painting two coats on back of chaukhats--	83.328	sq m	1145.95	/sq m	84994.56
	7.8	sq m	404	/sq m	3151.2

White washing and colour washing

white washing 3 coats walls (inside)	227.796	sq m	720	/sq m	164013.12
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colour washing two coats over one coat of white washing outside walls	17.81	sq m	1020	/sq m	18166.2
...)	1410501.964				

work	Labour required	no	per	quantity	ate(per da	amount rs
Concrete-						
	mazdoor	10	10 cu m	10.613	600	6367.8
	bhisti	2	10 cu m	10.613	450	955.17
	head mason	0.25	10 cu m	10.613	450	119.39625
	mason	2.5	10cu m	10.613	600	1591.95
Brick work						
	mazdoor	15	10 cu m	49.11	600	44199
	bhisti	2	10 cu m	49.11	450	4419.9
	head mason	0.5	10 cu m	49.11	450	1104.975
	mason	8	10 cu m	49.11	600	23572.8
Plastering						
	mazdoor	1.5	10 sq m	308.2873	600	27745.857
	bhisti	0.1	10 sq m	308.287	450	1387.2915
	head mason	0.2	10 sq m	308.287	450	2774.583
	mason	1	10 sq m	308.287	600	18497.22
Painting						
	painter	0.5	10 sq m	83.328	750	3124.8
flooring						
	mazdoor	1	10 sq m	81.128	600	4867.68
	bhisti	0.25	10 sq m	81.128	450	912.69
	head mason	1.5	10 sq m	81.128	450	5476.14

	mason	6	10 sq m	81.128	600	29206.08
wood work	carpenter	0.75	10 sq m	36.418	750	2048.5125

178371.8453

total 1588873.8

Add 8% for water supply and sanitary work 127109.9

Add 8% for Electrification work 127109.9

total 1843093.6

Add 3% for contingencies 55292.809

add 2% for work charged establishment 36861.872

Total 1935248.3

total 3524122.1

Plinth area rate of ground floor

Plinth area = 12.7 x 10.23

129.921

plinth area rate inclusive of water supply and sanitary work and electrifications work 520227/129.921

27125.1 per sq m

CHAPTER 9

DISCUSSIONS

In the course of this BIM project, the implementation of Building Information Modeling technology significantly enhanced the coordination and efficiency of the construction process. One of the key results observed was a marked reduction in the project's overall timeline by 15%, compared to traditional project management methods. This reduction was largely due to the improved accuracy in clash detection during the pre-construction phase, facilitated by BIM software, which decreased the number of on-site errors and rework. Moreover, cost savings were notable, with a reduction of approximately 10% in total project costs, primarily resulting from the optimization of material procurement and utilization. Discussions with project stakeholders highlighted their satisfaction with the increased transparency and continuous collaboration enabled by BIM. The real-time sharing of project modifications and updates through a unified model reduced communication lags and discrepancies between parties. However, some challenges were also noted, including the initial resistance to adoption from team members unfamiliar with BIM technologies and the upfront costs associated with BIM software and training. These issues were progressively mitigated through targeted training sessions and the demonstrable benefits of BIM in early project phases, which helped in building a positive consensus. Overall, the results affirm the substantial value of integrating BIM into construction projects, suggesting that further investments in BIM training and technology will be beneficial in enhancing project outcomes across future initiatives.

This paragraph synthesizes both the positive outcomes and challenges encountered, providing a balanced view of the project's execution and results, typical in a project report's discussion section.

CHAPTER 10

CONCLUSION

In conclusion, the implementation of Building Information Modelling (BIM) in the design and construction of G+1 residential buildings has shown significant advantages. Through a comprehensive methodology including soil bearing capacity analysis, utilization of BIM software for planning, designing, plumbing, and electrical installations, as well as quantity estimation and costing, this project has demonstrated the efficiency and accuracy of BIM in the construction industry.

The use of BIM has facilitated better collaboration among multidisciplinary teams, improved visualization and understanding of project components, and enhanced decision-making processes. The integration of BIM in the various stages of the project has resulted in improved efficiency, reduced errors, and better project outcomes.

Furthermore, the results obtained from this project underscore the potential of BIM to streamline construction processes, improve project management, and optimize resource utilization. As the construction industry continues to evolve, embracing technologies like BIM will be crucial for achieving sustainability, cost-effectiveness, and quality in building projects.

Overall, this project highlights the immense value of BIM in revolutionizing the construction industry and lays a foundation for future research and implementation of advanced technologies in building design and construction.

CHAPTER 11

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INTERNSHIP CERTIFICATES:



TO WHOM IT MAY CONCERN

This is to certify that Ms. **Banagani Deekshitha** studying Bachelor of Technology in Civil Engineering Roll number **12002004** under **Sri Venkateswara University College of Engineering** has successfully completed VIII Semester Project Internship for the academic year 2023-2024 on **Comprehensive Water Supply Scheme To Kamalapuram** under our organization "**ASM associates**". During the period of her Project Internship program with us she was found punctual, hardworking and inquisitive.

We wish her every success in her future endeavors.

Yours faithfully,


Managing Partner

Suresh Babu Anagondi

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TO WHOM IT MAY CONCERN

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We wish her every success in her future endeavors.

Yours faithfully,


Managing Partner

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TO WHOM IT MAY CONCERN

This is to certify that Ms. **T. Anusha** studying Bachelor of Technology in Civil Engineering Roll number **12002058** under **Sri Venkateswara University College of Engineering** has successfully completed VIII Semester Project Internship for the academic year 2023-2024 on **Comprehensive Water Supply Scheme To Kamalapuram** under our organization "**ASM associates**". During the period of her Project Internship program with us she was found punctual, hardworking and inquisitive.

We wish her every success in her future endeavors.

Yours faithfully,


Managing Partner

Suresh Babu Anagondi

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TO WHOM IT MAY CONCERN

This is to certify that Mr. **Pujari Sainath** studying Bachelor of Technology in Civil Engineering Roll number **12002046** under **Sri Venkateswara University College of Engineering** has successfully completed VIII Semester Project Internship for the academic year 2023-2024 on **Comprehensive Water Supply Scheme To Kamalapuram** under our organization "**ASM associates**". During the period of his Project Internship program with us he was found punctual, hardworking and inquisitive.

We wish him every success in his future endeavors.

Yours faithfully,



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This is to certify that Mr. **Kalesamudrum Venkat Ragheva** studying Bachelor of Technology in Civil Engineering Roll number **12002031** under **Sri Venkateswara University College of Engineering** has successfully completed VIII Semester Project Internship for the academic year 2023-2024 on **Comprehensive Water Supply Scheme To Kamalapuram** under our organization "**ASM associates**". During the period of his Project Internship program with us he was found punctual, hardworking and inquisitive.

We wish him every success in his future endeavors.

Yours faithfully,



Managing Partner
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TO WHOM IT MAY CONCERN

This is to certify that Mr. Uppu Ravi Kumar studying Bachelor of Technology in Civil Engineering Roll number 12002073 under Sri Venkateswara University College of Engineering has successfully completed VIII Semester Project Internship for the academic year 2023-2024 on **Comprehensive Water Supply Scheme To Kamalapuram** under our organization "**ASM associates**". During the period of his Project Internship program with us he was found punctual, hardworking and inquisitive.

We wish him every success in his future endeavors.

Yours faithfully,


Managing Partner

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